Omni-QC

AI-Driven Dynamic Defect Prediction and QC Optimization

Business Proposal

Prepared By: Omni-QC Team

Table of Contents

1.0 Introduction

This chapter outlines the need for advanced quality management in manufacturing by examining the cost of defects, market growth, and the limitations of reactive inspection systems. It begins by identifying the specific pain points experienced by manufacturers across IPC classification tiers, before introducing the Omni-QC project as a proactive solution focused on risk-scoring and prevention. It concludes by reviewing current research trends, including Predictive Quality, machine learning in acceptance sampling, and the gap in prescriptive decision-support systems.

1.1 Background

THE EXPANDING MARKET AND HIGH COST OF MANUFACTURING DEFECTS

Southeast Asia has emerged as a vital production hub within the economically significant global electronics manufacturing industry, which is projected to reach USD 1,201.16 billion by 2035. The global electronics manufacturing services (EMS) market is currently valued at approximately USD 617.90 billion in 2025 and is expanding at a compound annual growth rate (CAGR) of 6.87% (Narayan, 2026). The Southeast Asian region accounts for approximately 11% of global semiconductor market share and roughly one-fifth of global semiconductor equipment manufacturing. Countries such as Malaysia, Vietnam, Singapore, Thailand, and the Philippines each play a distinct and growing role in the global electronics supply chain, with Malaysia's electronics industry alone producing 13% of global back-end semiconductors and driving 40% of the nation's export output (Tan, 2024).

Maintaining consistent quality in Printed Circuit Board (PCB) manufacturing has become increasingly critical and difficult, with the cost of poor quality reaching up to 20% of a company's revenue in extreme cases. As production volumes and product complexity grow, ensuring the reliability of PCBs, the foundational building blocks of virtually every electronic device, is paramount. Losses arise from rework, scrap, production delays, and downstream failures, making these costs highly detrimental. For example, a production run of 10,000 PCBs with a 5% defect rate results in 500 boards requiring additional attention or replacement, translating to thousands of dollars in losses for a single batch (ALLPCB, 2025).

INEFFICIENCIES IN CURRENT DETECTION METHODS

Current defect detection methods suffer from severe inefficiencies, primarily due to uniform inspection protocols that generate massive volumes of false alarms and cause inspector fatigue. Manufacturers rely on automated systems like Automated Optical Inspection (AOI), Solder Paste Inspection (SPI), and X-ray inspection alongside human inspectors who handle verification and judgment calls. However, the prevailing approach applies all these resources uniformly across every unit produced, regardless of actual risk. On some production lines, as many as 70% of the units flagged as defective by AOI are ultimately deemed acceptable after manual re-inspection (Zhang, 2025). One industry estimate calculated the cost of false detections from AOI at approximately USD 7,000 per panel annually, a figure that scales linearly with additional false calls (Laba, 2026). Meanwhile, human inspectors often suffer from eye fatigue and other health issues caused by repetitive product inspection over long hours, allowing defective parts to pass or acceptable parts to be incorrectly rejected (Pyun et al., 2019).

There is a pressing industry need for a data-driven approach that assesses the defect risk of individual units prior to inspection, preventing the structural inefficiency of uniform testing. Because defects tend to cluster around specific process conditions, equipment states, and material inconsistencies rather than being evenly distributed across batches, applying intensive inspection to every unit consumes labor, machine time, and material resources without proportionate quality gains. Omni-QC is developed in direct response to this gap.

1.2 Problems and Pain Points

The PCB industry organises its products into three reliability classes under the IPC standard, and the pain points of regional manufacturers are most usefully understood through this lens because each class operates on fundamentally different inspection economics. Class 1 covers general consumer electronics where functional performance is the primary requirement and cosmetic imperfections are tolerated, encompassing low-complexity IoT devices, computing peripherals, and similar high-volume goods (IPC, 2020). Class 2 covers industrial controls, communications equipment, and business machines where extended service life and uninterrupted operation are desired without being absolutely critical (IPC, 2020). Class 3 covers mission-critical applications in aerospace, defence, and medical devices, where reliability requirements and downstream regulatory frameworks typically demand 100% inspection coverage and permit no statistical sampling. Within Omni-QC's target geography, Tier 2 mid-tier contract manufacturers predominantly serve Class 1 production with some Class 2 mix, while Tier 1 high-volume EMS providers concentrate on Class 2 output. Class 3 sits outside the primary commercial focus but presents a secondary opportunity worth examining. The pain points described below trace the structural inefficiencies experienced within each class. These inefficiencies presuppose a baseline of digital instrumentation; manufacturers without integrated machine data face a separate structural constraint.

CLASS 1: THE SCRAP TRAIN IN HIGH-VOLUME CONSUMER PRODUCTION

Class 1 production environments are defined by extreme volume and thin per-unit margins, conditions under which even small percentage losses translate into considerable absolute financial exposure. The dominant pain point in this class is what the industry refers to as the Scrap Train: the cumulative cost of detecting defects at the end of the assembly line rather than at their point of origin. Industry data confirms that 60% to 70% of Surface Mount Technology defects originate much earlier in the line, specifically at the solder paste printing stage (Saki, 2026). By the time a defect surfaces under end-of-line AOI or X-ray inspection, the board has already absorbed the full material cost of integrated circuits, the energy cost of thermal reflow, and the labour cost of placement. The unit is then scrapped wholesale, and at Class 1 production volumes, the compounding effect of this late-stage detection generates a structurally significant financial bleed for the Tier 2 manufacturers that dominate this class. The scrap train burdens Tier 1 manufacturers because of their big absolute scale production. However, its impact is most existential for the Tier 2 manufacturers serving Class 1 due to their direct hits on their revenue and their very thin per-unit margins that translate to the viability of the entire production run. Mid-tier specialists are actively pursuing strategic design partnerships to differentiate themselves (MordorIntel, 2026), which makes a documented reduction in inspection overhead commercially meaningful when bidding for new contracts.

This problem is amplified by the high-mix nature of Class 1 manufacturing. Tier 2 contract manufacturers serving consumer OEMs frequently cycle their production lines between distinct board types to accommodate diverse client portfolios and rapid new product introductions. Each new product invalidates the static defect baselines and rule-sets that traditional inspection systems depend on, causing rapid model drift at the very stage where defects are most likely to be born. The combination of upstream defect origin, downstream-only detection, and a high-mix production environment that erodes the accuracy of static inspection rules defines the structural inefficiency that pre-inspection process intelligence is designed to resolve.

CLASS 2: THE INSPECTION TAX AND THE LIMITS OF HUMAN RE-INSPECTION

This pain point lands on an audience already primed to act on it, Tier 1 providers are actively investing in fully automated SMT lines and digital production systems to boost yield and reduce defects (Narayan, 2026). Class 2 production carries a higher reliability mandate than Class 1, as the equipment produced, including industrial controls, communications hardware, and business machines, is expected to deliver extended service life under continuous operation. To meet this expectation, Tier 1 EMS providers and the Class 2 portions of Tier 2 facilities calibrate theirAOI systems aggressively, biasing detection thresholds toward over-flagging rather than underdetection. The result is what the industry refers to as the Inspection Tax: legacyAOI systems generate overwhelming volumes of false positives because their pre-programmed algorithms lack the contextual awareness to distinguish genuine defects from acceptable variation. On solder joint inspection part of the production lines, up to 70% of units flagged as defective are confirmed acceptable upon manual review (Zhang, 2025), and the cost of this false-detection burden has been estimated at approximately USD 7,000 per panel annually (Laba, 2026). KLA's own product literature implicitly acknowledges this dynamic, positioning its newer optical systems explicitly around the reduction of inspection escapes and overkill (KLA, 2023).

The financial weight of the Inspection Tax is compounded by the biological limits of the workforce that absorbs it. Manufacturers respond to false-positive volumes by maintaining manual re-inspection teams. However, human visual inspectors achieve only 70 to 80% defect detection accuracy even at peak performance (See, 2012), and that performance degrades sharply within the first 15 minutes of sustained inspection work due to the well-documented vigilance decrement (Mackworth, 1948). Class 2 production therefore exposes a structural mismatch: a class whose reliability bar demands the most rigorous re-inspection process is the same class where re-inspection is most heavily relied upon, performed by inspectors operating below the accuracy threshold the class itself requires. This mismatch translates directly into OEM contract penalties, recall exposure, and yield loss for the Tier 1 EMS providers whose absolute Inspection Tax burden scales linearly with their production volumes.

CLASS 3: AUGMENTING MANDATORY FULL INSPECTION IN PRODUCTION

Class 3 production governs aerospace, defence, and medical electronics, where regulatory frameworks mandate 100% inspection coverage and tolerate no statistical sampling. The structural pain points described in Class 1 and Class 2 do not translate cleanly into this environment: scrap reduction through inspection avoidance is not commercially relevant when inspection is regulatorily required, and aggressive false-positive reduction is constrained by the principle that a Class 3 facility would rather over-flag than risk a defect escape into a flight system or implantable device. Compounding this, Class 3 is the one segment of the PCB inspection market where AI-enabled platforms have already established a meaningful presence, with vendors such as LandingAI, Instrumental, and DarwinAI targeting Class 3 buyers whose compliance budgets justify premium software investment.

Within these constraints, however, a secondary role remains. The 100% inspection mandate places significant strain on inspection throughput and on the human reviewers who certify each unit, and the same vigilance decrement that limits Class 2 manual re-inspection applies with greater force in Class 3, where every board passes through human review. Risk-based prioritisation can determine the order in which boards enter the human review queue and flag units carrying elevated probability scores for closer scrutiny, without altering the underlying requirement that all units be inspected. The system can also generate the structured probability and feature-attribution records that Class 3 audit documentation increasingly demands. This role is supplementary rather than transformative, but it allows Omni-QC to participate in Class 3 environments as an augmentation layer rather than a direct competitor to the dominant vision based incumbents.

Across the three classes, the pain points are not parallel restatements of a single problem but stratified expressions of how inspection economics fail at different reliability tiers. In Class 1, the failure is temporal: defects are born upstream and detected too late, on lines whose high-mix volatility further erodes static rule-based detection. In Class 2, the failure is the cost of compensating for AOI false positives with a fatigued human workforce operating below the accuracy ceiling the class requires. In Class 3, the failure is one of throughput strain on a mandated full-inspection process that cannot be reduced. Each of these failures admits a different shape of intervention, which the further sections of this chapter develop into concrete commercial positioning for the Tier 1 and Tier 2 segments where the immediate addressable opportunity is concentrated.

1.3 Project Introduction

Omni-QC is an AI-driven quality control optimization system that evaluates each unit's defect risk using real-time machine data before it reaches the inspection queue. By introducing this system of intelligence between production and inspection, the system converts the previously identified structural inefficiencies into a targeted, data-informed process. Where the current approach treats every unit on a production line identically, Omni-QC leverages historical production records to predict anomalies dynamically.

The system transforms quality control into a risk-proportionate operation by directing existing inspection resources toward units that are statistically most likely to carry defects. Rather than applying 100% intensive AOI coverage uniformly, Omni-QC assigns a defect probability score to every unit processed through the monitored production stages and routes each unit to an inspection tier calibrated to its individual risk level. The result is a measurable reduction in total inspection workload, typically targeted between 30 and 50 percent of current uniform coverage, without a corresponding increase in the rate of defective units reaching downstream processes or end customers.

Beyond inspection routing, Omni-QC develops a prescriptive capability over time. As it accumulates production data from a facility, the system identifies the specific machine parameters and process conditions most strongly associated with elevated defect risk and generates targeted advisory outputs for the floor technician. These Co-Pilot notifications specify the implicated machine parameter and the calculated corrective offset, bridging the gap between anomaly detection and actionable correction that existing systems leave unaddressed. The system operates on a human-in-the-loop model throughout: all advisory outputs are presented as recommendations to the floor technician, preserving human authority over production equipment and eliminating the legal and operational liability that would arise from autonomous machine control.

The commercial design of Omni-QC reflects the specific adoption constraints of Southeast Asian manufacturing. The system is delivered as a Software-as-a-Service (SaaS) subscription with zero upfront capital expenditure requirements, eliminating the hardware procurement cycle that has historically been the primary barrier to AI adoption in the mid-tier EMS segment. A 90-day fully operational Proof of Concept program allows manufacturers to validate the system's performance on their own production data, under their own process conditions, before any long-term financial commitment is required.

The strategic timing of Omni-QC's entry into the Southeast Asian electronics manufacturing market reflects a structural convergence of conditions that make the current moment uniquely favorable. The region accounts for a significant and growing share of global SMT production, yet the AI adoption gap in the Class 1/2 segment remains almost entirely unaddressed by existing vendors. The maturation of MES infrastructure across mid-tier facilities, the collapse in ML inference costs documented by Stanford HAI (2025), and the proliferation of government manufacturing digitisation grants across Malaysia and Vietnam have simultaneously reduced the technical barriers, cost barriers, and financial barriers to deployment to their lowest historical point. Omni-QC is positioned to capture this opportunity as a software-first, data-driven quality orchestration platform whose value proposition is expressed entirely in verifiable operational outcomes.

1.4 Analysis of Current Research

The academic and applied research landscape that underpins Omni-QC's approach is organized around three converging streams: the emergence of Predictive Quality as a distinct field within manufacturing process control, the growing body of evidence supporting machine learning approaches to acceptance sampling and defect classification, and the identification of a persistent gap in prescriptive, human-in-the-loop decision-support systems for factory floor deployment.

Predictive Quality refers to the application of statistical and machine learning models to production process data for the purpose of forecasting quality outcomes before inspection occurs. The foundational insight of this field is that defects are not randomly distributed across production output but instead cluster around specific process conditions, equipment states, and material variations, is well established in the manufacturing quality literature and directly motivates the upstream integration model Omni-QC employs. The practical implication is that a system capable of ingesting real-time process telemetry and modeling its relationship with historical defect outcomes can assign probabilistic risk scores to units as they exit each monitored stage, allowing inspection resources to be directed toward units that are statistically most likely to carry defects rather than applied uniformly.

The machine learning literature on structured tabular manufacturing data has consistently identified gradient-boosted decision tree architectures, and XGBoost specifically, as high performing models for defect prediction tasks on sensor log data. Miedema et al. (2024) demonstrate that XGBoost achieves competitive recall performance on imbalanced manufacturing datasets, where defective units represent a small minority of total production output, while maintaining inference speeds compatible with real-time production line throughput. The system's configuration for recall-prioritised classification, achieved through threshold tuning and class weight adjustment rather than as an intrinsic property of the algorithm itself, directly aligns with the asymmetric cost structure of manufacturing quality control, where a false negative (an undetected defect reaching a customer) carries consequences an order of magnitude more severe than a false positive (an acceptable unit routed to additional inspection).

The Explainable AI (XAI) literature identifies SHAP (SHapleyAdditive exPlanations) as the most widely deployed industrial XAI approach for model-based quality systems, with documented deployments including DarwinAI's DVQI system, where SHAP-informed model refinement reduced AOI false positive rates from 1.7% to 0.11% (Chung et al., 2023; Maged et al., 2026). The ability to decompose each unit's risk score into per-feature contributions is a prerequisite for regulatory audit compliance and for the technician-facing Co-Pilot advisory function that OmniQC builds on top of the predictive engine.

The most significant gap in the existing literature and commercial landscape is the absence of prescriptive, human-in-the-loop advisory systems that translate model outputs into actionable corrective parameter recommendations at the factory floor level. Current deployed systems function as open-loop alarms: they identify that a process is drifting and flag the anomaly but stop short of specifying what corrective action should be taken. No commercially available system currently bridges this gap by computing parameter-level offsets and presenting them as targeted recommendations on the technician's dashboard. The design of Omni-QC's Co-Pilot module is a direct response to this identified gap in the applied research and product landscape.

2.0 Product Capabilities and Innovation

This chapter outlines the design and operational capabilities of the Omni-QC system. It covers the product overview, technical architecture, and deployment model, followed by its integration within the PCB manufacturing process. The chapter also highlights key functional capabilities such as predictive risk scoring and inspection optimization, and concludes with the system's approach to continuous model monitoring and retraining to ensure sustained performance.

2.1 Product Overview

Omni-QC is an AI-driven inspection optimization platform for PCB manufacturing that improves how existing quality control tools are used on the factory floor. It integrates with Manufacturing Execution Systems (MES) and equipment such as AOI and X-ray systems to analyse real-time and historical production data. By assigning each unit a defect risk score, it enables inspection resources to be allocated more efficiently through a risk-based prioritisation approach.

The primary goal of Omni-QC is to significantly reduce inspection inefficiencies while maintaining or improving defect detection accuracy across production lines. It aims to lower total inspection workload by approximately 30 to 50 percent by directing resources toward high-risk units and minimizing unnecessary checks on low-risk outputs. In addition, the system seeks to reduce scrap losses, false positives, and human inspection fatigue by introducing predictive intelligence and prescriptive recommendations into existing factory operations.

Usually, a standardized way of manufacturing performs the check for the quality after the production process and every product is gone from the inspection in the same level. This method results in operational inefficiency, higher expenses for the firm, and excessive usage of inspection systems. In the end, factories waste their effort, finance, and time. Li and Chen (2024) reveal an issue with poor distribution of inspection resources in conventional quality control practices, illustrating the significance of new, smarter systems as Omni-QC.

Omni-QC is designed to address the issue by implementing a forecasting system of quality control, which examines current and instant machine data as well as previous production information to identify the potential defects ahead of the inspection phase. Using this prediction, products are grouped depending on their different risk tiers, allowing adjustments to be made to the inspection process as necessary.

The long-term vision of Omni-QC is to transform manufacturing quality control from a reactive inspection model into a proactive, data-driven decision system that prevents defects before they occur.

It envisions a manufacturing ecosystem where predictive analytics and human-in-the-loop AI continuously optimize production parameters and inspection strategies in real time. Ultimately, Omni-QC aims to become a foundational layer of intelligent quality orchestration for global electronics manufacturing, particularly across high-growth Southeast Asian production hubs.

2.2 Product Details and Functional Capabilities

FUNCTIONAL CAPABILITIES

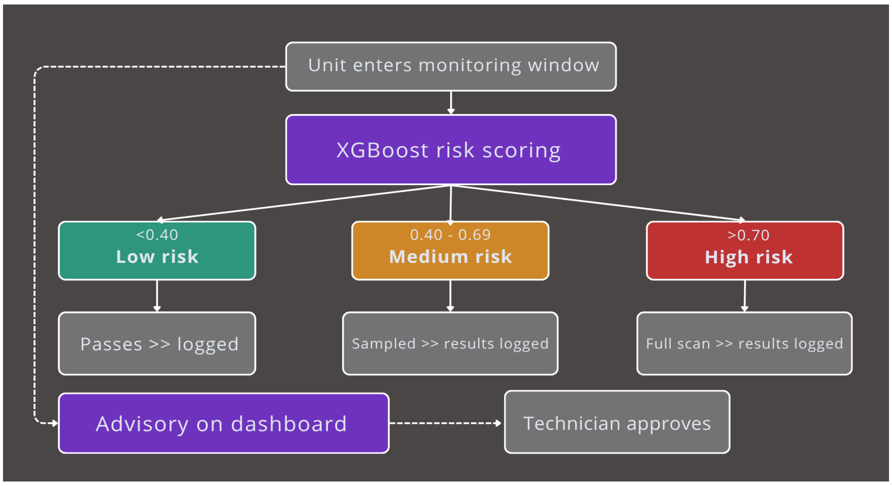


Figure 1. *Omni-QC Functionality General Flow Diagram*

Three-Tier Risk Classification

The primary output of Omni-QC is a defect probability score between 0.0 and 1.0 generated for every PCB unit that passes through the monitored production stages. This score is then used to route the unit into one of three inspection tiers: Units scoring below 0.40 are classified as Low Risk and routed to minimal inspection, typically a high-speed dynamic spot-check covering a small sample of the batch. Units scoring between 0.40 and 0.69 are classified as Medium Risk and directed to standard sampling protocols. Units scoring 0.70 or above are classified as High Risk and mandated for 100% intensive scanning through Automated Optical Inspection, X-ray inspection, or flying probe, depending on the facility's available equipment and the board type under production.

The default thresholds of 0.40 and 0.70 are initialised via cost-sensitive optimisation against the client's historical defect base rate, ensuring that the classification boundaries reflect the actual defect distribution of the specific facility rather than arbitrary round numbers. This calibration methodology is documented in the model validation report delivered to each client at onboarding.

This tiered structure allows the factory to concentrate its most resource-intensive inspection assets on the units that are statistically most likely to carry defects, while reducing the inspection burden on units that the model has assessed as low risk. The system's target objective is to reduce total inspection workload by 30 to 50% relative to the 100% uniform inspection baseline, without increasing the rate of defective units reaching downstream processes or end customers.

User-Configurable Risk Tolerance

The risk thresholds that define each tier are not fixed. Factory managers and quality engineers can adjust the sensitivity boundaries of the classification logic through the system's configuration interface to align with the specific quality standards of their production context. A facility producing boards for consumer electronics may tolerate a higher Low Risk threshold, directing more units to minimal inspection in order to maximise throughput. A facility producing boards where defect consequences are more severe, may lower the High Risk threshold to mandate intensive scanning for a broader range of units. This configurability ensures that Omni-QC can adapt to the full range of IPC classification standards and client-imposed quality requirements without changes to the underlying model.

Prescriptive Co-Pilot Advisory

Beyond unit-level risk scoring, Omni-QC develops over time the capability to attribute defect trends to specific, identifiable machine factors. When the system detects a statistically significant drift pattern, such as a recurring deviation in a particular reflow oven zone that correlates with elevated defect rates in the boards it processes, it generates a prescriptive advisory notification on the technician's dashboard. This notification specifies the nature of the drift and suggests a concrete corrective adjustment, for example a reduction of 1.2 degrees Celsius in a specific oven zone.

The system operates on a human-in-the-loop model. The advisory is presented to the floor technician as a recommendation, not an autonomous command. The technician reviews the suggested adjustment on the dashboard interface, which displays the underlying drift data alongside the proposed fix, and chooses whether to do further action. This architecture preserves human authority over production equipment, eliminates the legal and operational liability that would arise from autonomous machine control, and ensures that floor managers retain operational oversight at all times.

INTEGRATION IN THE PCB MANUFACTURING PROCESS

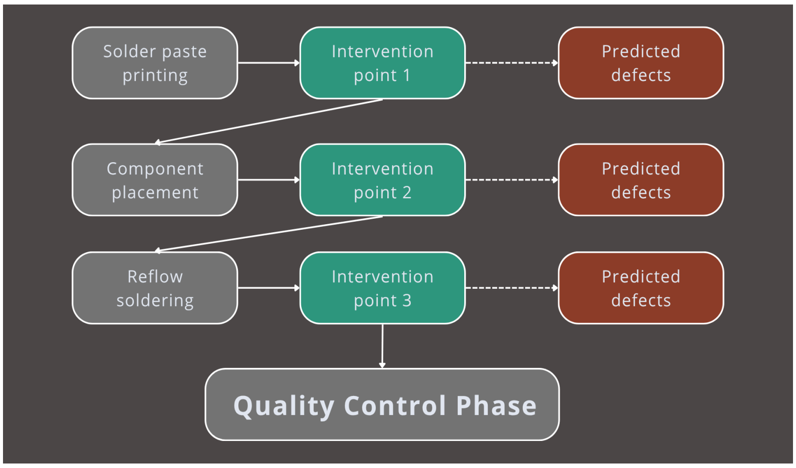


Figure 2. *PCB Assembly Integration Process Diagram*

Omni-QC integrates with both individual production machines and the facility's existing Manufacturing Execution System (MES). Data is captured at the machine level through existing sensor outputs and log files, then ingested by the system in real time. No additional dedicated sensing hardware is required at the point of installation, provided the facility's machines already generate structured data outputs, a prerequisite addressed in the market segmentation discussion.

The system monitors three specific stages of the PCB assembly process, each selected because they represent the highest-variance points in the production sequence, where process conditions are most likely to deviate in ways that produce downstream defects.

Solder Paste Printing Monitoring

The first integration point occurs at the conclusion of the solder paste printing stage. The system ingests data on stencil alignment precision, paste volume consistency, and paste viscosity measurements. The significance of this stage is considerable: over 60 to 70% of SMT defects originate from solder paste printing, making it the most defect-prone and consequential step in the surface mount assembly process (Saki, 2026). By evaluating paste deposition parameters before components are placed, Omni-QC can identify the probability of solder bridging or insufficient solder at the earliest possible moment in the assembly sequence, when intervention is least costly and most effective.

Component Placement Analysis

The second integration point occurs at the conclusion of the Pick-and-Place stage. The system ingests data on component placement coordinates, nozzle pressure readings, and component orientation relative to design file specifications. Deviation from these reference values is used to assess the probability of component misalignment, a category of defect that is frequently invisible to the naked eye at this stage but consistently detectable through the variance patterns in machine log data. By flagging placement anomalies before the board enters the reflow oven, the system allows intervention at a point where corrective action is still straightforward, rather than after the solder joint has been formed.

Reflow Oven Thermal Profiling

The third integration point occurs during the reflow soldering stage. The system monitors the thermal profile of the reflow oven across multiple zones, specifically reviewing soak time, ramp rate, and peak temperature zone by zone. A standard reflow profile progresses through a preheat phase, a soak phase, a reflow peak typically between 220 and 260 degrees Celsius for lead-free solder alloys, and a controlled cooling phase (PCBONLINE Team, 2021). If the system detects a thermal drift in which any zone deviates materially from the established profile, it flags the associated batch for elevated inspection and generates an advisory output for the floor technician.

2.3 Product User Dashboard

HOME - HERO LANDING VIEW

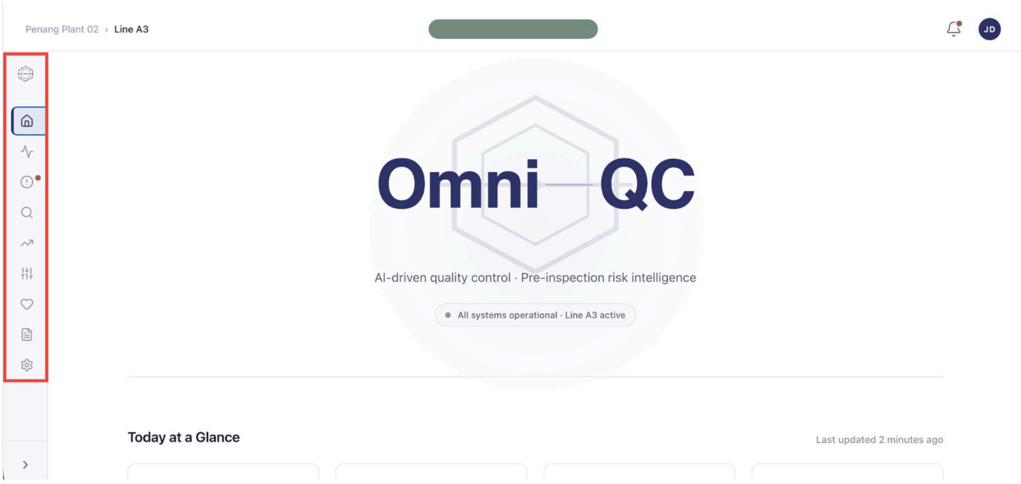


Figure 3. *Homepage (Brand Hero)*

This is the welcome screen the technical team sees the moment they sign in. The collapsed sidebar lists every section of the system: Home, Live Operations, Co-Pilot Advisories, Unit Inspector, Drift Analytics, Threshold Config, Model Health, Reports, and Settings. Each icon is one click away. The green pill confirms that all systems are operational and Line A3 is currently active, a quick at-a-glance signal before drilling into details. A red dot on the bell indicates new notifications waiting for review.

HOME - TODAY AT A GLANCE & QUICK ACTIONS

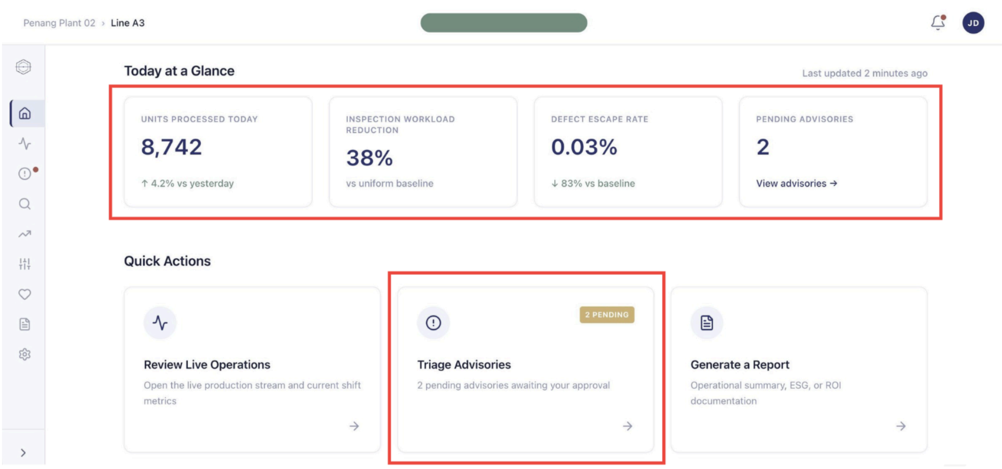


Figure 4. *Homepage (Daily Summary)*

Scrolling down from the hero reveals the day's headline numbers. These show units processed (8,742), inspection workload reduction (38%), defect escape rate (0.03%), and pending advisories (2). Each KPI is paired with a comparison line showing whether things are improving or declining. The Quick Actions area below routes the user directly to the most important destinations, Live Operations for real-time monitoring, Triage Advisories for pending approvals (notice the amber "2 PENDING" badge), and Generate a Report for management documentation.

HOME - TODAY AT A GLANCE & QUICK ACTIONS

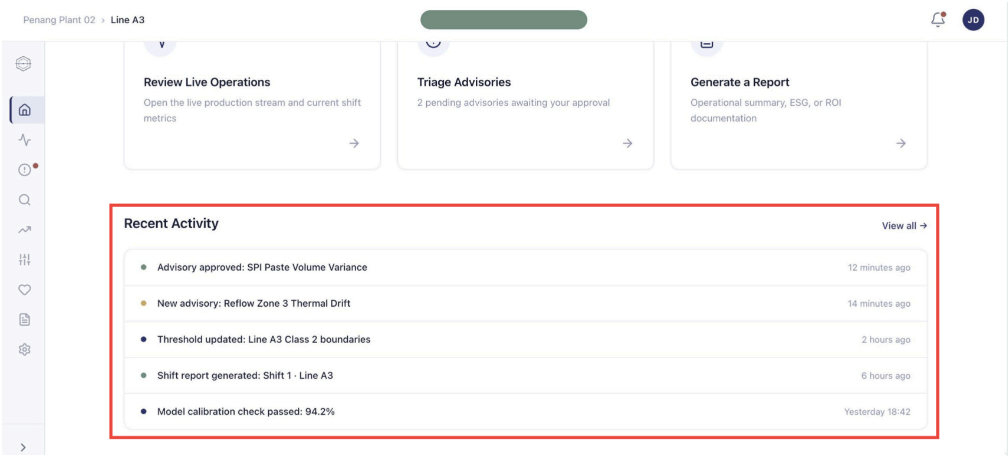


Figure 5. *Homepage (Recent Activity)*

At the bottom of the home view is a chronological feed of recent system events. Green dots indicate completed actions (advisory approved, shift report generated), amber dots indicate new advisories needing attention, and navy dots indicate routine system events (threshold updates, calibration checks). This gives the team a quick history of what's happened since they last logged in.

LIVE OPERATIONS - PRODUCTION STREAM DASHBOARD

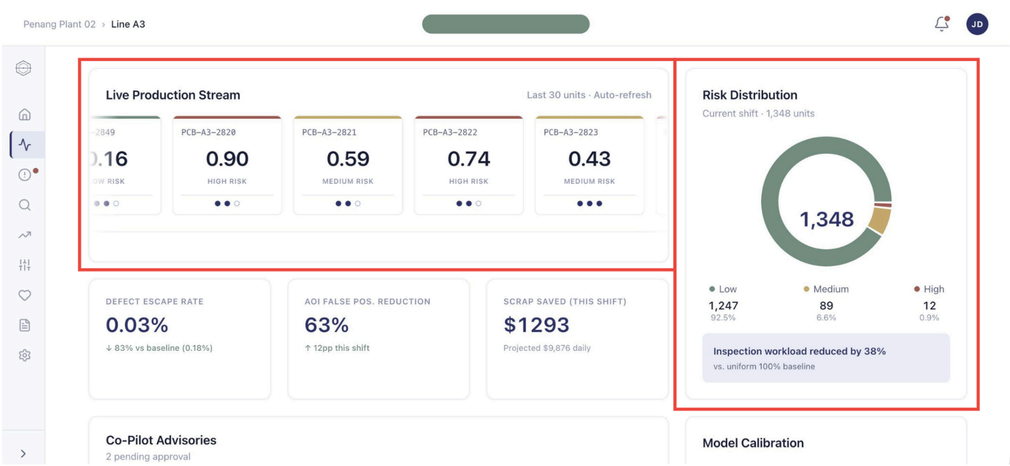


Figure 6. *Live Operations (Production Stream Dashboard)*

This is the primary monitoring screen the technician uses 90% of their shift. The top-left card shows the most recent 30 PCB units flowing through the line. Each unit displays its ID, risk score (color-coded: green for Low, amber for Medium, red for High), and stage progress dots. The donut chart on the right summarises the entire current shift, 1,348 units total, broken down into Low (1,247), Medium (89), and High (12) risk tiers. Below the donut is the headline operational outcome: "Inspection workload reduced by 38% vs. uniform 100% baseline." Defective escape rate, AOI false positive reduction, and scrap saved this shift, the three numbers that justify Omni-QC's value to management.

CO-PILOT ADVISORY - DRIFT VISUALIZATION & RECOMMENDED CORRECTION

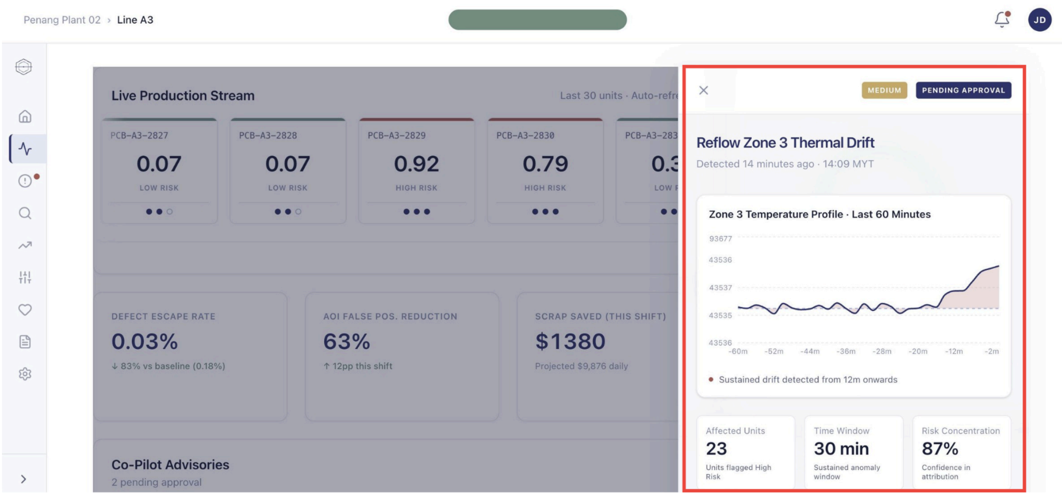


Figure 7. *Advisory Detail (Drift Diagnosis)*

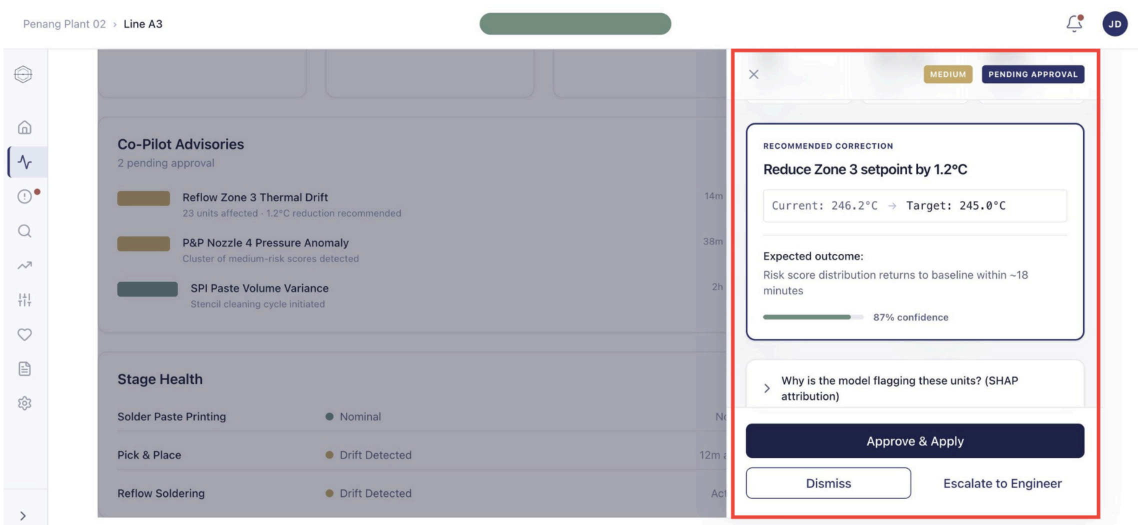


Figure 8. *Advisory Detail (Decision Approval)*

When the technician clicks an advisory, a side drawer slides in from the right. These tags identify severity and current status. The chart visualises Zone 3's actual temperature against its target profile over the last 60 minutes, the shaded area shows where the actual temperature has deviated from the target, making the drift pattern obvious at a glance. 23 units affected, 30-minute sustained anomaly window, 87% confidence in the model's attribution. These numbers establish how serious the drift is before the technician makes a decision.

The model's prescriptive recommendation is the most important element on this screen, "Reduce Zone 3 setpoint by 1.2°C, with the expected outcome ("risk score returns to baseline within ~18 minutes") and confidence level (87%) shown beneath. The technician has three choices: Approve & Apply (sends the correction to the equipment), Dismiss (rejects the advisory), or Escalate to Engineer (passes it up the chain). This is the human-in-the-loop architecture in action, the AI recommends, the technician decides.

CO-PILOT ADVISORY - SHAP FEATURE ATTRIBUTION

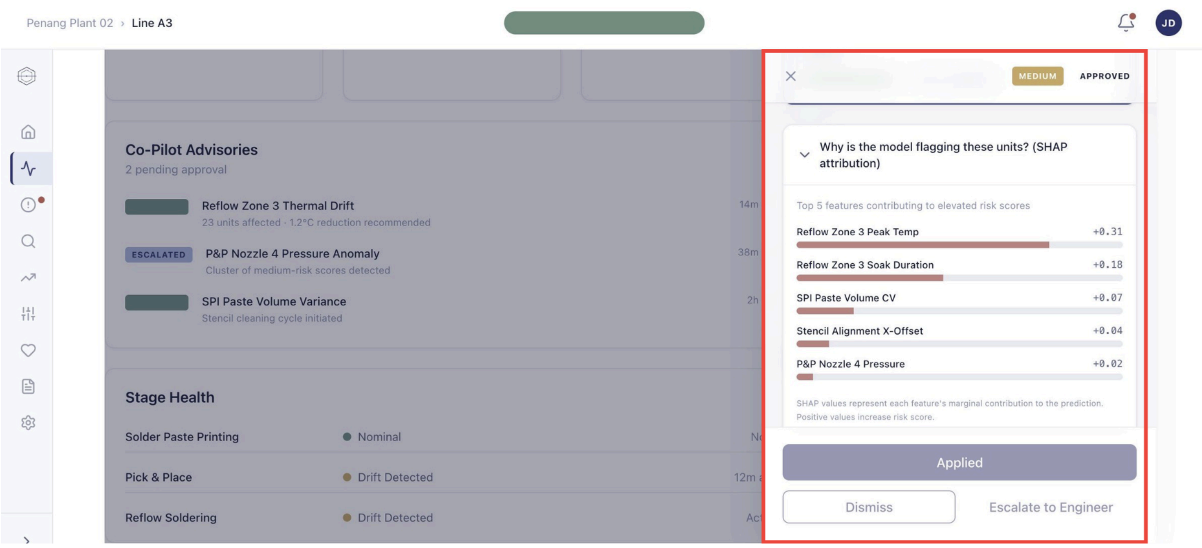


Figure 9. *Advisory Detail (SHAP Explanation)*

When a technician wants to understand *why* the model is flagging this drift, they expand the SHAP accordion. The chart breaks down which process variables contributed most to the elevated risk scores: Reflow Zone 3 Peak Temperature (+0.31) is the dominant contributor, followed by Reflow Zone 3 Soak Duration (+0.18), and so on. Notice that the second advisory in the background list now shows an "ESCALATED" status tag, demonstrating how the system tracks state changes across advisories.

CO-PILOT ADVISORY - TRIAGE PAGE

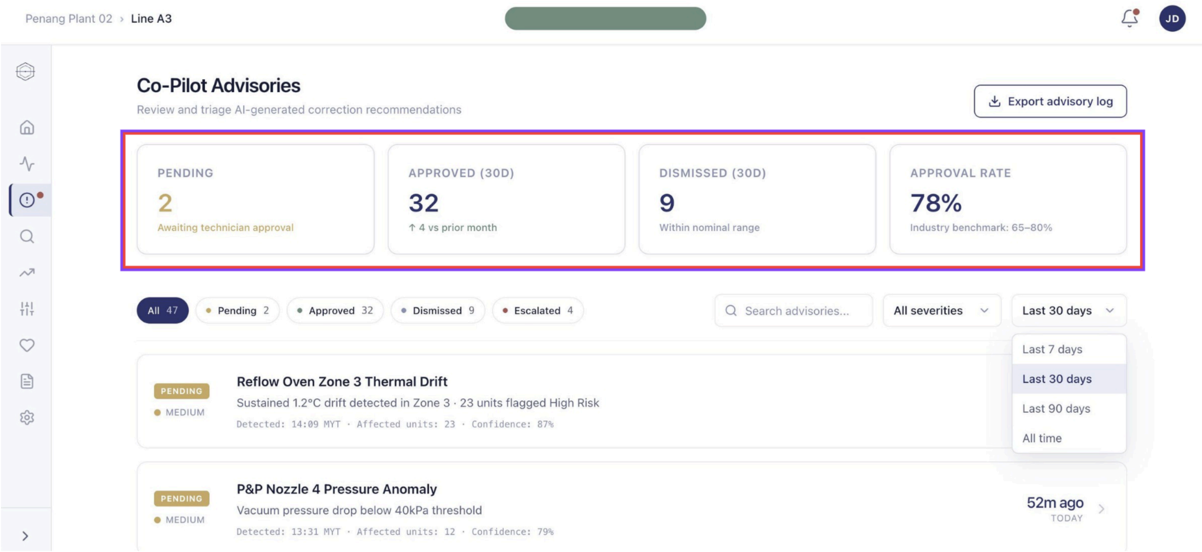


Figure 10. *Full Triage View*

This is the dedicated page where engineers triage and review advisory history. The summary strip shows pending count, 30-day approval count, dismissed count, and the team's overall approval rate (78%, within the industry benchmark of 65–80%). The filters below let the team narrow by status (All / Pending / Approved / Dismissed / Escalated). The dropdown demonstrates how the user can change the reporting window from Last 7 days up to All time. Below, individual advisory rows show full detail: title, description, detected timestamp, affected units, and confidence.

CO-PILOT ADVISORY - TRIAGE PAGE

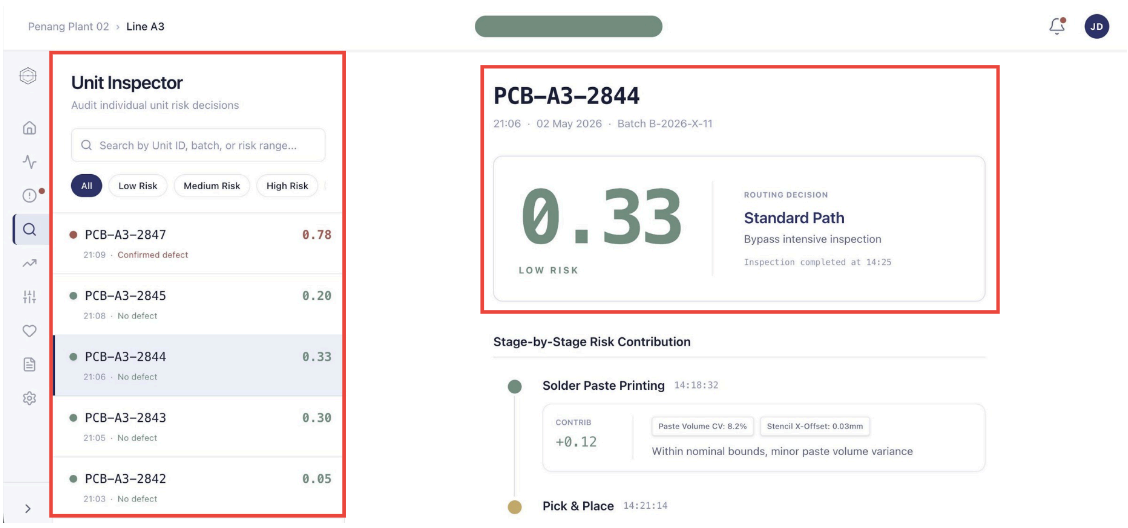


Figure 11. *Unit Inspector*

This is where engineers audit individual PCB units to understand exactly why each one received its risk score. The left panel lets users search by Unit ID, batch, or risk range, with quick filter chips for risk tiers. The selected unit is highlighted with a left border in primary color. The right panel shows the unit's final risk score, the routing decision the system made (Standard Path = bypass intensive inspection), and a stage-by-stage breakdown of how that score was computed.

DRIFT ANALYTICS - RISK DISTRIBUTION & TREND

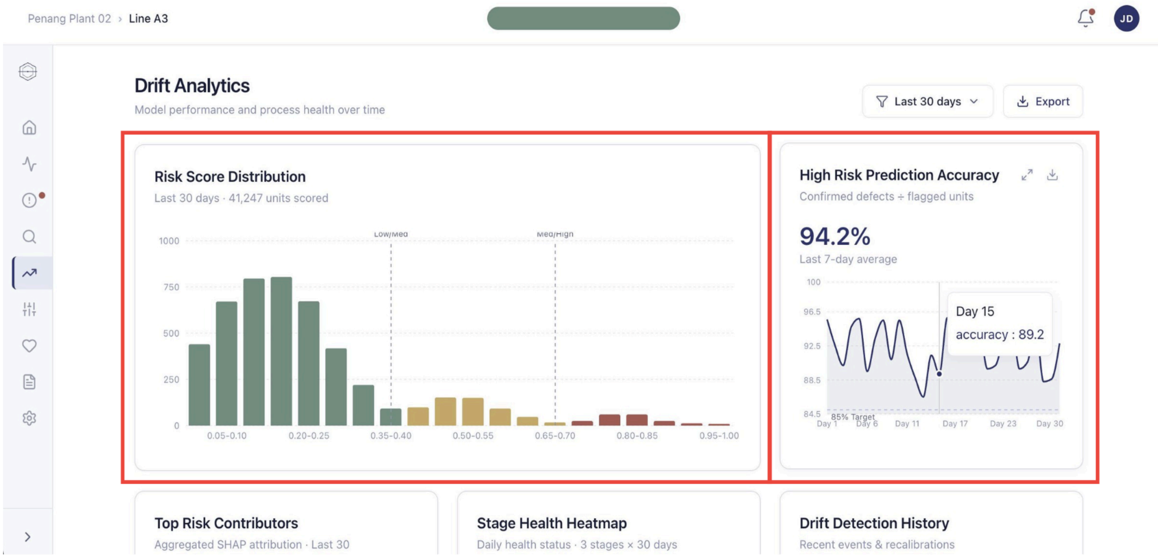


Figure 12. *Drift Analytics (Model Performance Trends)*

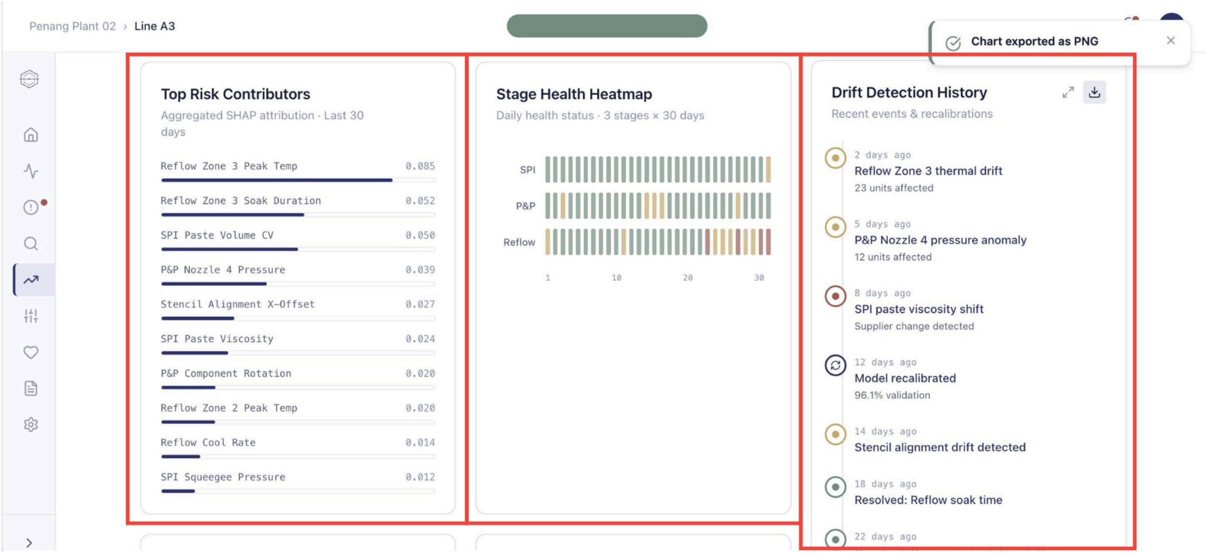


Figure 13. *Drift Analytics (Aggregated Insights)*

This is the engineer's analytics surface for reviewing how the model has been performing over time. The page filters by selectable time windows. The histogram shows how the model has distributed scores across the 41,247 units scored in the last 30 days, with vertical dashed lines marking the Low/Medium and Medium/High thresholds. Hovering on the accuracy chart reveals exact daily values, useful for spotting anomalies like the dip at Day 15.

Aggregated SHAP attribution across all units in the last 30 days, Reflow Zone 3 Peak Temp (0.085) is the largest single risk contributor across the production line. The heatmap shows daily health status for each of the three stages (SPI, P&P, Reflow) over 30 days, green = nominal, amber = drift detected, red = critical. A section confirming "Chart exported as PNG" demonstrates the export-to-file feature available on every chart.

DRIFT ANALYTICS - WORKLOAD REDUCTION & ADVISORY RESPONSE

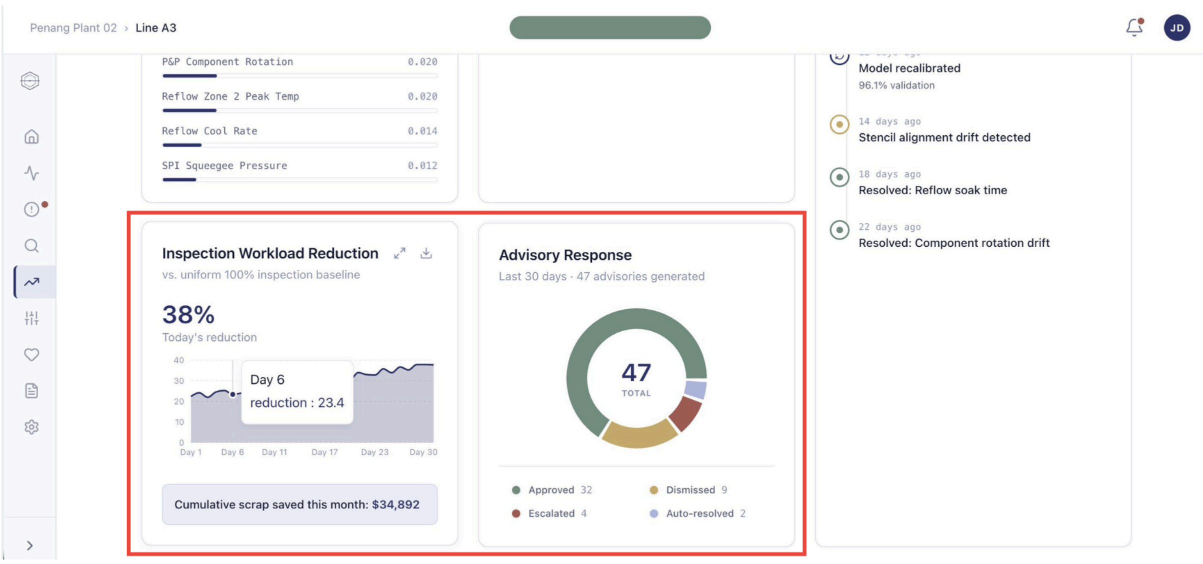


Figure 14. *Drift Analytics (Operational ROI)*

Today's inspection workload reduction is shown prominently. The dollar value of scrap saved this month, directly translates Omni-QC's value into financial terms. The chart breaks down how the team has responded to all 47 advisories in the last 30 days: 32 approved, 9 dismissed, 4 escalated, 2 auto-resolved.

THRESHOLD CONFIGURATION - STEP 1: PRODUCTION LINE

This is where quality engineers adjust the model's risk tier boundaries. A warning at the top makes clear the changes will affect future units only, current production cycle is not re-classified. The configuration is broken into 4 clear steps: Production Line -> IPC Class -> Adjust Thresholds -> Review & Confirm. Step 1 starts by selecting which production line the engineer wants to configure.

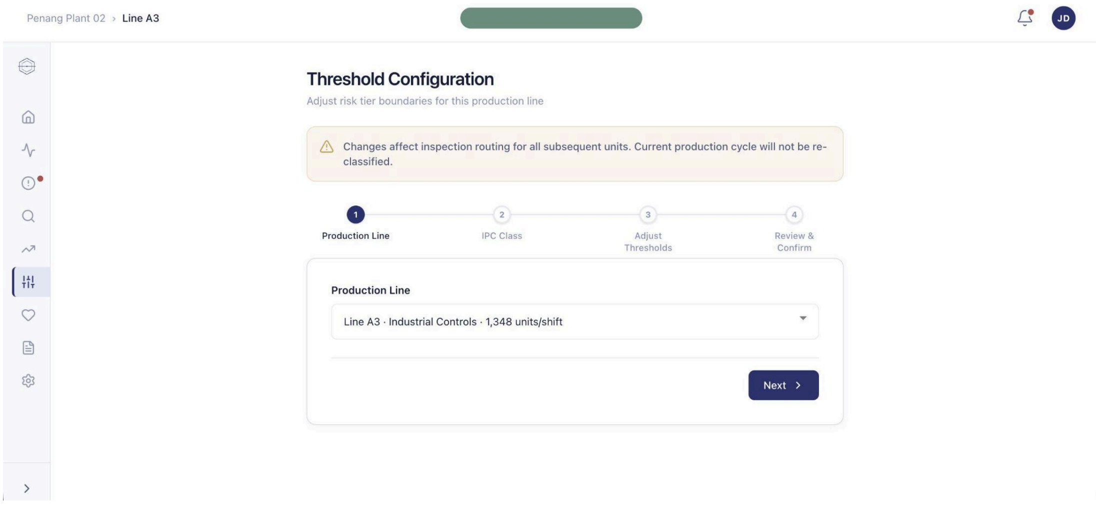


Figure 15. *Threshold Configuration (Part 1)*

THRESHOLD CONFIGURATION - STEP 2: IPC CLASS SELECTION

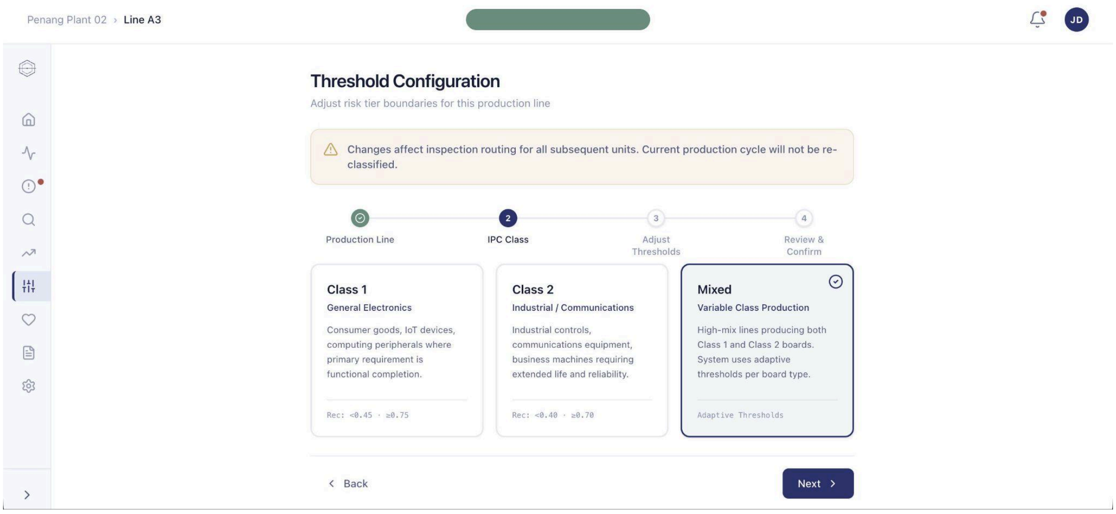


Figure 16. *Threshold Configuration (Part 2)*

Step 2 asks the engineer to select the IPC classification standard. Each card describes the use case (consumer goods, industrial controls, or mixed-class production) and shows the recommended threshold values. The engineer has selected Mixed in this example, which triggers adaptive thresholding behavior in the next step.

THRESHOLD CONFIGURATION - STEP 3: ADAPTIVE THRESHOLDS NOTICE

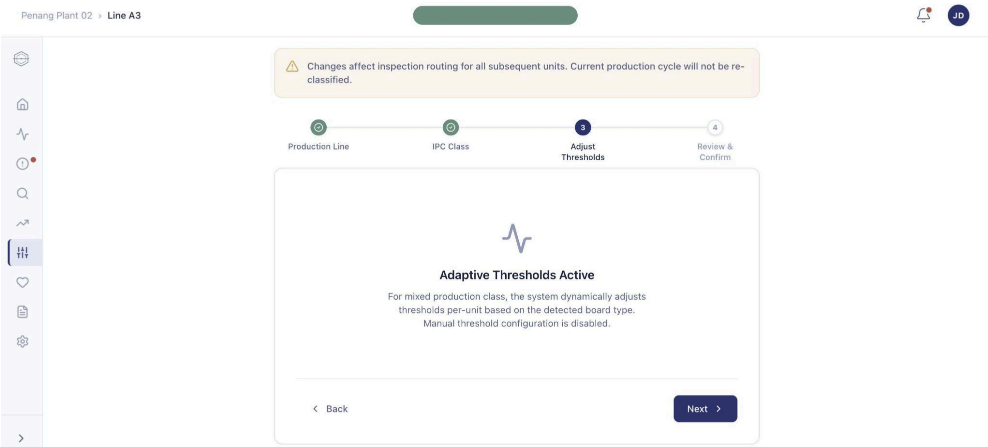


Figure 17. *Threshold Configuration (Part 3)*

Because the engineer chose the "Mixed" class in step 2, manual threshold configuration is disabled, the system instead uses adaptive thresholds that adjust per-board automatically. This protects engineers from misconfiguring a mixed-class line.

THRESHOLD CONFIGURATION - STEP 4: REVIEW AND CONFIRM

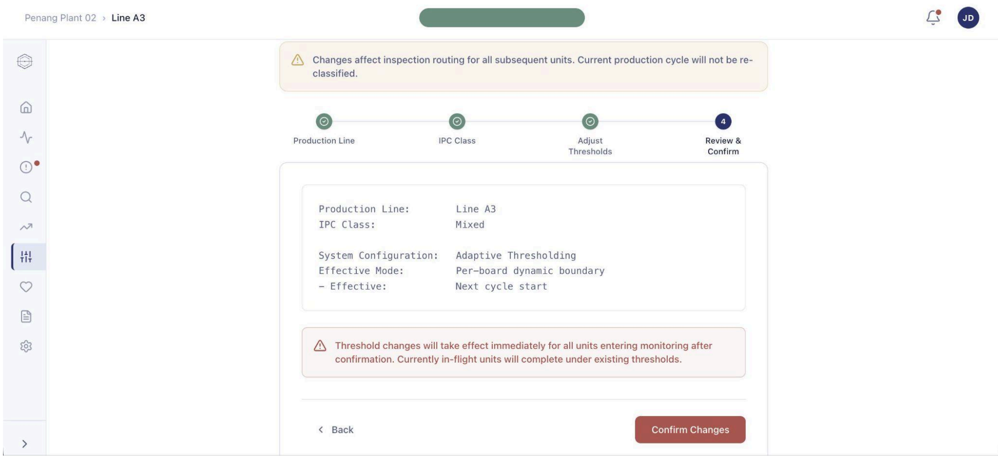


Figure 18. *Threshold Configuration (Part 4A)*

MODEL HEALTH - CALIBRATION GAUGE AND LIFECYCLE

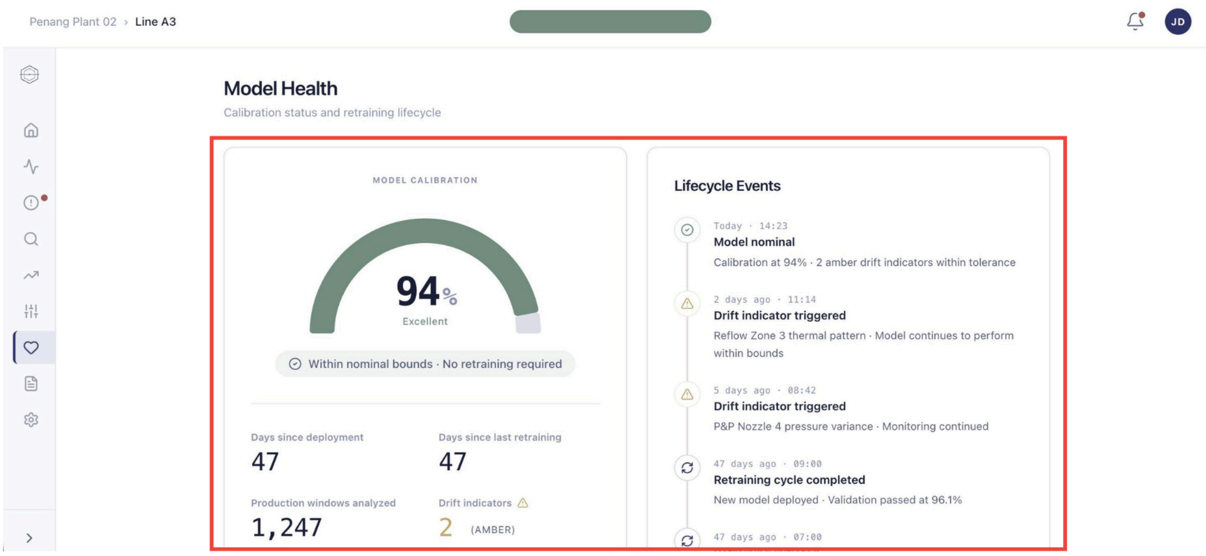


Figure 19. *Model health (Calibration Status)*

This is where engineers and managers monitor overall model health. The current calibration accuracy is 94%, labeled "Excellent" with the message "Within nominal bounds · No retraining required". Key vital signs: 47 days since deployment, 47 days since last retraining, 1,247 production windows analyzed, 2 amber drift indicators (within tolerance). Every meaningful event in the model's history: drift indicators triggered, retraining cycles completed, performance reviews, appears in chronological order.

MODEL HEALTH - INITIATE RETRAINING MODAL & TOAST

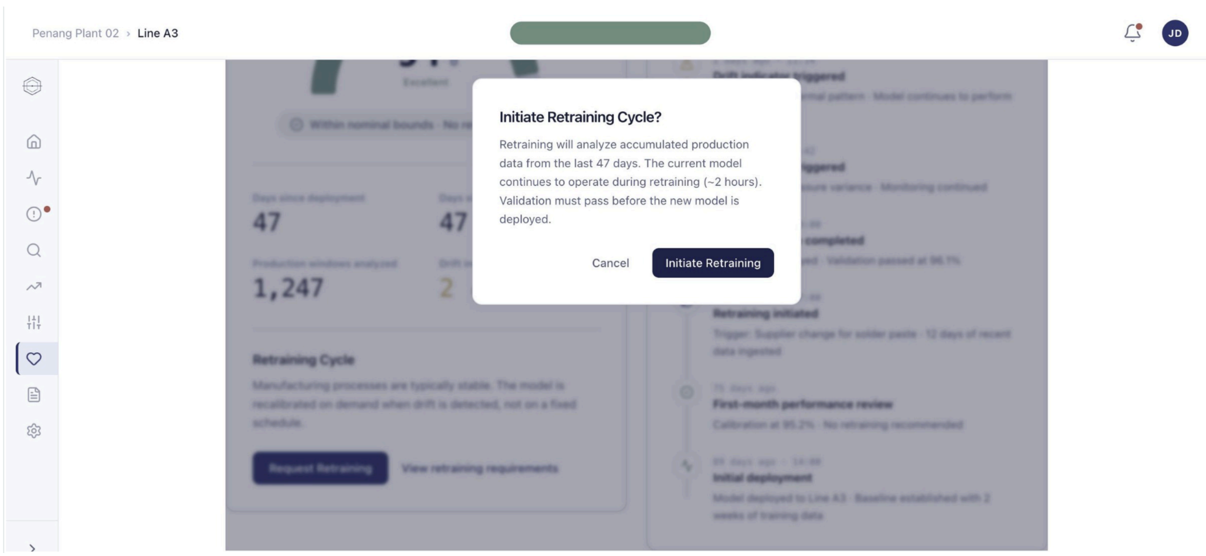


Figure 20. *Model Health (Retraining Confirmation)*

REPORTS - GENERATE NEW REPORT

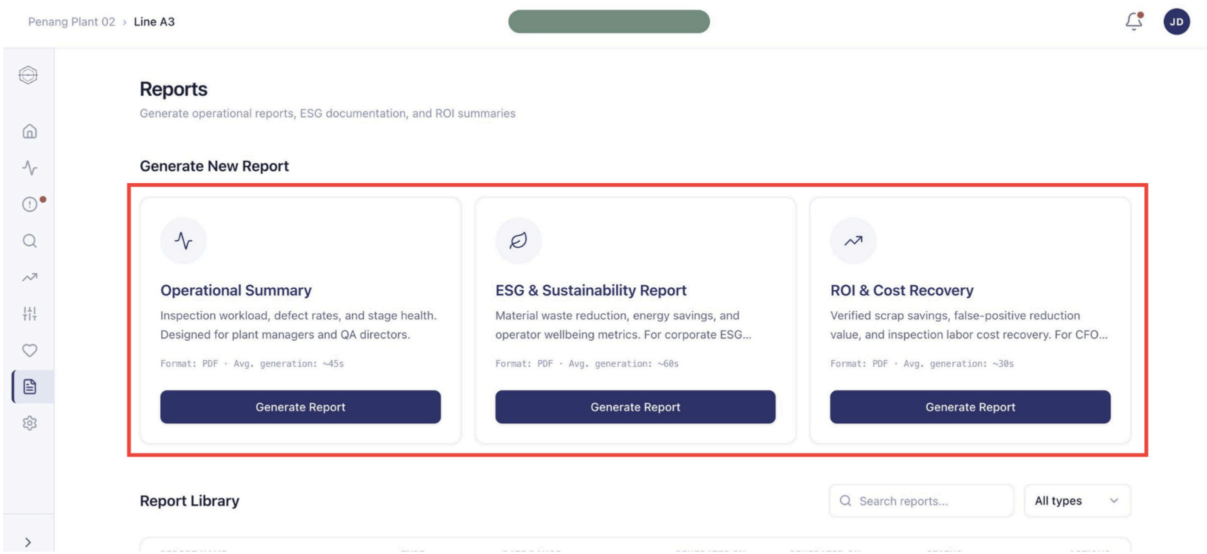


Figure 21. *Template Library*

This is the management-facing surface for generating formal reports. Three report types are available: Operational Summary (for plant managers and QA directors), ESG & Sustainability Report (for corporate ESG disclosures), and ROI & Cost Recovery (for CFO and finance teams). Each card shows the report's purpose, format (PDF), and average generation time.

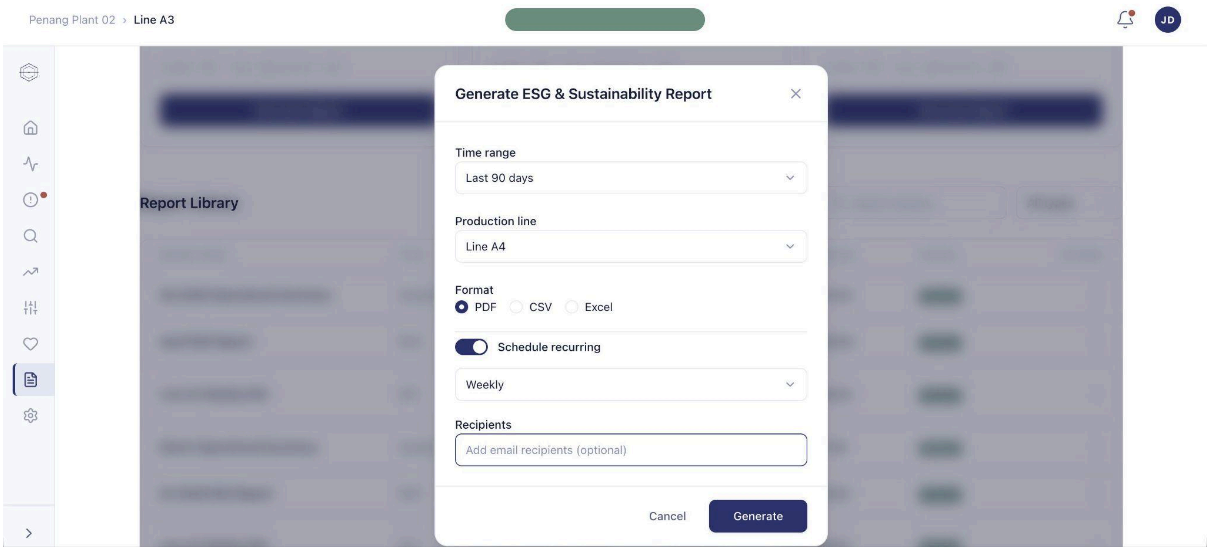


Figure 22. *Configuration Form*

Clicking Generate Report opens a configuration modal where the user picks the time range, production line, file format, recurring schedule (if needed), and email recipients. This makes it easy to set up automated weekly or monthly reports for stakeholders.

2.4 Product Innovation Overview

Omni-QC represents a fundamental architectural departure from the reactive inspection paradigm that has defined electronics manufacturing quality control for decades. Where conventional systems apply inspection resources uniformly after production, Omni-QC introduces intelligence upstream at the point where process data is generated, to determine where risk is concentrated before any physical inspection occurs. This shift from detection to prediction is the central innovation the system embodies.

The innovation is structural. Omni-QC does not introduce new hardware onto the factory floor; it derives value from data that manufacturers are already generating and largely ignoring for quality control purposes. Programmable Logic Controllers (PLCs), MES platforms, and inline SPI machines continuously produce structured telemetry that encodes the process conditions most likely to produce defects. By ingesting and modeling this data in real time, Omni-QC converts latent informational value into actionable risk intelligence that manufacturers can act upon before defects become embedded in finished boards.

The innovation also addresses a gap that purely academic or lab-based AI systems have failed to bridge: the prescriptive Co-Pilot. Most deployed factoryAI systems function as open-loop alarms, they identify that a machine is drifting but stop there, leaving the floor technician without a specific corrective path. Omni-QC closes this loop by computing parameter-level offsets and presenting them as targeted recommendations on the technician's dashboard, bridging the gap between anomaly detection and actionable correction without removing human judgment from the decision chain.

Three innovation principles underpin the system's design: non-invasiveness (no additional sensing hardware required), interpretability (all risk scores and advisory outputs are explainable through SHAP-based feature attribution), and incrementality (the system enhances rather than replaces existing capital investments in inspection infrastructure).

2.5 Product Innovation Highlights

Omni-QC's architecture creates a set of innovation outcomes that are measurable, verifiable, and directly tied to the operational and financial performance metrics that matter to manufacturing clients. The following highlights represent the most significant innovations from conventional QC system design.

PRE-INSPECTION RISK INTELLIGENCE

Omni-QC is one of the first commercially deployed systems to assign unit-level defect probability scores from structured MES telemetry before units enter the inspection queue. This pre-inspection intelligence layer shifts the timing of quality decision-making upstream by an average of 15 to 45 minutes in a standard SMT assembly line, depending on throughput, allowing intervention at the point of lowest remediation cost.

NON-INVASIVE DATA EXTRACTION

The system extracts all predictive signals from data that factories are already generating through existing PLCs and MES platforms. This eliminates the hardware retrofitting cost that has historically been the primary barrier to predictive quality adoption among mid-tier contract manufacturers. No external IoT sensors, vision cameras, or data acquisition modules are required at the point of installation.

RECALL-OPTIMIZED DEFECT PREDICTION

The system is configured for recall-prioritised classification through threshold tuning and class weight adjustment of the XGBoost model. This configuration directly aligns with the asymmetric cost structure of manufacturing quality control, where the cost of a false negative (an undetected defect reaching a customer) consistently exceeds the cost of a false positive (an acceptable unit routed to additional inspection) by an order of magnitude. This configuration choice is a deliberate innovation decision rather than an intrinsic property of the XGBoost algorithm.

INTERPRETABLE AI AT PRODUCTION SPEED

SHAP-based feature attribution is computed on demand for High Risk units and audit requests, providing real-time interpretability without latency impact on the bulk of scored units. This selective computation approach ensures that the system remains compliant with industrial transparency requirements and maintains the throughput necessary for production-line deployment, while accurately reflecting the computational cost structure used in the unit economics model.

HUMAN-IN-THE-LOOP PRESCRIPTIVE ADVISORY

The Co-Pilot's parameter-offset recommendations represent the first applied integration of prescriptive diagnostics with hardware-agnostic local AI in a commercially available PCB quality control platform. The advisory architecture bridges the gap between anomaly detection (which existing systems perform) and actionable correction (which none currently provide at the factory floor level) while preserving human accountability for all production equipment decisions.

EMBEDDED ESG VALUE CREATION

By concentrating inspection on high-risk units and reducing redundant processing of low-risk boards, Omni-QC structurally reduces cumulative energy consumption on AOI and X-ray equipment, extends the operational lifespan of capital infrastructure, reduces material waste from late-stage defect discovery, and improves workforce conditions by redirecting inspection effort away from repetitive false-positive verification. It is important to note that energy savings are proportional to the share of boards classified as Low or Medium Risk versus High Risk. Because the system is configured for recall-prioritised classification., deliberately conservative in the direction of over-flagging, the realized energy reduction will reflect the actual risk distribution of a facility's production output rather than a theoretical maximum. These outcomes are achieved through the system's operational logic and are verifiable through standard operational metrics.

3.0 Product Development and Technical Architecture

This chapter details the development, innovative architecture and underlying technical design that distinguish Omni-QC from conventional quality control approaches.

3.1 Product Development

PHASE 1: PROBLEM DISCOVERY AND REQUIREMENTS DEFINITION

The development of Omni-QC originates from a structured discovery process conducted across mid-tier EMS facilities in Malaysia and Vietnam. The initial phase involves direct engagement with quality engineers, production managers, and operations directors at Tier 1 and Tier 2 contract manufacturers to map the precise failure modes in existing inspection workflows. This field research surfaces three recurring structural inefficiencies: the temporal mismatch between the upstream origin of solder paste defects and their downstream detection under end-of-line AOI, the disproportionate burden of false-positive verification on manual inspection workforces, and the absence of any commercially available system capable of translating anomaly signals into specific corrective parameter recommendations at the factory floor level.

From these findings, the product requirements are formalised around five non-negotiable design constraints. First, the system must operate entirely on data that target facilities are already generating, requiring no additional sensing hardware at the point of installation. Second, it must produce interpretable outputs that quality engineers can audit and defend to clients and regulators. Third, it must integrate with the heterogeneous equipment ecosystems of mid-tier Southeast Asian facilities, which routinely combine machines from multiple vendors running distinct proprietary communication protocols. Fourth, it must be commercially deployable under a SaaS model with zero upfront capital expenditure, eliminating the hardware procurement cycle that has historically blocked AI adoption in the mid-tier segment. Fifth, the system must preserve human authority over all production equipment decisions, operating as an advisory layer rather than an autonomous controller.

PHASE 2: DATA INFRASTRUCTURE AND MODEL DEVELOPMENT

With requirements established, the development team moves into the data infrastructure and model development phase. This phase centers on two parallel workstreams: building the data pipeline architecture that ingests, normalises, and validates structured telemetry from heterogeneous factory equipment, and developing the predictive model that converts that telemetry into unit-level defect probability scores.

Data Pipeline Construction

The data pipeline is designed to accommodate the three primary communication protocols in use across target facilities: OPC-UA for Pick-and-Place machine telemetry, IPC-CFX for SPI data outputs, and MQTT for reflow oven thermal streams. Per-vendor adapter modules are engineered to translate each proprietary protocol into a normalised ingestion format, ensuring that the upstream data variability of mixed-vendor production environments does not propagate into the model's feature space. A validation layer is embedded at the ingestion stage to screen all incoming records against defined plausibility bounds, rejecting malformed or out-of-range readings before they reach the inference engine.

Feature Engineering

Raw sensor readings are transformed into predictive features through a combination of normalisation, rolling window aggregation to capture temporal trends in process conditions, and cross-parameter interaction terms encoding the relationships between process variables most strongly associated with defect outcomes in historical production data. Key features derived from each monitored stage include stencil alignment deviation and paste volume variance from the SPI stage, component placement coordinate offset and nozzle pressure from the Pick-and-Place stage, and zone-by-zone temperature profiles across the preheat, soak, reflow peak, and cooling phases of the reflow oven.

Model Selection and Training

Following evaluation against alternative architectures including logistic regression, Random Forest, and LightGBM on representative manufacturing datasets, XGBoost is selected as the predictive core based on its documented performance advantage on structured, tabular sensor log data and its inference speed compatibility with real-time production line throughput requirements. The model is configured for recall-prioritised classification through threshold tuning and class weight adjustment, reflecting the asymmetric cost structure of manufacturing quality control, where undetected defects reaching downstream processes carry consequences an order of magnitude more severe than false positives routed to additional inspection.

SHAP (SHapley Additive exPlanations) is integrated as the interpretability layer, providing per-feature attribution for each unit's risk score. To preserve computational efficiency, SHAP computation is scoped to High Risk units and explicit audit requests rather than applied to every scored unit, ensuring that the USD 2.40 per day inference cost estimate for a 10,000-panel production line remains accurate and that inference latency remains within production cycle time constraints.

PHASE 3: SOFTWARE BUILD AND INTEGRATION ENGINEERING

Dashboard and Interface Development

The technician-facing Co-Pilot dashboard is developed through an iterative design process informed by direct feedback from quality engineers and floor technicians at pilot facility partners. The interface is designed to surface three categories of output: the real-time risk tier assignment for each unit processed through the monitored stages, the SHAP-based feature attribution breakdown for High Risk units and audit requests, and the prescriptive parameter-offset advisory notifications generated when the system detects a statistically significant drift pattern attributable to a specific process variable. The dashboard architecture deliberately separates display from action, presenting all advisory outputs as recommendations accompanied by the underlying drift data, and requiring explicit technician confirmation before any corrective adjustment is logged, preserving the human-in-the-loop model throughout.

Deployment Architecture

Two deployment configurations are engineered in parallel to accommodate the range of data governance requirements across target facilities. The default cloud deployment routes securely transmitted factory telemetry to remote servers for processing, enabling scalable compute allocation, centralised model updates, and remote accessibility across multi-site clients. The edge deployment configuration packages the complete software stack as a compiled, encrypted container installed on a local industrial server within the facility's own network, activated through a digital license key, with all computation and model inference remaining within the facility perimeter. The edge option is developed specifically for clients with data sovereignty or intellectual property mandates that preclude cloud transmission of production data.

PHASE 4: VALIDATION AND PILOT DEPLOYMENT

Technical Validation

Model performance is evaluated on held-out subsets of historical production data sourced from pilot facility partners, with evaluation metrics weighted toward recall to reflect the asymmetric cost structure of the target application. The validation process confirms that the deployed model achieves the recall threshold necessary to support the system's core commercial claim of reducing total inspection workload by 30 to 50 percent relative to uniform coverage without increasing the rate of defective units reaching downstream processes. Threshold calibration is validated as a facility-specific rather than universal procedure, confirming that the cost-sensitive optimisation approach against each client's historical defect base rate produces materially better classification boundaries than generic round-number defaults.

Pilot Deployment

A structured 90-day Proof of Concept program is designed and executed with a cohort of pilot manufacturing partners. The pilot program is sequenced to deliver a fully operational deployment within 14 business days from the completion of a preparatory integration phase, during which infrastructure readiness is confirmed and per-vendor adapters are configured and tested against the facility's live equipment outputs. Pilot results are used to refine the onboarding documentation, validate the MLOps retraining pipeline under live production conditions, and calibrate the performance monitoring thresholds that govern automatic drift detection. Feedback collected from quality engineers and floor technicians during the pilot phase informs the final iteration of the Co-Pilot dashboard interface prior to commercial release.

PHASE 5: PRODUCTIZATION AND COMMERCIAL RELEASE READINESS

The final development phase converts the validated technical system into a commercially deployable product with defined onboarding, support, and retraining service structures. A standardised 14-business-day onboarding SLA is established, covering the completion of infrastructure readiness confirmation, per-vendor adapter configuration and testing, initial model training on the client's historical production data, threshold calibration against the facility's defect base rate, and delivery of a model validation report documenting predictive performance on held-out production data. This report serves as the technical foundation for the client's internal sign-off and, where applicable, audit documentation.

The retraining service is structured on a demand-triggered basis rather than a fixed calendar cadence, reflecting the documented stability of manufacturing process environments and the finding from pilot deployments that imposing uniform retraining schedules generates unnecessary overhead for clients whose production conditions remain relatively stable. The pricing architecture is designed to ensure that facilities with stable environments are not cross-subsidising the retraining costs of high-drift clients, with retraining billed as a tiered on-demand engagement detailed in Section 6.5.

The MLOps pipeline is engineered to support automated few-shot retraining at scale, with the transition from manual to semi-automated to fully automated MLOps operations planned across the three deployment phases described in Section 3.6. This architectural roadmap ensures that the per-client engineering overhead required to onboard and maintain each deployment decreases progressively as the installed base grows, preserving the unit economics of the SaaS model at the deployment volumes projected in the financial analysis.

3.2 System Architecture

TECHNICAL ARCHITECTURE AND DEPLOYMENT MODEL

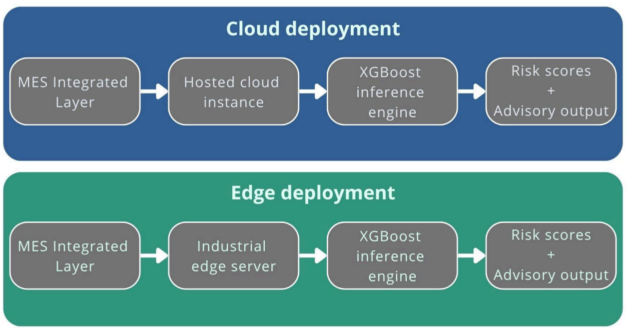


Figure 25. *Technical architecture and deployment model*

Omni-QC supports both cloud and edge deployment models. The default deployment is cloudbased, where data collected from factory systems is securely transmitted to remote servers for processing, enabling scalable processing power, centralised model updates, and remote accessibility across multiple production sites. For facilities with uncompromising data sovereignty or intellectual property mandates, Omni-QC can alternatively be deployed on a local industrial server within the factory's own network. In this edge configuration, the factory receives the software as a compiled, encrypted container installed on the local industrial server and activated through a digital license key; all computation and model inference remain within the facility perimeter at all times. The default recommendation is cloud deployment for most clients; edge deployment is the exception for environments where data governance requirements make cloud transmission infeasible. The "data remains within the facility" guarantee applies specifically and exclusively to edge-deployed instances.

It is important to note that while SPI, P&P, and Reflow machines are often depicted as feeding a unified PLC/MES bus, in practice these stages frequently run on separate vendor-managed networks. Omni-QC handles this complexity through per-vendor data adapters that translate proprietary protocol outputs: OPC-UA for P&P telemetry, IPC-CFX for SPI data, and MQTT for reflow thermal streams ,into a normalised ingestion format. This per-vendor adapter architecture is what allows the system to operate across the heterogeneous equipment ecosystems typical of Southeast Asian mid-tier EMS facilities without bespoke integration work for each equipment vendor

The predictive core of Omni-QC is built on XGBoost, the Extreme Gradient Boosting algorithm. XGBoost operates through a gradient-boosted decision tree structure, in which successive trees are built to correct the residual errors of those that precede them, progressively refining the model's predictive accuracy. This architecture is particularly well-suited to the structured, tabular nature of manufacturing process data, where sensor readings, machine parameters, and historical defect logs form rows and columns rather than images or sequences. In direct comparisons against other machine learning models on manufacturing defect prediction tasks, XGBoost consistently demonstrates high predictive performance against multidimensional production and quality metrics (Miedema et al., 2024).

The system is configured for recall-prioritised classification through threshold tuning and class weight adjustment, which is desirable in high-risk manufacturing settings where missing a defective product is more critical than over-inspecting a non-defective one (Miedema et al., 2024). This configuration, rather than any intrinsic property of XGBoost itself, directly aligns with the system's safety objective of minimising false negatives, that is, preventing any genuinely defective unit from being incorrectly classified as low risk and cleared without adequate inspection.

Omni-QC is delivered under a Software-as-a-Service (SaaS) model. The underlying model weights and training logic remain encrypted and are not accessible to the facility. License validity is maintained on a subscription basis, meaning that if a subscription lapses, the system ceases to generate risk scores and advisory outputs. This architecture protects the intellectual property of Omni-QC while simultaneously creating the conditions for ongoing model maintenance, which is addressed in later sections.

Omni-QC's system architecture is structured as a four-layer data pipeline that transforms raw machine telemetry into risk-scored inspection routing decisions and Co-Pilot advisories. Each layer performs a distinct function and operates independently, allowing the system to be updated or recalibrated at any layer without disrupting the others.

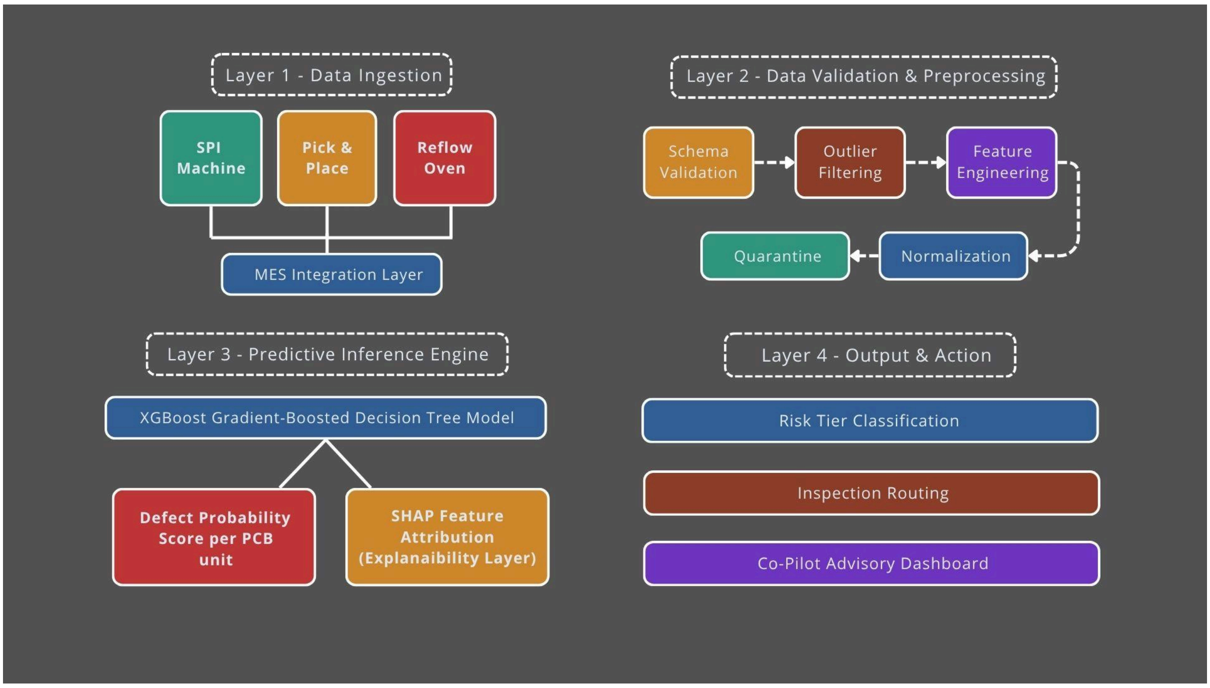


Figure 26. *Omni-QC four-layer system Architecture*

LAYER 1 - DATA INGESTION

Data Ingestion connects Omni-QC to the three monitored production stages (solder paste printing, pick-and-place, reflow soldering) via the facility's existing MES platform. The system ingests structured telemetry through standard IPC-2591 CFX 2.0, CSV, or XML data formats, ensuring vendor-agnostic compatibility across equipment from manufacturers such as Koh Young, Omron, and ASM. No proprietary data connectors or additional sensing hardware are required.

LAYER 2 - DATA VALIDATION AND PREPROCESSING

Data Validation and Preprocessing applies schema validation, plausibility bound checking, outlier filtering, and feature engineering before any data reaches the inference engine. Records failing validation are quarantined and flagged to the client's quality engineering team with a remediation checklist, preventing corrupted inputs from degrading model outputs.

LAYER 3 - PREDICTIVE INFERENCE ENGINE

Predictive Inference Engine executes the XGBoost model against validated feature vectors to produce a defect probability score between 0.0 and 1.0 for each PCB unit. SHAP (SHapleyAdditive exPlanations) values are computed alongside each score to attribute the prediction to specific process variables, providing the interpretability layer required for regulatory audit and technician trust

LAYER 4 - OUTPUT AND ACTION

Output and Action translates probability scores into three operational outputs: a risk tier classification (Low, Medium, or High) that routes the unit to the appropriate inspection channel; a structured inspection routing signal to the facility's AOI, X-ray, or spot-check equipment; and a Co-Pilot advisory notification to the technician dashboard when a statistically significant drift pattern is detected that needs corrective parameter adjustment.

INTEGRATION TOPOLOGY

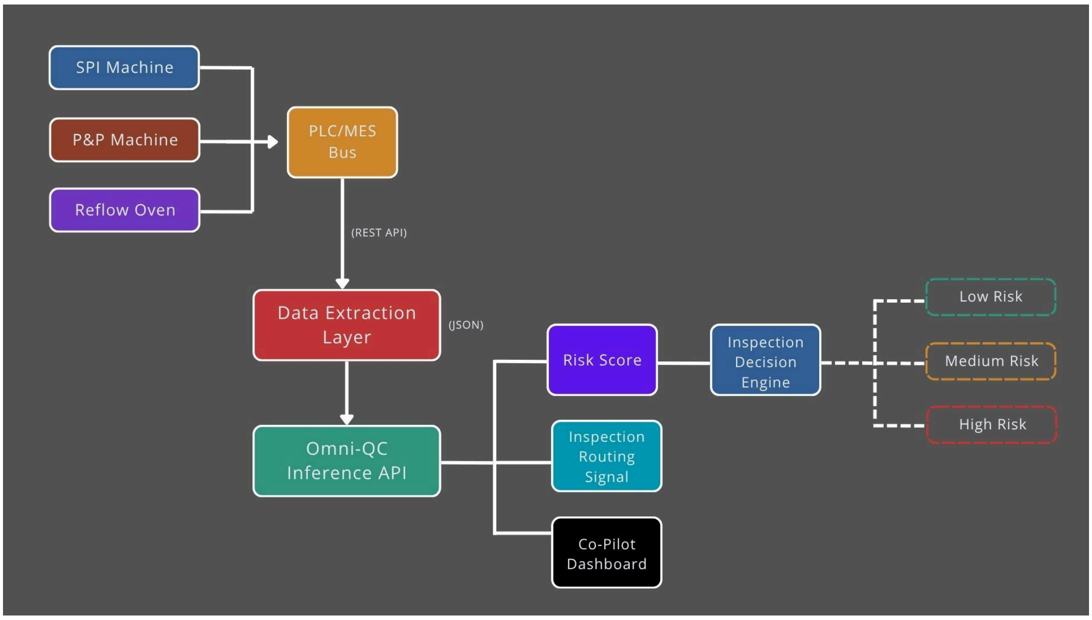


Figure 27. *Integration Topology Diagram*

The integration topology of Omni-QC maps the physical and logical flow between production machines, the PLC/MES communication layer, the data extraction layer, the Omni-QC inference API, and the downstream inspection-routing outputs.

- SPI Machine > PLC/MES Bus
- P&P Machine > PLC/MES Bus
- Reflow Oven > PLC/MES Bus
- PLC/MES Bus > REST API > Data Extraction Layer

- Data Extraction Layer > JSON-formatted telemetry > Omni-QC Inference API
- Omni-QC Inference API > Risk Score + Inspection Routing Signal + Co-Pilot Dashboard
- Risk Score → Inspection Decision Engine > Low / Medium / High Risk classification

1. Multi-Protocol Data Acquisition: The system interfaces with heterogeneous machine types using standardized industrial communication protocols. Solder Paste Inspection (SPI) data is extracted through IPC-CFX, capturing solder volume deviations and paste consistency metrics. Pick-and-Place (P&P) telemetry utilises OPC-UA to monitor component centroid offsets, placement accuracy, and feeder anomalies. Reflow oven controllers publish thermal profile and zone-temperature data through MQTT-based telemetry streams.
2. PLC/MES Bus: The PLC/MES Bus acts as the central aggregation layer for raw machine telemetry before transmission to the Omni-QC platform. Machine-level data is consolidated and exposed through a secure REST API interface to the Data Extraction Layer. This intermediary architecture decouples shop-floor operations from the inference system, ensuring that production continues even during temporary cloud or network latency.
3. Data Extraction Layer: The Data Extraction Layer standardises heterogeneous machine outputs into a unified JSON structure suitable for inference processing. This layer performs telemetry parsing, timestamp alignment, feature normalization, and schema harmonisation before forwarding the processed payload to the Omni-QC Inference API.
4. Inference & Risk Scoring: The Omni-QC Inference API evaluates incoming telemetry against trained defect-prediction models and historical production yield patterns. The system continuously updates a normalized risk score between 0.0 and 1.0 as telemetry becomes available from each production stage. In parallel, the inference layer generates an inspection routing signal and forwards operational insights to the Co-Pilot Dashboard for technician visibility.
5. Inspection Decision Engine: The Inspection Decision Engine converts the generated risk score into actionable routing outcomes:
   - Low Risk > Routed to high-speed statistical spot-check inspection
   - Medium Risk > Routed to standard AOI inspection flow
   - High Risk > Routed to mandatory full inspection or operator review

   This logic-based routing architecture enables adaptive inspection coverage while preserving manufacturing throughput and maintaining quality assurance requirements for Class 1 and Class 2 production environments.
6. Co-Pilot Dashboard: The Co-Pilot Dashboard provides operators and process engineers with real-time monitoring visibility. The dashboard displays live risk scores, telemetry anomalies, routing decisions, and process-drift alerts generated by the inference engine. In cases of abnormal sensor readings or elevated process drift, technicians are presented with the specific telemetry source contributing to the elevated risk condition.

PER-UNIT DATA FLOW

To illustrate the operational workflow, the following sequence tracks a single PCB panel (Unit ID: B2026-X) through an integrated production line:

1. 08:00 - SPI Entry: Panel B2026-X completes Solder Paste Inspection. The SPI machine transmits volumetric deviation telemetry through IPC-CFX to the PLC/MES Bus. The Data Extraction Layer retrieves the payload through the REST API, converts the telemetry into normalized JSON features, and forwards it to the Omni-QC Inference API. Based solely on SPI-stage telemetry, the system generates a preliminary risk score of 0.12 (Low Risk).
2. 08:02 - Pick & Place: During component placement, the P&P machine publishes OPC-UA telemetry confirming all component placements remain within 20 µm of nominal centroid alignment. The Omni-QC model updates the existing risk assessment using the newly received placement telemetry. The recalculated score remains stable at 0.11 (Low Risk).
3. 08:15 - Reflow Monitoring: As the PCB panel progresses through the reflow oven, MQTT-based thermal telemetry is streamed zone-by-zone into the PLC/MES Bus and subsequently processed by the Data Extraction Layer. All thermal zones remain within acceptable profile tolerances. The Omni-QC Inference API produces a final consolidated risk score of 0.12, incorporating all three production-stage inputs.
4. 08:15:30 - Routing Decision: The final risk score is transmitted to the Inspection Decision Engine, which generates an inspection routing signal for downstream AOI handling.
5. 08:16 - Outcome: Because the final score remains below the configured Low Risk threshold, the panel is routed to a high-speed spot-check inspection lane rather than full AOI coverage. This reduces inspection cycle time while maintaining statistical quality assurance coverage appropriate for the board classification.

**Note: The risk score is incrementally updated at each production stage as new telemetry becomes available. The inspection routing decision is finalized only after the reflow stage, when the complete three-stage feature set has been evaluated. This staged-update architecture is what gives the multi-stage integration its predictive value, rather than relying on a single static score generated at SPI alone.*

DEPLOYMENT PREREQUISITES AND TIMELINE

Category	Requirement
Infrastructure	Existing MES aggregation layer; machines capable of structured data output (SPI, P&P, Reflow)
Connectivity	Internal network access (LAN); API communication capability (REST)
Data Readiness	Access rights to machine telemetry; historical defect and inspection logs

IT & Security	Approval for containerised software deployment; firewall rules allowing secure API communication

Figure 28. *Omni-QC Deployment Prerequisites*

14-Day Deployment Timeline

Disregarding the further sales program described in later sections such as SLA programs, the general technical deployment timeline of Omni-QC is as follows

1. Day 1-2 [IT setup]: Network and container deployment
2. Day 3-5 [Integration]: MES connection and data mapping
3. Day 6-8 [Data validation]: Schema checks and feature calibration
4. Day 9-11 [Model calibration]: Initial scoring alignment
5. Day 12-13 [Pilot run]: Shadow mode with no routing control
6. Day 14 [Go-Live]: Full routing and Co-Pilot activation

**For Band 2 manufacturers with partial MES instrumentation, a preparatory integration phase is scoped and completed before the 14-day onboarding SLA clock begins, ensuring that infrastructure readiness is confirmed before deployment commitments are made.*

3.3 Core Technologies

Omni-QC's technical stack is assembled to maximize predictive performance on manufacturing data while minimizing computational overhead, deployment complexity, and IT security friction. Each technology selection is determined by a specific engineering rationale tied to the constraints of industrial deployment

Technology Component	Implementation	Rationale
Predictive Engine	XGBoost (Extreme Gradient Boosting)	Superior performance on structured tabular data; recall-biased architecture minimizes false negatives; computationally efficient relative to deep learning
Explanability Layer	SHAP (SHapleyAdditive exPlanations)	Industry-standard XAI method; provides perprediction feature attribution; satisfies regulatory audit requirements
Data Integration Protocol	IPC-2591 CFX 2.0 / CSV / XML	Vendor-agnostic; compatible with all major SMT equipment manufacturers; no proprietary connectors required

Deployment Runtime	Encrypted SaaS container (DRM-protected)	Protects model IP; enables remote license management; supports both cloud and edge deployment
Cloud Infrastructure	Serverless cloud inference (AWS / GCP)	Collapsed inference costs (~USD 2.40/day for 10,000- panel line under standard SHAP-on-demand policy); automatic scaling; no on-premise server required
Edge Compatibility	Partner-provided industrial edge servers	Supports air-gapped, data-sovereign deployments; hardware-agnostic encrypted container
MLOps Pipeline	Demand-triggere d retraining with held-out validation	Recalibrates to current process conditions without fixed-schedule overhead; validates accuracy before deployment
Performance Monitoring	Rolling-window drift detection	Continuous alignment tracking between risk score distributions and confirmed defect outcomes

Figure 29. *Omni-QC core technology stack*

XGBoost was selected as the predictive engine after comparative evaluation against alternative model architectures including Random Forest, Gradient Boosting Machines, and convolutional neural networks applied to tabular data. In manufacturing defect prediction contexts, XGBoost consistently achieves top-tier recall performance on imbalanced datasets, where defective units represent a small minority of total production output, while operating at inference speeds compatible with real-time production line throughput (Miedema et al., 2024). Its gradientboosted tree structure also avoids the "black box" opacity problem characteristic of deep neural networks, making model behaviour more amenable to the interpretability requirements of regulated manufacturing environments.

SHAP (SHapleyAdditive exPlanations) is applied at inference time to decompose each unit's risk score into contributions from individual process variables, providing the mathematical transparency required for quality engineers to audit and validate the model's decision logic. Maged et al. (2026) identify SHAP as the most widely deployed industrial Explainable AI (XAI) approach, with documented deployments including DarwinAI's DVQI system, where SHAPinformed model refinement reduced AOI false positive rates from 1.7% to 0.11% (Chung et al., 2023). SHAP computation is performed on demand for High Risk units and audit requests rather than for every scored unit, ensuring that the USD 2.40/day inference cost estimate for a 10,000- panel line remains accurate and that inference latency remains within production cycle time requirements.

IPC-2591 CFX 2.0 provides the standardized data communication framework that allows OmniQC to interface with factory floor equipment without bespoke integration work for each equipment vendor. By leveraging this existing industry standard, the system achieves the vendor-agnostic compatibility required to deploy across the heterogeneous equipment ecosystems typical of Southeast Asian EMS facilities.

3.4 Predictive Quality Control Mechanism

The predictive quality control mechanism is the operational core of Omni-QC. It defines how raw machine telemetry is transformed into unit-level defect probability scores and how those scores drive inspection routing decisions. The mechanism operates across five sequential steps.

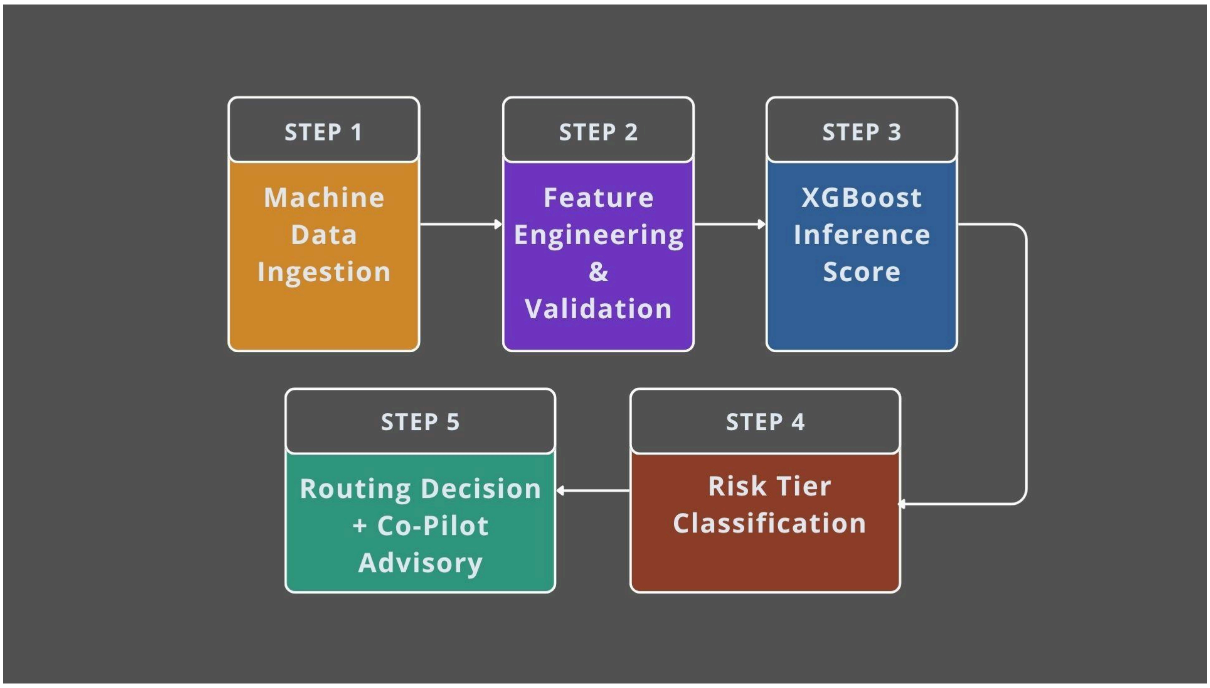


Figure 30. *Five-step predictive quality control mechanism*

MACHINE DATA INGESTION

At the conclusion of each monitored production stage, structured telemetry is extracted from the facility's MES platform. Key data fields include stencil alignment deviation and paste volume variance (SPI stage), placement coordinate offset and nozzle pressure (Pick-and-Place stage), and zone-by-zone temperature profiles across preheat, soak, reflow peak, and cooling phases (Reflow stage).

FEATURE ENGINEERING AND VALIDATION

Raw sensor readings are transformed into predictive features through normalisation, rolling window aggregation (to capture temporal trends), and cross-parameter interaction terms that encode the relationships between process variables most strongly associated with defect outcomes in the training data. The validation layer simultaneously screens all incoming records against defined plausibility bounds before they proceed to inference

XGBOOST INFERENCE

The validated feature vector for each PCB unit is passed through the deployed XGBoost model, which produces a continuous defect probability score between 0.0 and 1.0. The score represents the model's estimated probability that the unit carries a latent defect that would be identified under intensive inspection. This inference step is performed for every unit scored; SHAP values are computed selectively for High Risk units or on explicit audit request.

RISK TIER CLASSIFICATION

The probability score is mapped to one of three inspection tiers according to the thresholds configured for the facility. Default thresholds are initialised via cost-sensitive optimisation against the client's historical defect base rate

Risk Tier	Score Range	Default Threshold	Inspection Protocol
Low Risk	0.0 – 0.39	< 0.40	High-speed dynamic spot-check (sample of batch)
Medium Risk	0.40 – 0.69	0.40 – 0.69	Standard statistical sampling protocol
High Risk	0.70 – 1.0	≥ 0.70	100% intensive scanning (AOI / X-ray / flying probe)

Figure 31. *Three-tier risk classification thresholds and inspection protocols*

Thresholds are user-configurable through the factory manager interface, allowing facilities to adjust inspection rigor based on board classification, production context, and client quality requirements.

ROUTING DECISION AND CO-PILOT ADVISORY

Each unit is routed to the designated inspection channel based on its risk tier. Concurrently, the Co-Pilot module monitors aggregate risk score trends and SHAP attribution patterns across all units processed within a rolling production window. When a statistically significant pattern emerges, for example, a sustained cluster of elevated risk scores attributable to Zone 4 thermal drift in the reflow oven, the module generates a targeted advisory notification to the technician dashboard specifying the implicated process variable and the calculated corrective offset.

3.5 Model Retraining System

CONTINUOUS MODEL MONITORING AND RETRAINING

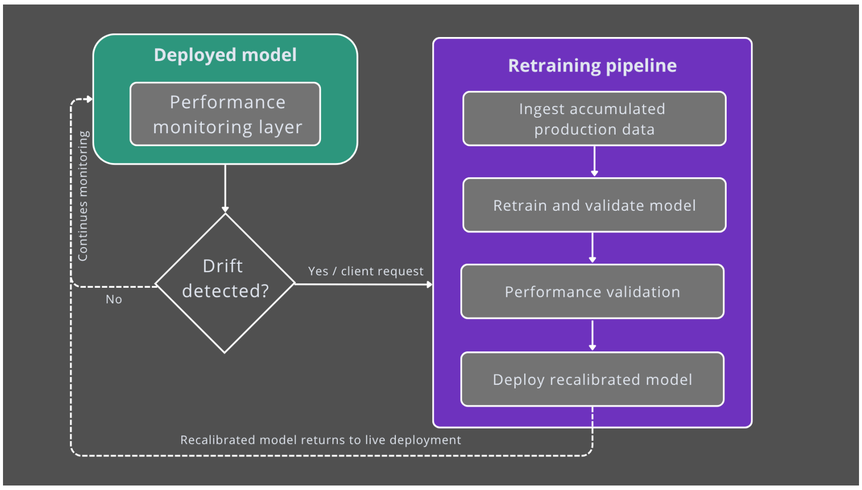


Figure 32. *Continuous model monitoring and demand-triggered retraining*

The predictive accuracy of Omni-QC is not a static property of the system at the point of deployment. It is a function of how closely the model's training data continues to reflect the live process conditions of the factory in which it operates. A model trained on one set of production conditions will progressively lose its alignment with actual defect outcomes as those conditions evolve over time.

This phenomenon, commonly referred to as model drift, arises because the relationships between input features and defect outcomes that the model learned during training shift as the factory environment changes. These changes individually introduce small inaccuracies into the risk scoring output, and their collective effect over time is a progressive miscalibration in which the model's defect probability scores no longer correspond reliably with actual outcomes on the factory floor.

To address this, Omni-QC incorporates a dedicated performance monitoring layer that operates continuously across all deployed instances. This layer tracks a set of predictive performance indicators over rolling production windows, including the distribution of risk scores across inspection tiers, the alignment between High Risk classifications and confirmed defect outcomes from the facility's inspection records, and the rate at which Medium Risk units escalate to confirmed defects during standard sampling. When these indicators drift beyond defined acceptable bounds, the monitoring layer generates an internal alert flagging that the deployed model's calibration has degraded to a degree that warrants a retraining cycle. In addition to system-triggered alerts, retraining may also be initiated proactively at the client's request when the quality team observes operational signals, such as an uptick in defect escapes or an unusual clustering of High Risk classifications, that suggest the model is no longer scoring consistently with the factory's current process state.

When a retraining cycle is initiated, Omni-QC's team ingests the accumulated production and inspection data from the client's facility since the last training cycle. This data is validated for completeness and structural integrity before being used to retrain the model, ensuring that the updated model is calibrated against the client's current process conditions rather than the historical state captured at initial deployment. The retrained model undergoes performance validation against a held-out subset of the client's recent production data before it is cleared for deployment, confirming that predictive accuracy has been restored to an acceptable threshold.

It is worth noting that this retraining process does not operate on a fixed calendar schedule. Manufacturing processes tend to be stable, meaning a quality prediction model can last a whole year, or until the first external change such as a new raw materials supplier (EvidentlyAI, 2025), which means that imposing a uniform retraining cadence regardless of actual drift would add unnecessary cost and operational complexity for clients whose production environments remain relatively stable. Omni-QC therefore structures its retraining service on a demand-triggered basis, activated by monitored performance degradation or client request, rather than as a recurring scheduled activity. This retraining service is billed as an on-demand engagement at tiered rates detailed in Section 6.5 , not bundled into the subscription fee, ensuring that facilities with stable production environments are not subsidising the retraining costs of highdrift clients. This ensures that the model remains accurate over the life of the deployment without generating retraining overhead disproportionate to the actual pace of change in any given facility

3.6 Technical Feasibility and Scalability

Omni-QC's technical feasibility is based on three validated conditions: the infrastructure prerequisites for deployment already exist in the target market segment, the computational cost of the system is commercially viable at production scale, and the architecture scales predictably with deployment volume without requiring proportionate increases in engineering overhead.

INFRASTRUCTURE PREREQUISITES

The system's data dependency, such as structured telemetry from SPI, Pick-and-Place, and Reflow machines, is met by all Band 1 manufacturers in the target market (digitally mature facilities with functional MES platforms). The Asia-Pacific MES market is projected to reach USD 56 billion by 2035 (Fortune Business Insights, 2025), reflecting the maturation of this infrastructure across the manufacturing base Omni-QC targets. For Band 2 manufacturers (partial MES instrumentation), a preparatory integration phase is scoped before the 14-day onboarding SLA clock begins, ensuring that infrastructure readiness is confirmed before deployment commitments are made.

COMPUTATIONAL VIABILITY

Per-board inference on an XGBoost model operating on structured tabular features carries a computational cost several orders of magnitude lower than image-based deep learning alternatives. At serverless cloud rates reflecting the 280-fold inference cost reduction documented between 2022 and 2024 (Stanford HAI, 2025), a 10,000-panel-per-day production line costs approximately USD 2.40 per day to score under a SHAP-on-demand policy (where SHAP values are computed only for High Risk units and explicit audit requests). This validates the system's unit economics at industrial throughput volumes.

ARCHITECTURAL SCALABILITY

The four-layer pipeline architecture is designed for horizontal scalability. Each deployed instance operates independently, with model weights stored in encrypted client-specific partitions that are not shared across deployments. As the client fleet grows, cloud inference capacity scales automatically through serverless compute allocation, and the MLOps retraining pipeline is designed to process multiple concurrent retraining cycles without sequential bottlenecks. The performance monitoring layer aggregates metrics across all deployed instances, providing Omni-QC's engineering team with a centralized view of model health without requiring dedicated monitoring infrastructure per client.

SCALABILITY PROJECTION BY DEPLOYMENT PHASE

Phase	Deployed Lines	Infrastructure Mode	Engineering Overhead
Phase 1 (Yr 1)	1 – 5	Cloud-primary; manual MLOps Hi	High (baseline calibration)
Phase 2 (Yr 2)	5 – 20	Cloud-primary; semi-automated MLOps	Moderate (playbook-driven)
Phase 3 (Yr 3+)	20 – 100+	Cloud + Edge; automated few-shot retraining	Low (pipeline-automated)

Figure 33. *Scalability Projection by Deployment Phase*

The transition from manual to automated MLOps is a planned architectural development funded within the Year 2 and Year 3 R&D budgets, it enables the system to onboard new clients with progressively lower per-client engineering hours as the installed base grows.

3.7 Comparison with Traditional QC Systems

The table below provides a structured comparison between Omni-QC and the three prevailing QC system categories in electronics manufacturing: traditional uniform inspection (100% AOI/X-ray), reactive computer vision platforms, and static statistical sampling protocols

Criterion	Traditional 100% Uniform Inspection	Reactive Computer Vision	Static Statistical Sampling	Omni-QC
Inspection Stage	Post-assembly	Post-assembly	Post-assembly	Pre-inspection (upstream)
Data Modality	Image / optical	Image / optical	None (sampling tables)	Structured MES telemetry (tabular)
Defect Prediction	None	None	None	Unit-level probability score
Inspection Coverage	100% of all units	100% of selected units	Fixed statistical sample	Risk-proportionate (30–50% workload reduction target)
False Positive Rate	High (up to 70% reinspection)	High (image-based misclassification)	N/A	Reduced (recall optimized scoring)
Actionable Guidance	None	Flag only	None	Prescriptive CoPilot parameter offsets
Interpretability	N/A	Limited	N/A	SHAP feature attribution per unit
Hardware Requirement	AOI / X-ray (existing)	Additional vision hardware	None	No additional hardware (MES integration only)
CapEx to Deploy	High (hardware bound)	High (camera infrastructure)	None	Zero (SaaS subscription)
MES Integration	None	Limited	None	Full (SPI, P&P, Reflow)
Human-in-Loop	Yes (manual reinspection)	No	Yes (sampling decisions)	Yes (Co-Pilot advisory model)
Scalability	Linear cost scaling	Linear cost scaling	Fixed	Sub-linear (cloud inference)
ESG Impact	High energy / waste	High energy	Minimal	Reduced energy, waste, and operator fatigue

Figure 34. *Comparative analysis: Omni-QC vs traditional C system categories*

The comparison illustrates that Omni-QC does not compete directly within any single existing category. It operates as an orchestration layer above all three, determining how existing inspection infrastructure (whetherAOI, X-ray, or sampling-based) should be deployed based on unit-level risk intelligence. This positions the system as complementary to rather than competitive with a facility's existing capital equipment investment, removing the adoption objection most commonly raised against new quality technology: the risk of rendering prior investments obsolete.

The quantitative performance differentials are most pronounced in three dimensions. First, inspection workload: the 30 to 50% reduction target compresses the total volume of units requiring intensive processing, recovering the labor and machine time currently consumed by uniform 100% coverage. Second, false positive overhead: by pre-filtering low-risk units before they enter the AOI queue, the system directly reduces the volume of flagged units that require manual re-inspection, addressing the USD 7,000-per-panel-per-year false detection cost identified in previous sections. Third, prescriptive guidance: no competing system currently bridges the gap between anomaly detection and parameter-level corrective action, leaving floor technicians without a structured path from "the machine is drifting" to "here is the specific adjustment that will resolve it."

4.0 Capability and Performance Analysis

This chapter evaluates Omni-QC's competitive standing and operational strengths through two structured analytical frameworks. A SWOT analysis first maps the system's internal capabilities against external market conditions, identifying the strategic positions and vulnerabilities that will shape its trajectory in the Southeast Asian electronics manufacturing sector. This is followed by a strength-based capability analysis that examines each of Omni-QC's core technical and commercial advantages in detail, demonstrating how they reinforce one another to form a cohesive and defensible value proposition. Together, these analyses establish the performance foundation upon which the market strategy in Chapter 5 is built.

4.1 SWOT Analysis

SWOT ANALYSIS

<table>
<tr><th>STRENGTHS (S)</th><th>WEAKNESSES (W)</th><th>OPPORTUNITIES (O)</th><th>THREATS (T)</th></tr>
<tr>
<td>Upstream risk prediction rather than post-assembly detection

XGBoost model architecture with recall optimization

SaaS cloud delivery model with low adoption barrier

User-configurable risk thresholds adaptable to diverse quality standards

Preserves existing infrastructure

Structurally embedded ESG and sustainability outcomes</td>
<td>Data dependency creates a structurally limited addressable market at entry

No installed production-floor track record at market entry

Sensitivity to data drift weakening model performance</td>
<td>Expanding EMS market driving higher inspection volumes and costs

Geopolitical production rebalancing increasing site complexity in Southeast Asia</td>
<td>Well-capitalized competitors with established manufacturing client bases

Client security sensitivity of cloud-hosted deployment extending procurement cycles</td>
</tr>
<tr>
<td colspan="2">S-O STRATEGY
● Position predictive risk-scoring as the standard for scaling EMS operations
● Target high-growth Southeast Asian manufacturing hubs first
● Exploit SaaS model to accelerate adoption among cost-sensitive firms
● Promote configurability as a cross-industry advantage</td>
<td colspan="2">W-T STRATEGY:
● Develop strong onboarding and data integration support
● Position as a complementary layer rather than a replacement
● Strengthen data security</td>
</tr>
</table>

• Leverage the product's ESG outcomes to capture demand for sustainability-aligned manufacturing technology	

Figure 35. *SWOT Analysis Table*

Strengths

Omni-QC's core strength lies in its upstream, predictive approach to quality control, shifting inspection from a reactive to a proactive process. By assigning defect probabilities before boards reach inspection, it enables more efficient allocation of inspection resources such as AOI and X-ray systems. This approach enhances, rather than replaces, existing infrastructure, reducing adoption resistance while improving operational efficiency.

This capability is supported by an XGBoost-based architecture that performs effectively on structured manufacturing data and is optimized for recall, aligning with the industry's priority of minimizing undetected defects. Combined with a SaaS delivery model, the system lowers upfront investment requirements and allows cost-sensitive manufacturers to adopt advanced analytics without significant capital expenditure. The inclusion of user-configurable risk thresholds further extends its applicability across production environments with varying quality standards.

A distinct strength is the system's embedded ESG contribution. By prioritizing high-risk units and reducing unnecessary inspection, Omni-QC lowers energy consumption, reduces material waste from late-stage defect discovery, and extends the lifespan of inspection equipment. These benefits are achieved directly through its operational logic, without requiring additional system modifications.

Opportunities

On the opportunity side, the expanding EMS market is increasing both production volume and process complexity, amplifying the inefficiencies of uniform inspection approaches. This creates strong demand for risk-based optimization solutions. This demand is further reinforced by the ongoing shift of manufacturing capacity toward Southeast Asia, where rapidly scaling facilities face higher cost pressure and operational variability.

Weaknesses

However, Omni-QC's effectiveness depends heavily on the availability of structured, high-quality production data, limiting adoption among less digitally mature manufacturers. As a new entrant, it also lacks a proven production track record, which can slow adoption in a risk-averse industry. Additionally, model performance may degrade over time due to data drift, requiring continuous monitoring and retraining.

Threats

Externally, the presence of well-capitalized competitors with established client relationships creates competitive pressure, even if their solutions differ technically. At the same time, concerns around data security in cloud-based deployments can extend procurement timelines, particularly among large manufacturers handling sensitive production data.

S-O STRATEGIES

The Strength–Opportunities (S-O) strategy for Omni-QC focuses on leveraging its technical and operational advantages to capture value in a rapidly expanding and evolving EMS market. At the core of this approach is positioning predictive risk-scoring as a new standard for scaling manufacturing operations. By combining upstream defect prediction with a recall-optimized XGBoost architecture, Omni-QC aligns closely with the industry's increasing need to minimize undetected defects while managing rising inspection costs. As production volumes grow, especially in high-mix, high-complexity environments, traditional uniform inspection becomes less efficient, making risk-based prioritization both economically and operationally compelling.

This positioning is further strengthened by targeting high-growth manufacturing hubs in Southeast Asia, where geopolitical production shifts and rapid industrial expansion are creating more complex and cost-sensitive production environments. These regions provide ideal conditions to demonstrate the system's value, as inefficiencies in conventional inspection approaches are more pronounced at scale. By entering markets where the pain points are most acute, Omni-QC can establish early proof of value and build momentum through measurable performance improvements.

In addition, the SaaS-based delivery model enables the product to appeal to cost-sensitive firms, particularly Tier 2 manufacturers that may lack the capital for large upfront investments. By lowering the barrier to entry and shifting costs toward a more flexible operational expenditure model, Omni-QC accelerates adoption while maintaining scalability. This is complemented by the system's configurability, which allows users to adjust risk thresholds based on their specific quality requirements. Such flexibility makes the solution applicable across multiple industries with varying standards, from consumer electronics to more stringent sectors like automotive, without requiring extensive customization.

The ESG and sustainability dimension of Omni-QC's strength profile opens a complementary strategic avenue as procurement organizations and institutional investors apply growing scrutiny to the environmental and social credentials of their technology vendors. Manufacturers facing pressure from multinational OEM clients or institutional shareholders to demonstrate measurable reductions in material waste and energy consumption can credibly point to Omni-QC's inspection allocation logic as a verified operational mechanism for achieving those outcomes. This framing strengthens the product's commercial case beyond pure cost efficiency, adding a dimension that well-capitalized incumbents focused on post-assembly detection cannot replicate through their existing architectures.

W-T STRATEGIES

On the other hand, the Weakness–Threats (W-T) strategy is designed to mitigate the product’s inherent limitations while addressing external competitive and market risks. A key priority is the development of strong onboarding and data integration support to overcome the product’s dependence on structured manufacturing data. By assisting clients in centralizing, cleaning, and preparing their data, Omni-QC can expand its addressable market and reduce friction during early adoption. This not only improves implementation success rates but also strengthens client confidence in the system’s reliability.

Another critical strategic direction is positioning Omni-QC as a complementary layer rather than a replacement for existing inspection systems. Given the presence of well-established competitors and the significant capital already invested in traditional inspection infrastructure, emphasizing enhancement over disruption reduces resistance from potential clients. This approach reframes the product as a value-adding optimization tool that works alongside current systems, making adoption less risky from both an operational and financial perspective.

Finally, strengthening data security is essential to addressing one of the most significant barriers to cloud-based deployment. By implementing robust data governance frameworks, secure data handling practices, and compliance with industry standards, Omni-QC can alleviate concerns among manufacturers handling sensitive production data. This not only helps to build trust but also shortens procurement cycles that are often prolonged due to security evaluations

4.2 Strength-Based Capability Analysis

The technical and commercial strengths of Omni-QC form a cohesive set of capabilities that reinforce one another across its architecture, deployment model, and operational design. Rather than relying on a single differentiating feature, the system is built around a layered value structure in which each capability amplifies the effectiveness of the others.

Upstream risk prediction rather than post-assembly detection

At the foundation of the system is its upstream, predictive approach to quality control. Conventional inspection architectures assess units after the assembly process has concluded, meaning that any defect identified at that stage has already consumed the full cost of production. Omni-QC intervenes earlier, ingesting structured process data from solder paste printing, component placement, and reflow profiling stages to assign a defect probability score before a unit enters the inspection queue. This temporal shift from reactive detection to pre-inspection prediction fundamentally changes the economics of quality assurance, as flagging a high-risk unit before reflow allows for corrective intervention at the lowest possible cost in the production sequence. According to Saki (2026), over 60 to 70% of surface mount technology defects originate at the solder paste printing stage, which makes early-stage data ingestion not merely advantageous but structurally significant.

XGBoost model architecture with recall optimization

The predictive engine is built on an XGBoost gradient-boosted decision tree architecture, selected specifically for its alignment with the nature of manufacturing process data. Unlike deep learning models trained on image tensors, XGBoost operates on structured, tabular inputs, matching the row-and-column format of machine sensor logs, parameter records, and historical defect outcomes without requiring a visual dataset. Miedema et al. (2024) demonstrate that XGBoost consistently achieves high predictive performance on multidimensional manufacturing datasets and exhibits a conservative bias toward defect recall, prioritizing the detection of genuine defects over the avoidance of false alarms. This recall-oriented behavior directly serves the system's safety objective: ensuring that no genuinely defective unit is incorrectly cleared as low risk and routed away from adequate inspection coverage.

SaaS cloud delivery model with low adoption barrier

A further strength lies in the system's deployment model. Omni-QC is delivered as a Software-as-a-Service (SaaS) platform, eliminating the capital expenditure requirement that has historically been one of the most significant barriers to AI adoption in manufacturing. The subscription-based structure converts adoption from a capital investment decision into an operational expenditure, lowering the financial barrier for cost-sensitive manufacturers, particularly mid-tier contract manufacturers and original design manufacturers operating in Southeast Asia.

Preserves existing infrastructure

Beyond access, the system preserves the value of existing inspection infrastructure rather than competing with it. Omni-QC functions as an orchestration layer that determines how AOI, X-ray, and solder paste inspection systems should be deployed, concentrating their coverage on units the model has scored as high risk while reducing redundant inspection of units assessed as low risk. This compatibility with existing capital equipment removes a common adoption objection, namely the concern that deploying a new quality system will render prior investments obsolete, and repositions Omni-QC as an efficiency multiplier for infrastructure already in place.

User-configurable risk thresholds adaptable to diverse quality standards

The system's user-configurable risk threshold interface extends this adaptability further. Factory quality engineers and production managers can adjust the sensitivity boundaries of the three-tier classification logic to reflect the specific quality standards and risk tolerances of their production environment without modifying the underlying model. A facility producing consumer electronics may operate with a higher low-risk threshold to maximize throughput, while a facility producing industrial or safety-adjacent boards may lower the high-risk threshold to mandate intensive inspection across a broader range of units. This configurability allows Omni-QC to comply with the full range of IPC classification standards and client-imposed quality requirements across industry segments without bespoke model development for each deployment.

Structurally embedded ESG and sustainability outcomes

A structurally distinct strength is the system's embedded ESG and sustainability profile. As electronics manufacturing faces mounting institutional pressure to demonstrate measurable reductions in material waste, energy consumption, and workforce health burdens, Omni-QC's core operational logic directly addresses all three dimensions without requiring any modification to its technical architecture. By concentrating inspection resources on high-risk units and reducing unnecessary processing of low-risk boards, the system lowers the cumulative energy load on AOI and X-ray equipment, extends the operational lifespan of existing capital infrastructure, and reduces the volume of raw materials wasted through late-stage defect discovery. Critically, earlier defect interception at the solder paste printing and component placement stages prevents the full material and energy investment of completing assembly on units that would have been rejected downstream. The system also improves workforce conditions by redirecting inspection effort away from repetitive false-positive verification tasks, reducing the eye fatigue and monotony that Pyun et al. (2019) identify as a persistent occupational burden on manual inspection personnel. These outcomes are verifiable through standard operational metrics and are therefore credible to procurement committees and institutional investors applying ESG scrutiny to manufacturing technology vendors, without requiring any additional product development to substantiate them.

4.3 Weakness-Based Limitations Analysis

While Omni-QC offers significant operational advantages, a rigorous technical evaluation identifies three structural weaknesses that impose specific constraints on its deployment and long-term performance. Each limitation is paired with a concrete mitigation, either a roadmap item that addresses it directly or a bounded acceptance that defines where the product does not and does not intend to compete

DATA DEPENDENCY CREATES A STRUCTURALLY LIMITED ADDRESSABLE MARKET AT ENTRY

The core functionality of Omni-QC is fundamentally predicated on the availability of highfidelity, machine-level telemetry data. This creates a structural limitation in its addressable market: the system cannot operate on legacy production lines that lack digitised Solder Paste Inspection (SPI) or Pick-and-Place (P&P) data export capabilities. In facilities where equipment utilises proprietary, closed-loop protocols without modern API or industrial bus (OPC-UA/IPCCFX) support, Omni-QC's value proposition is nullified. This dependency means the system is a "digital-first" solution, excluding a significant portion of mid-tier manufacturers in Southeast Asia who have not yet reached Industry 3.0 maturity. Furthermore, the accuracy of the riskscoring mechanism is directly proportional to the granularity of the ingested data; if a factory provides only pass/fail binary data rather than continuous variable measurements, the XGBoost model's predictive precision degrades significantly.

Mitigation: Omni-QC's go-to-market strategy explicitly prioritises Band 1 manufacturers, digitally mature facilities with functional MES infrastructure, for the initial commercial phase. Band 2 manufacturers (partial instrumentation) are approached as a second wave. This staged sequencing converts the data dependency from a market exclusion into a market prioritisation, ensuring that early deployments occur in the environments where the system can demonstrate maximum performance. Expansion into less-instrumented facilities is addressed in the Year 3 product roadmap through a lightweight telemetry bridging module that reduces the minimum data readiness threshold for onboarding.

NO INSTALLED PRODUCTION-FLOOR TRACK RECORD AT MARKET ENTRY

Omni-QC faces a cold-start challenge during the initial 14-day deployment phase. Because the supervised learning model requires a baseline of site-specific historical data to calibrate its weightings for local environmental variables, the system cannot provide peak accuracy on Day 1. During this window, the model operates on generalised weights which may lead to a higherthan-average rate of "Neutral" risk scores, necessitating a temporary reliance on traditional 100% inspection protocols. This initial performance lag represents a commercial weakness, as it requires the customer to invest in integration and data collection for approximately two weeks before the system begins to deliver measurable ROI through inspection optimisation.

Mitigation: The 90-day POC program is specifically designed to absorb this cold-start period. By the end of the POC, the model has accumulated sufficient site-specific production data to demonstrate calibrated performance against the client's own defect history, converting the Day 1 limitation into a measurable improvement trajectory that forms the basis of the subscription conversion conversation.

SENSITIVITY TO DATA DRIFT WEAKENING MODEL PERFORMANCE

The reliability of the predictive QC mechanism is sensitive to "data drift," where changes in the physical production environment occur without corresponding updates to the model's feature set. For example, if a manufacturer switches to a different brand of solder paste or changes the nozzle type on a P&P machine without re-calibrating the Omni-QC model, the established risk thresholds may become obsolete. Unlike a human technician who can intuitively account for a physical change on the shop floor, the algorithmic nature of Omni-QC treats these changes as statistical anomalies or, worse, fails to recognise them as new variables entirely.

Mitigation: The continuous performance monitoring layer is the primary architectural response to this risk. By tracking the alignment between risk score distributions and confirmed defect outcomes across rolling production windows, the system detects drift before it causes meaningful miscalibration and triggers a retraining alert. The demand-triggered retraining service, billed per engagement at the rates described in Section 6.5, then recalibrates the model to current process conditions, closing the gap between environmental change and model response.

5.0 Market Analysis and Strategy

This chapter defines the external environment in which Omni-QC operates and identifies the specific market segments it is positioned to capture. It opens with a market landscape analysis and PEST examination of the political, economic, social, and technological forces actively shaping demand for predictive quality control. From this macro foundation, the chapter narrows into target market segmentation, targeting and beachhead strategy, product differentiation, and competitive positioning, following the standard Segmentation > Targeting > Differentiation > Positioning framework.

5.1 Market Landscape

Omni-QC enters a market defined by a structural paradox: a multi-billion dollar inspection industry that is simultaneously saturated at its high end and almost entirely unserved at its midtier. Understanding this paradox is the foundation of the entire commercial strategy. The sections below map the market from its competitive ownership structure, through the macrolevel opportunity space, down to the specific addressable beachhead Omni-QC is positioned to capture

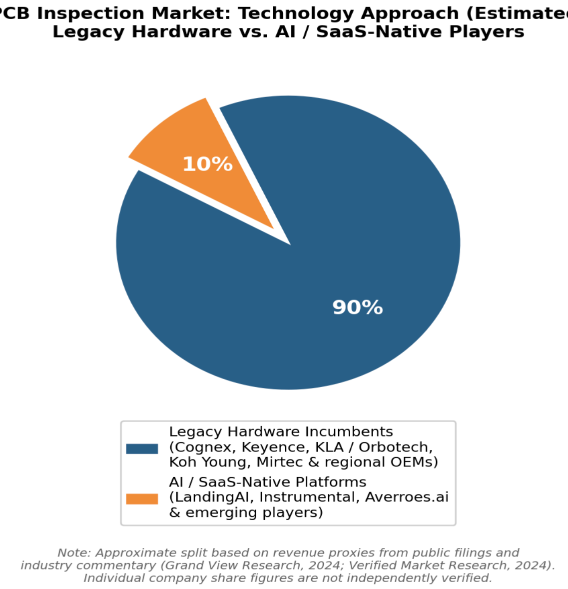
MARKET SHARE AND THE COMPETITIVE LANDSCAPE

The global PCB inspection equipment market, valued at USD 4.41 billion (Verified Market Research, 2024), provides the starting point for understanding the commercial opportunity. Its current ownership structure reveals not only who dominates the space, but, more importantly, which segments they have left entirely uncontested.

Who Owns the Market

Based on revenue proxies from public filings and industry commentary, market participation is dominated overwhelmingly by legacy hardware incumbents. Cognex, Keyence, KLA (which includes the Orbotech PCB inspection division), Koh Young, Mirtec, and a range of regional OEM suppliers collectively account for the vast majority of the market. Their products are optical inspection machines, sensor systems, and post-assembly defect-detection tools, hardware sold into manufacturing facilities on capital expenditure cycles, with multi-year replacement horizons.

AI and SaaS-native platforms (the most visible of which is LandingAI's LandingLens, alongside Instrumental and Averroes.ai) account for a small fraction of the total market by revenue. This is corroborated by the broaderAI-in-manufacturing market sizing: the entire global AI manufacturing sector was valued at USD 5.32 billion in 2024 (Grand View Research, 2024), while the PCB inspection hardware market alone sits at USD 4.41 billion, indicating that hardware incumbents still generate multiples of the revenue of all AI software players combined. Note: Individual company market share percentages for the PCB inspection segment are not published in a single verified public source. The chart below presents a directional split: Legacy Hardware vs. AI/SaaS-Native, derived from revenue proxies and industry commentary (Grand View Research, 2024; Verified Market Research, 2024). It is presented as an illustrative estimate, not a verified market research figure.



Note: Individual company market share percentages for the PCB inspection segment are not published in a single verified public source. The chart below presents a directional splitLegacy Hardware vs. AI/SaaS-Native, derived from revenue proxies and industry commentary (Grand View Research, 2024; Verified Market Research, 2024). It is presented as an illustrative estimate, not a verified market research figure.

Figure 36. *Estimated PCB Inspection Market by Technology Approach*

The IPC Class Bifurcation: A Structural Market Gap

The more strategically significant insight emerges when market participation is mapped against the IPC board classification system. The global PCB market is governed by three classification tiers: Class 1 (general consumer electronics), Class 2 (industrial controls, communications, and business machines requiring extended reliability), and Class 3 (mission-critical applications in aerospace, defence, and medical devices). These classifications define entirely different commercial ecosystems, with different buyers, different procurement processes, and critically, different levels of AI penetration.

Class 3 manufacturers operate under extreme compliance pressure and carry the budget and mandate to invest in sophisticated vision AI systems. AI-powered platforms have made meaningful inroads into this segment: LandingAI, Instrumental, and DarwinAI, acquired by Apple in March 2024 and no longer commercially available as a standalone product, vacating a segment of the Class 3 AI inspection market and leaving no credible startup-tier successor.

Class 1 and Class 2 manufacturers, which constitute the bulk of the ASEAN electronics manufacturing ecosystem, producing consumer IoT, telecommunications hardware, and industrial controls represents an entirely different situation. This is the segment responsible for the majority of global SMT production volume. It is also the segment where AI penetration is functionally non-existent. As shown in previous figures, AI-served approaches account for less than 4% of the Class 1/2 inspection market, estimated based on the stated target segments of publicly documented AI inspection vendors. Legacy AOI hardware, reactive, post-assembly, and applied uniformly regardless of unit risk constitutes virtually the entire operational infrastructure.

This figure reflects standalone AI software vendors only. Hardware-embedded AI features, such as Koh Young's KPO process optimizer or ASMPT's WORKS Process Expert are bundled into capital equipment cycles and are thus inaccessible to facilities running mixed-vendor lines, which constitute the majority of mid-tier ASEAN SMT facilities.

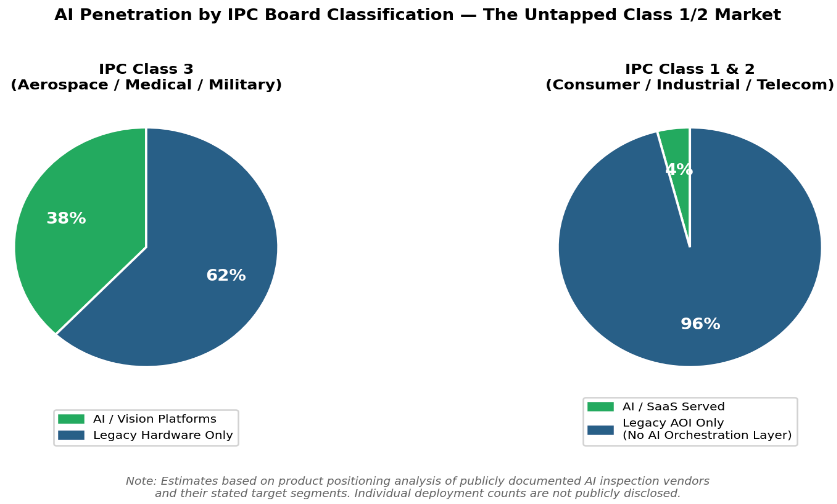


Figure 37. *Estimated AI Penetration by IPC board classification*

TOTAL ADRESSABLE MARKET (TAM)

Omni-QC is, fundamentally, an AI software system that generates revenue through subscriptions priced against measurable manufacturing outcomes. Its commercial success depends not only on the readiness of individual facilities to adopt it, but on the broader willingness of the manufacturing sector to treat AI software as a recurring operational expenditure rather than a capital investment in hardware. The appropriate TAM frame is therefore the global integration of artificial intelligence within manufacturing environments.

The global AI in the manufacturing sector was valued at USD 5.32 billion in 2024 and is projected to reach USD 47.88 billion by 2030, at a compound annual growth rate of 46.5% (Grand View Research, 2025). This trajectory reflects a structural transition across the industrial base, from reactive detection-and-repair quality models toward predictive, data-driven prevention. Three convergent trends define the conditions under which this transition is accelerating.

The first is the emergence of predictive quality as a distinct procurement category. Facilities are increasingly distinguishing between post-assembly defect detection and upstream defect prevention driven by process telemetry. This is creating a software budget line (process intelligence) that did not exist as a standalone category five years ago and into which OmniQC's subscription revenue model fits directly.

The second is the growing readiness of structured machine data. PLC outputs, MES logs, and inline SPI measurements are now standard data streams in mid-to-large SMT facilities. This infrastructure predates the AI use cases now being built on top of it, which means the integration cost for process telemetry-based AI is substantially lower than for solutions that require new sensor hardware.

The third is the declining cost of ML inference. Compute efficiency gains across both hardware and software have reduced the unit economics of always-on AI systems to a point where mid-tier manufacturers can justify subscription-priced AI tools that would have been economically unviable three years ago.

The startup ecosystem within this TAM is predominantly concentrated in computer vision: postassembly defect detection, visual quality control, and image-based anomaly classification. Companies including LandingAI, Instrumental, and Averroes.ai have attracted venture investment on the strength of this use case. What is structurally absent from this landscape is the sub-segment Omni-QC occupies: pre-inspection process intelligence for Class 1/2 production environments, operating on structured telemetry rather than images, priced for mid-tier operational budgets. That absence is the commercial entry point.

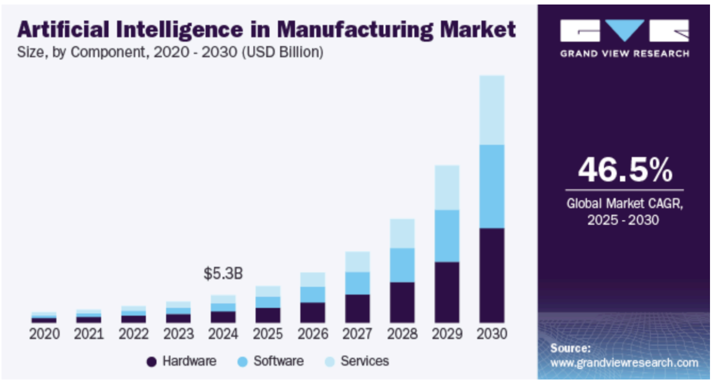


Figure 38. *AI in Manufacturing Market Value (Grand View Research, 2025)*

SERVICABLE ADDRESSABLE MARKET (SAM)

The structural moat Omni-QC builds over time is not the algorithm, XGBoost and SHAP are open-source and reproducible, but the proprietary dataset accumulated across active deployments. Each production line added to the fleet contributes labelled production telemetry and confirmed defect outcomes specific to ASEAN mid-tier process conditions, a dataset that vendors serving Class 3 customers or operating in North America and Europe cannot replicate. A market entrant arriving after Omni-QC has instrumented 100+ ASEAN lines would face a model calibrated on manufacturing conditions that are structurally different from those in their existing training base.

Omni-QC's direct operational market is the Global PCB Inspection and Electronics Manufacturing Services (EMS) sector**.** The PCB inspection equipment segment is currently valued at USD 4.41 Billion, within a global EMS market projected to reach USD 1,201.16 Billion by 2035 (Precedence

Research, 2025). Several conditions make this segment specifically addressable rather than merely large:

- False-Alarm Overhead: Industry practitioners report that on some production lines, the majority of AOI-flagged units are cleared as acceptable after manual re-inspection (Zhang, 2025), a pattern AOI vendors themselves market against (KLA, 2023).
- Revenue Exposure: The cost of poor quality can drain up to 20% of a company's revenue in extreme cases, with false detection costs estimated at approximately USD 7,000 per panel annually (Laba, 2026).
- Process Vulnerability: Over 60–70% of SMT defects originate specifically at the solder paste printing stage, creating a concentrated opportunity for upstream predictive intervention (Saki, 2026).
- Data Readiness: These manufacturers already collect high volumes of structured machine data from PLCs and log files that Omni-QC requires for its XGBoost-based risk scoring, no additional sensor hardware is needed. Provided the facility's machines generate structured data outputs via MES, SPI machine logs, or IPC-CFX-compatible protocols.
- Zero AI Coverage: As established in previous sections, no software-first AI orchestration layer exists for this classification tier. Omni-QC enters structurally thin competitive coverage.

The commercial mechanics through which Omni-QC enters this market, including the zero-capital 90-day POC, quantified success thresholds, and grant-funded pilot economics are detailed in Chapter 6.

SERVICABLE OBTAINABLE MARKET (SOM)

Characteristics of the SOM

The SOM is geographically concentrated in three high-density electronics manufacturing corridors: Penang and Johor in Malaysia, and the Ho Chi Minh City and Hanoi corridors in Vietnam. These locations share four characteristics that define an addressable SOM rather than a theoretical market figure:

- Digital Infrastructure Readiness: Facilities in these corridors overwhelmingly operate with functional MES infrastructure, inline SPI machines, and structured machine data outputs the prerequisite for Omni-QC integration without needing a hardware upgrade.
- Production Volume and Complexity: These hubs process the highest volumes of Class 1/2 SMT production in the region, making the financial case for inspection optimisation most immediate and most quantifiable.
- Competitive Void at the Software Layer: No AI orchestration product exists in this space. The SOM captures a genuinely uncontested market position rather than a share of an existing competitive field.

- Regulatory and Policy Tailwinds: Malaysia's NIMP 2030 and Vietnam's Digital Industry Law create institutional momentum and direct subsidy access that accelerate procurement of AI-enabled manufacturing tools.

How Omni-QC Utilizes the SOM

First, capturing the ASEAN beachhead builds the data flywheel that is Omni-QC's most defensible long-term asset. Each production line added to the active deployment base contributes labelled production and defect data that continuously improves model accuracy across the installed fleet. A competitor entering the market after Omni-QC has established 300+ active lines would face a model trained on hundreds of thousands of hours of ASEAN factory production data, a dataset impossible to replicate from outside the market.

Second, the SOM geography provides the reference account density required for enterprise B2B sales at scale. Procurement decisions for quality management systems in manufacturing are heavily reference-driven. A Penang-based EMS provider considering Omni-QC is significantly more likely to proceed when the system is already deployed at comparable facilities within the same industrial corridor. Geographic concentration within the SOM accelerates this reference-building cycle.

Third, the SOM functions as a Series A validation instrument. Demonstrating meaningful active line penetration in the ASEAN beachhead, with documented inspection workload reductions and scrap savings, this provides the empirical foundation for expansion into the broader Asia-Pacific market and eventually North America and Europe, where the Class 1/2 untapped AI void exists at far larger scale.

Rationale for the 380-Line Target

The ASEAN electronics manufacturing corridor, spanning Malaysia, Vietnam, Thailand, and Singapore, hosts a large number of active SMT production lines across Tier 1 and Tier 2 EMS facilities. Penang alone hosts over 50 EMS operators, including domestic and regional Tier-2 manufacturers such as Dufu Technology, Nationgate, Globetronics, and SCI Manufacturing, alongside larger multinationals whose local contract manufacturing arms operate independently of group procurement, and runs multiple SMT lines per facility. The Johor corridor serves cross-border Singapore OEM clients. Vietnam's Ho Chi Minh City and Hanoi hubs are among the fastest-growing EMS locations in the region due to supply chain rebalancing away from China (MordorIntel, 2026). Thailand, Singapore, and the Philippines contribute additional manufacturing capacity through established contract manufacturers. Together, these corridors represent a substantial population of SMT lines that meet Omni-QC's data readiness criteria.

Geography	Key Operators	Est. facilities	Avg. Lines / Facility	Target Lines (Yr 5)
Penang, Malaysia	Dufu Technology, Nationgate, Globetronics, SCI Manufacturing	50+structurally uncontested white space	5-8	95
Johor, Malaysia	VS Industry, NN PCB	30+	4-6	55
Vietnam (HCMC & Hanoi)	Nidec, Jabil, Sanmina, and emerging domestic operators	60+	5-7	140
Thailand / Singapore / Philippines	Hana Microelectronics, Venture Corporation, IMI	40+	3-5	65
Broader ASEAN (pipeline seeding)	Various	Variable	3-5	25
Total	-	**180+**	-	**380**

**Note: Lines-per-facility estimates are based on standard SMT line configurations for the facility size mix in each region. Operator examples are representative of the Tier-2 and mid-tier domestic EMS profile that constitutes Omni-QC's primary commercial target; Tier-1 multinationals operating in the same corridors are treated as aspirational reference accounts rather than initial pipeline targets, given their longer procurement cycles and internal engineering capacity.*

Figure 39. *Facility counts derived from publicly reported EMS operator presence in each corridor (MordorIntel, 2026; Tan, 2024).*

- Penetration Rate Conservatism: 380 lines represent a deliberately conservative penetration rate relative to the total addressable line population in these corridors. ASEAN manufacturing is a risk-averse procurement environment, which is why the target is calibrated to what is achievable with a direct enterprise sales team of realistic size, without assuming viral adoption or channel distribution that does not yet exist.

- Annual POC Pipeline Capacity: The POC pipeline scales from 4 initiations in Year 1 to 380 in Year 5, a rate consistent with the planned growth of the Technical Account Management team. A Year 5 team of 8-10 TAMs, each managing 30-40 POC relationships annually, is standard for an enterprise SaaS operation targeting USD 15 million ARR.

- POC-to-Line Conversion Logic: Because each POC engagement targets a single SMT line for the 90-day validation cycle, the 380-line SOM also represents the maximum POC throughput capacity in the final year of the plan. The base-case conversion rate of 45% is calibrated against published enterprise B2B pilot-to-paid conversion benchmarks for industrial software, which typically range from 30 to 50%, and is weighted toward the midpoint to reflect the longer evaluation cycles characteristic of quality-function procurement in manufacturing environments. At this rate, cumulative active lines reach 292 by Year 5. The SOM ceiling of 380 is only realised at 100% conversion, which is deliberately not assumed. This construction ensures the SOM figure is a mathematically defensible bound rather than an optimistic projection.

5.2 PEST Analysis

The adoption of Omni-QC is driven by external shifts that transform predictive inspection from a luxury into an industrial requirement. Politically, tightening IPC standards and aggressive ASEAN subsidies provides the compliance pressure and capital for digital upgrades. Economically, inflation in manual labor and the exponential escalation of defect-related expenses (Crosby, 1979) require proactive quality management. Socially, the mandate for sustainable manufacturing and safety-critical reliability aligns with scrap reduction and ESG reporting (Zhang et al., 2025). Finally, the growth of MES infrastructure and collapsed inference costs (Stanford HAI, 2025) remove technical barriers to adoption. This PEST analysis shows how these forces combine to establish AI-driven risk triage as the essential operational baseline.

P, Political / Regulatory	
Lead claim	*As global IPC standards expand beyond human biological accuracy, predictive risk triage has transitioned from a luxury into a strict regulatory necessity. This shift intersects with a massive RCEP-driven supply chain diversification into Southeast Asia, where regional governments are deploying billions in targeted subsidies to fund smart manufacturing upgrades. By capturing greenfield facilities during their commissioning phase, these state-sponsored funds neutralize adoption barriers, establishing AI-driven inspection as the permanent operational baseline for the region's modernized manufacturing sector.*
Move 1 Increasing PCB quality standards	Updated IPC-A-610 Revision J and IPC-6012 Revision F standards significantly expand inspection requirements for complex board geometries. As human accuracy is capped at 80%, manual inspection fails to meet the 100–500 PPM defect targets demanded by modern OEMs. This compliance pressure transforms risk-based triage into a regulatory necessity. Omni-QC resolves this by automating risk-proportionate inspection, focusing resources on high-risk units to ensure compliant statistical process control.
Move 2 RCEP Trade Agreement & Supply Chain Diversification	The Regional Comprehensive Economic Partnership (RCEP) is aggressively reducing intra-ASEAN trade barriers, accelerating cross-border EMS expansion. Driven by the need to mitigate geopolitical risks, multinational OEMs are actively rebalancing their supply chains away from China into Malaysia and Vietnam. These new, high-complexity facilities are being built from the ground up with specific budget mandates for AI-ready quality control (QC) infrastructure. Strategic Response: Omni-QC is positioned to enter the sales cycle precisely during this facility commissioning phase. By integrating directly into the factory blueprint before legacy behaviors are entrenched, we maximize platform adoption rates and secure long-term vendor lock-in.
Move 3 ASEAN subsidies lower the adoption barrier	ASEAN governments are aggressively subsidizing the transition to smart manufacturing through multi-billion dollar matching grants and fiscal incentives. Malaysia's NIMP 2030 targets 3,000 smart factories via RM 500,000 matching grants, while Indonesia offers a 300% R&D super-deduction. Simultaneously, 2026 stimuli in Thailand and Vietnam grant multi-decade tax exemptions and 50% cost coverage for high-tech projects. These state-backed frameworks structurally lower entry barriers for industrial automation, transforming digital infrastructure into a subsidized national strategic priority across the region.
Closing insight	**Tightening IPC standards and RCEP-driven supply chain shifts have made automated, risk-based inspection a structural mandate for meeting elite PPM targets. Aggressive regional subsidies further de-risk this transition, transforming digital quality control from a discretionary upgrade into a subsidized regulatory necessity for ASEAN's modernized greenfield facilities.**
E, Economic	

Lead claim	*With manual inspection rapidly accelerating into the most expensive quality-control line item across Southeast Asia, the reliance on human visual verification is becoming financially unsustainable. Because human accuracy possesses a biological ceiling, and every undetected defect compounds in cost by an order of magnitude at each subsequent production stage, reactive inspection is now a severe financial liability. Omni-QC's upstream risk triage intercepts this value-bleed at the source, transforming quality control from a heavily inflated operational expense into a proactive margin expander.*
Move 1 The labour cost burden	Manual visual inspection constitutes a significant and growing share of total appraisal cost in SMT production. ALLPCB (2025) benchmarks manual inspection at approximately USD 0.40 per board (RM20/hr labour at 50 boards/hr), compared to under USD 0.10 per board for automated AOI at full utilisation, a 4x cost differential that widens as labour costs rise. A single AOI machine replaces 5-8 manual inspectors, and Twisted Traces (n.d.) estimates inspection charges add 10-15% to total PCB manufacturing project cost. Across ASEAN, direct assembly labour already accounts for 10-15% of total PCB manufacturing cost, with dedicated testing and quality control adding a further 5-10% (Instrumental, n.d.). As wage inflation accelerates across previously low-cost manufacturing hubs, Malaysia, Vietnam, and Thailand are all experiencing tight technical labour markets (DigiconAsia, 2025), the financial viability of sustaining manual inspection floors as the primary quality control mechanism is structurally eroding.
Move 2 The Rule of Ten cost escalation	The economic urgency of upstream interception is governed by a well-established quality cost escalation pattern: defects that cost roughly RM1 to prevent at source cost approximately RM10 to remediate internally and RM100 once they reach the customer, a principle rooted in Crosby's (1979) Cost of Poor Quality framework, which places total quality costs at 10-15% of revenue for well-managed organisations and up to 20-40% for poorly controlled ones. For a RM50M revenue PCB facility, the COPQ exposure range is RM5–20M annually, dwarfing the cost of a SaaS subscription. For electronics manufacturing specifically, over 60% of electronic failures originate from soldering defects, component placement errors, or insufficient process control (IPC field data, as cited in SCS PCBA, 2026), all variables captured upstream by MES and predictable before the board reaches the AOI machine.
Closing insight	**The economic math of modern PCB manufacturing leaves no room for reactive quality control. Because over 60% of defects originate upstream, and the labor required to manually inspect them is structurally inflating, Omni-QC's predictive triage functions not as an additional software expense, but as a massive cost-recovery engine. By actively neutralizing the multi-million dollar exposure of the Cost of Poor Quality (COPQ), the platform delivers a mathematically verifiable payback period measured in weeks, transforming a typical enterprise software purchase into an irrefutable financial mandate.**
S, Social	

Lead claim	*In the Southeast Asian manufacturing landscape, the primary drivers for AI adoption are the societal mandate for sustainable production and the requirement for rigorous technical legitimacy. By converting factory-floor efficiency into compliance-ready ESG currency and aligning industrial outputs with safety-critical consumer trust standards, Omni-QC transforms industry-wide social constraints into a structurally defended recurring revenue engine.*
Move 1 Societal Push for Sustainable Electronics	Driven by social movements and multinational OEM green procurement scorecards, the societal push for sustainable resource usage is pressuring contract manufacturers to urgently adopt green practices. In PCB assembly, environmental liability is heavily concentrated in the "Scrap Train," where late-stage defect detection results in the severe waste of embodied carbon from scrapped raw materials and the needless expenditure of highly polluting thermal reflow energy. Omni-QC strategically capitalizes on this social shift by converting factory-floor waste reduction directly into sustainability metrics. As its upstream predictive triage systematically prevents material scrap, the platform aligns with green consumer expectations, providing a powerful secondary purchase justification that transforms Omni-QC into a strategic asset for corporate sustainability (Zhang et al., 2024).
Move 2 Industrial Safety & Consumer Trust Standards	Increasing social pressure for zero-defect manufacturing in safety-critical sectors necessitates rigid technical legitimacy. Omni-QC addresses these societal expectations by strictly adhering to IPC compliance and safety standards required by Class 2 and 3 manufacturers. This commitment to high safety standards ensures that industrial outputs remain reliable and compliant with evolving global standards for consumer protection. This technical legitimacy is essential for maintaining social trust in complex electronics.
Closing insight	**To summarize, Omni-QC is strategically engineered to resolve two escalating social pressures: the mandate for environmental sustainability and the demand for technical safety. By dismantling the "Scrap Train," the platform converts operational efficiency into the ESG data required for green procurement. Simultaneously, its adherence to IPC compliance provides the legitimacy needed to secure consumer trust in safety-critical electronics. This dual alignment bridges the gap between the factory floor and the boardroom, equipping executives to navigate modern environmental and safety compliance.**
T, Technological	
Lead claim	*MES maturation means the process data already exists; cloud inference costs have collapsed to make per-unit scoring economically trivial; and XAI tools like SHAP have made model decisions fully auditable, removing the three technical prerequisites that previously made AI-driven QC unviable on the factory floor.*
Move 1 The data infrastructur	The maturation of Manufacturing Execution Systems (MES) provides the essential structured data for machine learning, with the Asia-Pacific market projected to reach RM56 billion by 2035 (Fortune Business Insights, 2025; MarketsandMarkets, 2025). While MES ensures compliance, integration

e already exists	occurs through standard CSV or XML files, maintaining vendor-agnostic compatibility across Koh Young, Omron, and ASM equipment. Using IPC-2591 CFX 2.0, the system leverages existing server data including thermal profiles and placement pressures to train dynamic quality models (I-Connect007, 2025).
Move 2 Inference costs have collapsed	The cost of cloud machine learning inference have fundamentally inverted since 2022. The Stanford HAI AI Index Report (Stanford Human-Centered AI Institute, 2025) documented a 280-fold reduction in inference cost for GPT-3.5-level performance between November 2022 and October 2024. a16z's LLMflation analysis (a16z, 2024) measured costs declining at approximately 10x per year for equivalent performance. In June 2025, AWS reduced its H100 GPU instance pricing by 44% to approximately USD 3.90 per hour (ARK Invest, 2026). For computer vision inference on structured tabular manufacturing data, considerably less computationally demanding than image-based LLM tasks, per-board scoring on a 10,000-panel/day production line costs in the region of USD 2.40/day at serverless rates.
Move 3 XAI has solved the black-box trust problem	Historically opaque manufacturing ML now uses SHAP (Lundberg & Lee, 2017) to quantify feature attribution, allowing quality managers to audit variables like temperature and humidity. As the most widely deployed industrial XAI approach (Maged et al., 2026), SHAP enabled DarwinAI's DVQI to lower false positive rates from 1.7% to 0.11% (Chung et al., 2023). SolderNet further leverages visualizations for precise defect identification (Gunraj et al., 2023; Haigh, 2023), while 2025 research confirms XAI significantly improves detection accuracy over non-explainable systems (Applied Sciences, 2025).
Closing insight	**The convergence of mature MES infrastructure, collapsed inference costs, and Explainable AI (XAI) has transitioned predictive defect detection into an immediate industrial reality. By leveraging standard IPC-2591 data and edge-based inference, systems bypass the financial and latency bottlenecks of traditional cloud models. Ultimately, SHAP-based attribution provides the mathematical transparency required to audit decisions against real-time variables, ensuring a deployment-ready asset that scales across diverse, vendor-agnostic production environments.**

Figure 40. *Comprehensive PEST Analysis Table*

5.3 Target Market Segmentation

GEOGRAPHIC SEGMENTATION

Southeast Asia constitutes the primary geographic target for Omni-QC. As established in the first section, the region accounts for approximately 11% of global semiconductor market share and roughly one-fifth of global semiconductor equipment manufacturing, with Malaysia, Vietnam, Thailand, Singapore, and the Philippines each operating distinct and growing roles within the global electronics supply chain. This concentration of PCB assembly activity means that the population of manufacturers experiencing the quality control inefficiencies Omni-QC addresses is dense, geographically accessible, and operating at production volumes where efficiency gains translate directly into measurable financial returns.

The region has also become a more strategically significant manufacturing location in recent years for reasons that increase rather than reduce the pressure on quality systems. Multinationals have been rebalancing global production footprints to mitigate geopolitical and tariff risks, with major providers scaling up their Thai and Malaysian sites specifically for AI networking hardware and export-oriented production under the Regional Comprehensive Economic Partnership (MordorIntel, 2026). As production volumes and board complexity at these expanded sites increase, the cost of uniform 100% inspection scales with them, making the case for a targeted, risk-proportionate QC system more pressing at precisely the moment when Omni-QC is entering the market.

Within Southeast Asia, Malaysia represents the most immediate entry point. The country's electrical and electronics industry is the most mature in the region, with Penang in particular functioning as a high-density hub for both large-scale EMS operations and a supporting ecosystem of component suppliers, engineering talent, and industrial infrastructure. Benchmark Electronics opened a new manufacturing facility in Penang, Malaysia, in late 2024 to support growing demand for EMS in the Asia-Pacific region, with a specific focus on medical and industrial electronics (Benchmark Electronics, Inc., 2024). Beyond Southeast Asia, the broader Asia-Pacific EMS market and global markets in North America and Europe constitute a longer-term expansion surface, to be approached once a Southeast Asian track record has been built. .

FIRMOGRAPHIC SEGMENTATION

Within the geographic markets identified above, Omni-QC targets two distinct firmographic tiers defined by manufacturer scale, production volume, and board complexity.

Tier 1: High-Volume EMS Providers

The primary firmographic target consists of large-scale EMS providers operating high-throughput SMT assembly lines at industrial capacity. These organizations process millions of PCB units annually, handle complex multilayer board designs, and serve major OEM clients across sectors including consumer electronics, telecommunications, and automotive electronics. They represent the segment where the financial case for Omni-QC is most immediate and most quantifiable.

From a product classification standpoint, Tier 1 facilities predominantly manufacture IPC Class 2 boards. IPC Class 2 covers general electronic products, including communications equipment, sophisticated business machines, and instruments where extended life is required and uninterrupted service is desired, though not absolutely critical (IPC, 2020).

At this production scale, the cumulative cost of uniform 100% inspection compounds in ways that smaller manufacturers do not encounter. As noted in the first section, the cost of false detections from AOI systems has been estimated at approximately USD 7,000 per panel annually, a figure that scales linearly with volume (Laba, 2026). For Tier 1 providers processing hundreds of panels per day across multiple shifts, this represents a material and recurring operational liability. A system that redirects inspection effort toward genuinely high-risk units, as Omni-QC does through its three-tier risk classification, addresses this liability directly.

Companies in this segment are already investing heavily in fully automated SMT lines and digital production systems to boost yield and reduce defects (Narayan, 2026), which signals a demonstrated organizational readiness to adopt AI-driven process intelligence when a credible return on investment is demonstrated. Omni-QC enters this environment not as a disruptive replacement but as an optimization layer that makes existing capital investments in AOI, SPI, and X-ray infrastructure perform more efficiently. Within this tier, the product specifically targets facilities that still deploy human inspectors alongside automated systems for re-inspection and judgment calls, as these are the operations carrying the highest combined burden of false alarms, inspector fatigue, and resource misallocation.

Tier 2: Mid-Tier Contract Manufacturers and ODMs

The secondary firmographic target comprises mid-size contract manufacturers and original design manufacturers operating in Southeast Asia. These firms typically run multiple SMT lines, have implemented basic automated inspection capabilities, and serve a mix of regional OEMs and global brands at medium production volumes, often across a wider variety of board types within a single facility.

Tier 2 manufacturers predominantly produce IPC Class 1 and Class 2 boards. IPC Class 1 covers general-purpose electronic products where the primary requirement is the function of the completed assembly, encompassing consumer goods, certain computing peripherals, and low-complexity IoT devices where cosmetic imperfections are acceptable provided core functionality is maintained (IPC, 2020). Class 2 boards also feature prominently in this segment, particularly for industrial controls, point-of-sale systems, and networking hardware where reliability expectations are higher but the production mix across a single facility remains broad. This high-mix, multi-class production environment is precisely where uniform 100% inspection creates the most disproportionate overhead, as inspection protocols calibrated for the most demanding board type in a batch are indiscriminately applied to every unit regardless of complexity or risk level.

Mid-tier specialists are actively pursuing strategic design partnerships as a means to differentiate themselves and avoid being drawn into purely price-driven competition (MordorIntel, 2026). For these manufacturers, a documented reduction in inspection cycle times and defect escape rates carries direct commercial value when bidding for new client contracts, particularly as OEM clients themselves raise quality requirements. Omni-QC's SaaS delivery model is well-suited to this segment's budget structure, as it eliminates the upfront capital commitment of a hardware-heavy solution and allows the system's cost to be treated as an operational expense tied to measurable output. The user-configurable risk thresholds further allow mid-tier clients to calibrate the system to their specific production mix, rather than applying parameters built for a single high-volume product line to a more varied board portfolio.

TECHNOLOGICAL READINESS SEGMENTATION

Omni-QC is a data-dependent system. Its predictive accuracy relies on the availability of structured, consistent process data from machine sensors at the solder paste printing, component placement, and reflow soldering stages. A manufacturer can only derive full value from the product if their production environment already generates and retains this data in a form the system can ingest. Technological readiness segmentation therefore divides the addressable market into three bands.

The first and most immediately addressable band consists of manufacturers whose facilities are already digitally instrumented. These are operations with functional Manufacturing Execution Systems (MES), inline SPI machines, and machine-level sensor logging that produces structured data outputs. For these manufacturers, integration with Omni-QC is primarily a configuration exercise rather than an infrastructure investment, and the system can begin generating risk scores within a defined onboarding period. This band is where initial commercial efforts should be concentrated.

The second band consists of manufacturers who have some digital instrumentation in place but have not yet fully integrated their machine data into a centralized system. These facilities may have standalone AOI machines whose outputs are reviewed manually rather than logged systematically, or reflow ovens that record temperature profiles locally without feeding that data into a plant-wide MES. For this band, onboarding Omni-QC requires a preparatory integration phase to establish the data pipelines the system depends on. These manufacturers are viable customers but require a longer sales and implementation cycle, and are more appropriately approached as a second wave once Tier 1 references have been secured.

5.4 Targeting & Market Entry Beachhead

Building upon the geographic and firmographic segmentation established, Omni-QC's market entry beachhead strictly targets both Tier 1 (High-Volume EMS) and Tier 2 (Mid-Tier CMs) manufacturers operating across Southeast Asia. While enterprise software rollouts traditionally sequence from smaller clients up to larger ones, Omni-QC will execute a parallel market entry. Capturing Tier 2 facilities acts as a vital accelerator for rapid market penetration and data diversity, while hunting massive Tier 1 enterprise customers serves as our ultimate, scalable growth engine.

TARGET COMPANIES AND STRATEGIC RATIONALE

For Tier 1, the primary targets are global EMS giants with rapidly expanding footprints in Malaysia and Vietnam, such as Foxconn, Flex, Jabil, and Benchmark Electronics (Komaspec, 2026). These organizations process millions of boards annually, meaning their most acute operational pain point is the late-stage material waste known as the "Scrap Train." By deploying Omni-QC upstream of legacyAOI machines, the system directly intercepts these losses. Reducing material scrap by even a fraction of a percent at a Tier 1 facility yields mathematically irrefutable savings, establishing definitive market authority for Omni-QC .

For Tier 2, the targets are regional mid-size contract manufacturers and ODMs. The target pilot customer profile is a Tier-2 EMS operator in Penang with 3-6 SMT lines, 150-400 employees, existing SPI and AOI equipment, no in-house data science team, and a defect rate above 800 PPM. Annual revenue RM 30-80M. Pain point: monthly scrap cost estimated at RM 40-90K. Budget owner: Operations Director with capex sign-off up to RM 150K without board approval. Rising regional wage inflation has turned the Inspection Tax into an unsustainable labour burden. Targeting Tier 2 allows for faster sales cycles and immediate regional footprint expansion. Furthermore, the high-mix nature of Tier 2 lines exposes the machine learning models to a vast array of varying component geometries and anomaly signatures, accelerating the data flywheel.

BEACHHEAD ACQUISITION PLAN

To penetrate these highly guarded manufacturing environments, Omni-QC will employ a direct enterprise sales motion targeting VP of Operations and Quality Assurance Directors. Initial outreach will leverage localized technical authority, engaging prospects at premier regional industry events such as SEMICON Southeast Asia in Malaysia and NEPCON Vietnam to generate inbound B2B leads.

Once engagement is established, the acquisition plan relies on a zero-friction Proof of Concept (POC) program. To overcome the notoriously protracted procurement cycles and risk aversion of factory operators, Omni-QC will not require integration that halts the production line. Instead, the system will be deployed in "Shadow Mode" (Jidoka Tech, 2025). The AI Co-Pilot operates passively alongside manual inspectors, analyzing telemetry and capturing images without triggering the physical reject mechanisms or disrupting live operations.

During this POC phase, Omni-QC conducts a baseline data audit, comparing the AI's logs against human findings to verify accuracy. By proving the system can successfully reduce the manual re-inspection workload or intercept panel-level scrap in real-time, the sales motion transitions from a subjective negotiation into a mathematical demonstration of value. Upon hitting these predefined success metrics, the POC formally converts into a long-term enterprise integration.

5.5 Differentiation

PRODUCT POSITIONING STATEMENT

This positioning is grounded in operational language rather than abstract AI terminology. The target buyers, large-scale EMS providers and mid-tier contract manufacturers, are operationally sophisticated organizations whose procurement and quality leadership evaluate vendors on measurable production outcomes, not technology novelty. Omni-QC's marketing communications therefore lead with the metrics that matter to these buyers: inspection reduction rate, false negative rate, and the downstream cost consequences of defect escapes and AOI false alarms.

Co-Pilot advisory function operates on a human-in-the-loop model, This architectural choice is a meaningful marketing asset in an industry where quality engineers and production managers carry direct accountability for output. Communications directed at these stakeholders emphasize that Omni-QC equips them with better information and clearer sight lines into process drift, while leaving operational decisions to the people responsible for them.

Omni-QC 's sustainability and ESG outcomes are structurally embedded in its core operational design rather than delivered as a separate compliance layer. The system's predictive inspection allocation logic directly reduces material waste, lowers cumulative energy consumption on AOI and X-ray equipment, and improves workforce conditions by redirecting inspection effort away from repetitive false positive verification tasks. These outcomes position Omni-QC as a credible ESG-aligned vendor for procurement committees and institutional investors applying growing scrutiny to the environmental and social credentials of manufacturing technology partners, without requiring any architectural modification or additional development cost to substantiate those claims.

ERRC GRID - ELIMINATE, RAISE, REDUCE, CREATE

Eliminate	Raise
● **Reliance on reactive, post-assembly detection:** Omni-QC removes the dependency on downstream after-production defect detection. Instead of identifying defects only after they manifest, the system shifts quality control upstream by predicting defect likelihood before inspection begins.	● **Efficiency of the current capital investments:** Omni-QC increases the utilization efficiency of existing AOI and X-ray infrastructure. Rather than replacing installed equipment, it operates as an orchestration layer that ensures inspection resources are allocated where they generate the highest marginal value, extending asset

● **Massive CapEx and hardware lock-in:** The system eliminates the need for large upfront capital investment typically associated with AI-enabled inspection systems with the onboarding fee integrated in its subscription fee. Its cloud-based SaaS architecture decouples intelligence from hardware, allowing manufacturers to deploy advanced quality optimization without committing to inflexible, equipment-bound solutions.	lifespan and improving ROI. ● **Flexibility in a wide range of quality requirements:** The system raises operational adaptability by introducing user-configurable risk thresholds. This allows quality teams to dynamically adjust inspection rigor across different product lines, production volumes, and customer requirements without retraining or redeploying models.
Reduce	**Create**
● **Dependence on heavy image datasets:** Omni-QC reduces the implementation burden associated with computer vision systems by leveraging structured process data through an XGBoost-based architecture. This removes the need for large-scale image labeling pipelines and shortens deployment timelines while maintaining predictive performance. ● **Sustainability Through Reducing Inefficient Resource Allocation** Omni-QC structurally embeds sustainability outcomes into its core workflow. By reducing redundant inspections, lowering machine runtime, and minimizing false-positive verification effort, the system decreases energy consumption, material waste, and operator fatigue. These improvements are realized as direct consequences of operational optimization, not as separate compliance features, strengthening its positioning with ESG-focused procurement and investment stakeholders.	● **A new category: Predictive Inspection Allocation** Omni-QC establishes a distinct operational category by integrating process data ingestion with probabilistic scoring to determine inspection routing. This shifts the role of AI from defect detection to inspection decision-making, redefining how quality control is executed on the factory floor. ● **Actionable Co-Pilot Advice:** The system creates an operational Co-Pilot that translates predictions into concrete actions, including parameter adjustments and inspection routing decisions. This preserves human accountability while enhancing decision quality, aligning with how quality engineers and production managers operate in practice.

Figure 41. *ERRC Grid*

COMPETITOR BASED DIFFERENTIATION

Omni-QC distinguishes itself from the prevailing competitive landscape by shifting the value proposition from detection precision to process orchestration. The differentiation is categorized into three primary pillars:

1. Temporal Differentiation: Upstream Prediction vs. Post-Assembly Detection
   Most market incumbents, including VisionPro, LandingLens and Averroes.ai, operate as a post-production layer. They look at a finished (or semi-finished) board to find a defect that has already occurred. In contrast, Omni-QC intervenes at the process level. By ingesting structured data from solder paste printing and component placement, the system assigns a risk score before a unit even enters the inspection queue.
2. Resource Orchestration
   A significant pain point for EMS providers is the expense of uniform inspection which applies 100% AOI or X-ray coverage to every unit regardless of risk, which creates massive bottlenecks. Omni-QC creates a new category: Predictive Inspection Allocation. It reduces the volume of units requiring intensive inspection. By filtering low-risk boards out of the primary inspection flow, the system preserves the lifespan and increases the throughput of existing capital equipment from vendors like KLA or Orbotech.
3. Human-Centric Execution (Co-Pilot Advisory)
   While specialized systems like Averroes.ai focus on autonomous unsupervised learning to replace human judgment, Omni-QC's Co-Pilot function treats the floor technician as the ultimate decision-maker. By transforming raw probability scores into actionable advisory insights, Omni-QC differentiates itself as an augmentation tool that empowers the existing workforce rather than a disruptive technology that seeks to replace it.

Criterion	Cognex VisionPro	LandingLens	KLA Orbotech	OMNI-QC
Deployment Stage	Post-Assembly	Post-Assembly	Fabrication Layer	**PRE-INSPECTION (Upstream)**
Data Modality	Image/Vision	Image/Vision	Optical/Image	**Tabular Process Data (XGBoost)**
Deployment Model	On-Premise Hardware	Cloud or Edge	On-Premise Hardware	**Cloud-based**
CapEx Requirement	High	Low	Very High	**Zero (SaaS)**

MES Integration	Limited	Partial	No	**Full (SPI, P&P, Reflow)**
Human-in-Loop	No	No	No	**Yes (Co-Pilot)**
Target Segment	All Manufacturing	SMB to Enterprise	Advanced Fabs	**SEA EMS & CMs**
Primary Pain Solved	Post-defect detection	Post-defect detection	Fab-layer defects	**Inspection Tax + Scrap Train**

Figure 42. *Comprehensive Competitor Analysis*

5.6 Positioning

Omni-QC is positioned as the pre-inspection process intelligence layer for IPC Class 1 and Class 2 PCB manufacturers in Southeast Asia. The product is positioned not against existing AOI vendors or computer vision platforms, but as a complementary orchestration layer that intervenes earlier in the production sequence than any commercially available alternative. The target buyer perceives Omni-QC not as a replacement for their existing inspection infrastructure, but as the intelligence that determines how that infrastructure should be deployed.

This positioning is anchored in three claims that the maps below substantiate:

1. The price-feature ratio Omni-QC offers is structurally inaccessible to existing competitors due to their architectural choices, not their commercial choices.
2. The deployment-stage and hardware-dependency combination Omni-QC occupies has no commercially deployed alternative as of Q1 2026.
3. The positioning is defensible long-term because the operating cost structure derives from architecture rather than promotional pricing.

The two maps that follow plot Omni-QC against three named competitors selected to represent the prevailing approaches to PCB inspection: a hardware-bundled fab-tier vendor (KLA Orbotech), a hardware-bundled industrial machine vision vendor (Cognex VisionPro), and a software-first computer vision platform (LandingLens). Plotting against this representative cross-section is intentional, these three competitors collectively occupy the dominant architectural patterns in the market, and a positioning argument that holds against all three holds across the broader landscape. Coordinates are based on publicly available product specifications, vendor positioning materials, and industry analyst commentary current as of Q1 2026, and are intended to communicate strategic positioning rather than precise market measurement.

COMPETITIVE POSITIONING MAPS

The following positioning map Omni-QC against KLA Orbotech, Cognex VisionPro, and LandingLens by SaaS pricing accessibility (vertical axis) against feature breadth from basic detection to predictive plus prescriptive Co-Pilot (horizontal axis). Omni-QC alone occupies the bottom-right quadrant

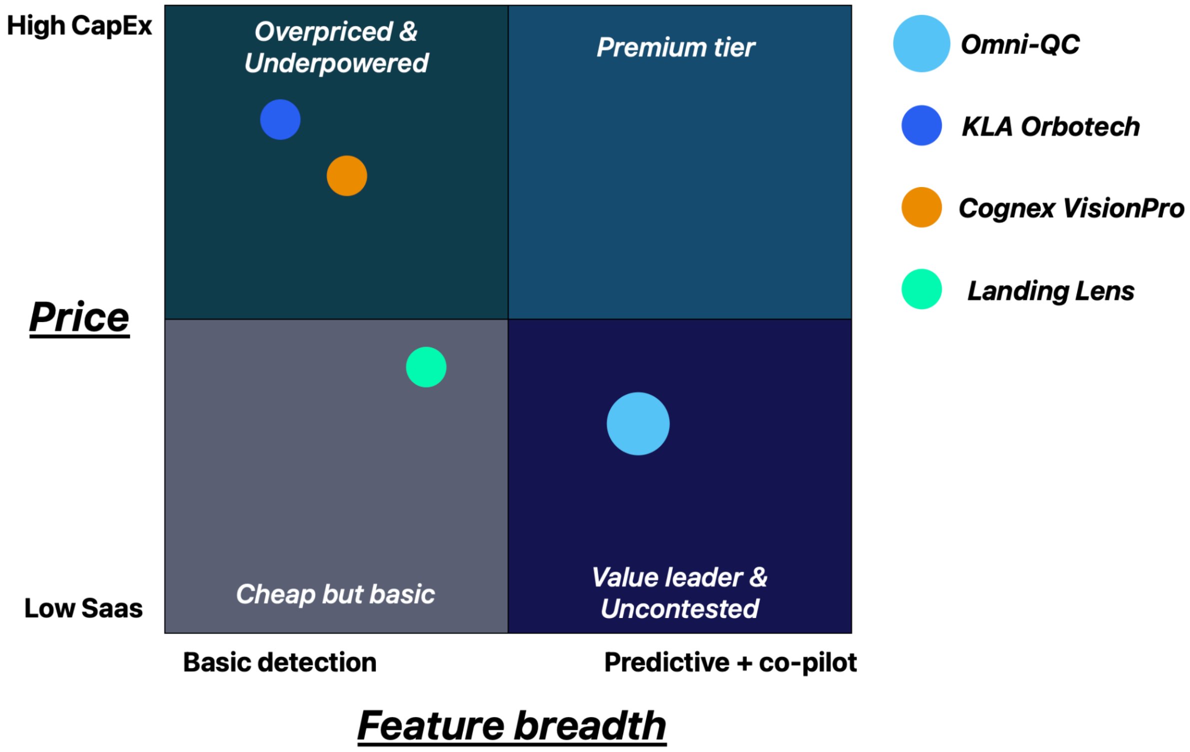


Figure 43. *Competitive Positioning Maps (Map A: Price vs Feature Richness)*

The vertical axis on Map A separates participants by procurement model. Hardware vendors at the top of the map require capital expenditure approval cycles measured in quarters. SaaS-delivered platforms in the lower bands clear plant-manager budgets and proceed without board-level review. The horizontal axis separates participants by the temporal scope of their intelligence: products on the left identify defects after they have been produced, products further right add interpretive or predictive layers, and products on the far right couple prediction with prescriptive operator guidance.

The three competitors trace a diagonal descent from top-left to mid-band: KLA Orbotech anchors the top-left as a fab-tier capital purchase with detection-only intelligence; Cognex VisionPro sits slightly below KLA with marginally broader features bundled with industrial cameras; LandingLens drops into the mid-band as a SaaS platform but remains constrained to image-based post-production analysis. None of the three approach the right side of the map.

Omni-QC's position in the bottom-right is the structural consequence of its architectural choices. XGBoost on structured MES telemetry carries an inference cost approximately three orders of magnitude lower than image-based deep learning at equivalent throughput. The RM28,000 per line annual price point for Tier 2 manufacturers is not a market-entry tactic but a rational price given the underlying cost structure, a price that competitors operating image-bound or hardware-bundled architectures cannot match without absorbing margin destruction. The feature breadth on the right of the map reflects the combination of unit-level risk scoring (predictive) and Co-Pilot parameter offset recommendations (prescriptive), a combination not present in any competitor on the map.

The following positioning map plots the same four market participants by deployment stage (horizontal axis: post-assembly detection to pre-inspection prediction) against hardware dependency (vertical axis: heavy hardware-bound to software-only / cloud). Omni-QC alone occupies the top-right quadrant.

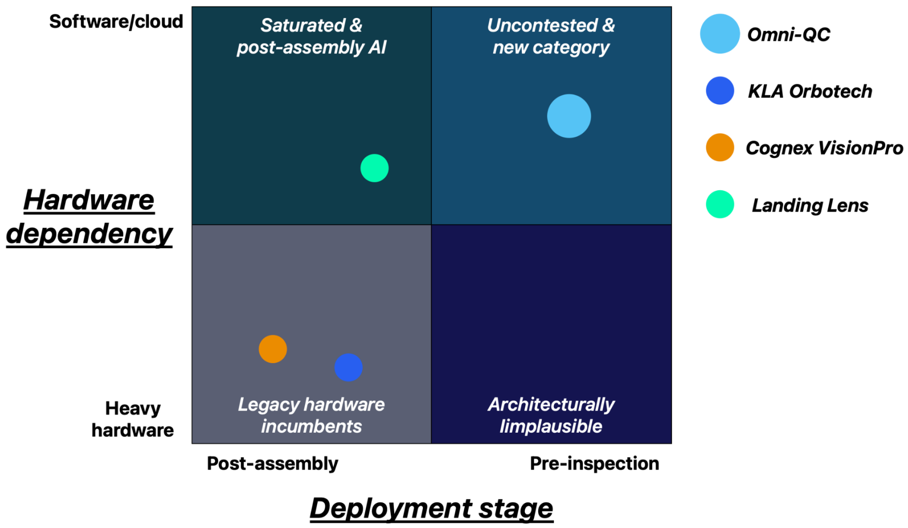


Figure 44. *Competitive Positioning Maps (Map B: Deployment Stage vs Hardware Dependency)*

Map B removes pricing and feature considerations to expose the architectural distribution of the market. The visual result is a vertical column on the left side of the map: KLA Orbotech and Cognex VisionPro at the bottom because they ship inspection systems as bundled hardware, LandingLens in the upper-left band because it operates as a SaaS platform but remains image-input dependent. The three competitors differ markedly in their hardware footprint, yet share the same horizontal position. All three operate strictly post-assembly.

The top-right quadrant is the Omni-QC position. Pre-inspection prediction means the system intervenes before a board enters the AOI queue, scoring each unit's defect probability based on telemetry generated at the SPI, pick-and-place, and reflow stages. The software-only profile is achieved by ingesting structured data the facility's MES already produces, rather than introducing new sensor hardware at any monitored stage.

The bottom-right quadrant is unoccupied not because no vendor has noticed it, but because it is architecturally incoherent. Building pre-inspection intelligence using new dedicated hardware would require sensor retrofits at every monitored stage of every deployed line, which would reintroduce the capital expenditure profile of the bottom-left quadrant and eliminate the operating-cost advantage that makes upstream prediction commercially viable. The only architecturally consistent way to occupy the right half of Map B is the path Omni-QC has chosen: software-only on existing telemetry.

STRATEGIC IMPLICATION OF THE TWO MAPS

Read together, the maps establish that Omni-QC's positioning is not a marketing claim but a structural fact. Map A demonstrates that the price-feature combination Omni-QC offers cannot be matched by competitors operating heavier-cost architectures. Map B demonstrates that the deployment-stage and hardware-dependency combination Omni-QC occupies has no current alternative across the dominant architectural patterns in the market.

The competitive risk implied by Map B is not that the position disappears before Omni-QC reaches it, no current competitor is publicly building toward this specification. The risk is one of timing and reference density. A larger competitor recognizing the gap and capitalizing it within the next 24 to 36 months could erode the first-mover advantage before Omni-QC accumulates the installed base needed to defend it. Section 5.4 (Targeting & Market Entry Beachhead) addresses this through the 380-line ASEAN deployment target, which is calibrated to convert the temporary architectural lead into a durable data flywheel and reference-account moat before that competitive window opens.

The positioning Omni-QC claims is therefore not "better detection" or "cheaper inspection AI." It is the creation of a product category, pre-inspection process intelligence on existing telemetry, and the capture of the operating space that category opens before incumbents can architecturally restructure to enter it.

6.0 Marketing and Business Strategy

This chapter outlines the commercial architecture through which Omni-QC converts technical performance into sustained revenue. It begins with the two foundational marketing strategies that define how the product is brought to market, then details the 7P marketing mix calibrated for B2B enterprise sales in the Southeast Asian manufacturing sector, the partner-provided edge ecosystem, and the after-sales service framework. Sales promotion strategy and the threestream profit model conclude the chapter.

6.1 Basic Marketing Strategy

STRATEGY 1: ZERO-CAPEX ADOPTION MODEL

The first foundational strategy is the complete elimination of upfront capital expenditure as a condition of deploying Omni-QC. Under this model, a client signs an annual subscription agreement and gains immediate access to the full system, encompassing real-time risk scoring across all three monitored production stages, the three-tier classification engine, the user-configurable threshold interface, and the Co-Pilot advisory dashboard, without paying any separate implementation fee, integration charge, or hardware cost. All deployment and onboarding activities are absorbed internally by Omni-QC as an operational cost of the business. The client's first financial commitment is the annual subscription itself, billed upon contract execution, and every subsequent engagement with the system is governed by that single, predictable subscription relationship.

This decision is grounded in a structural reality of the Southeast Asian manufacturing market. Cost remains a significant factor for small and medium enterprises in the region, though the mindset is gradually shifting toward evaluating long-term returns rather than short-term expenses (Swaminathan, 2026). Tier 2 manufacturers, the mid-tier contract manufacturers and original design manufacturers that represent Omni-QC 's primary entry cohort, operate on tighter capital budgets than their Tier 1 counterparts and are therefore disproportionately sensitive to large upfront expenditure requirements. An implementation or integration fee assessed at the point of sale functions as a filter that systematically excludes the firms most likely to generate long-term subscription value for Omni-QC, not because they cannot eventually justify the cost, but because the upfront charge forces a capital allocation decision at exactly the moment of maximum uncertainty about the system's performance in their specific environment.

The SaaS model converts large capital expenditures into manageable operational costs, improving cash flow and reducing financial risk, and this lower entry barrier is particularly attractive to small and medium-sized businesses with limited budgets. Omni-QC exploits this structural property of cloud-delivered SaaS. By anchoring the commercial relationship entirely within operational expenditure, the procurement question shifts from "can we justify this capital investment?" to "does this subscription generate more value than its annual cost?", a question the 90-day POC program is designed to answer definitively before the client is ever asked to commit.

The Zero-CAPEX model is also consistent with Omni-QC s cloud-only deployment architecture. Because all inference, data processing, and model management occur on Omni-QC's cloud infrastructure. There is no physical installation footprint on the factory floor that would traditionally trigger a capital procurement process. The client receives an encrypted software container that integrates with its existing Manufacturing Execution System, and all subsequent computation is handled remotely. This eliminates the hardware procurement cycle entirely, further compressing the time between commercial agreement and first operational output.

Critically, the Zero-CAPEX model is not a loss-leader mechanism or a discounting instrument. The annual subscription price, set at RM54,000 per line for Tier 1 high-volume EMS providers and RM28,000 per line annually for Tier 2 mid-tier clients, remains consistent and uncompromised. What the model removes is not revenue but friction. A client who reaches full operational deployment quickly, without encountering unexpected financial obstacles early in the relationship, represents a substantially more durable subscription than one who enters the engagement with a sense of financial grievance or budget overextension.

A one-year minimum contract term is enforced from the point of subscription execution. This term serves a dual function. It protects Omni-QC's recovery of onboarding costs absorbed under the Zero-CAPEX model, as the first subscription cycle provides the working capital that funds integration engineering. Simultaneously, it provides the client with a sufficient operational window to move through the full performance learning curve: initial deployment, baseline calibration, first retraining cycle if required, and measurable ROI recognition. A shorter term would not allow the system's risk scoring accuracy to compound meaningfully against the factory's production data, undermining the client's ability to evaluate the system fairly and reducing the probability of renewal.

For qualifying Malaysian facilities, this already-accessible price point is further reduced through Malaysia's Industry4WRD Intervention Fund (MITI, Industry4WRD Intervention Fund, 2024), administered under the National Investment Manufacturing Policy 2030 (NIMP 2030). The fund provides a 70:30 matching grant of up to RM 500,000 for eligible AI and digital quality control implementations, covering 70% of qualifying costs with the applicant contributing only 30%. At Omni-QC's Tier-2 subscription rate of RM28,000 per line annually, approximately RM 126,000 at prevailing exchange rates. A qualifying facility's net first-year outlay reduces to approximately RM 37,800. The POC period, which precedes subscription execution, also allows a facility to complete the grant pre-assessment process in parallel with the trial, so that the subscription agreement and grant approval can be timed to coincide. This means the commercial decision a Tier-2 Penang or Johor manufacturer is actually being asked to make is not whether to commit RM 28,000, it is whether to commit RM 8,400 to a system whose ROI has already been demonstrated on their own production floor during the preceding 90 days.

STRATEGY: 90-DAYS FULL ACCESS PROOF OF CONCEPT PROGRAM

The second foundational strategy is a structured, fully operational 90-day Proof of Concept (POC) program offered to qualified manufacturers before the annual subscription agreement is executed. During the POC, the client receives unrestricted access to every functional capability of Omni-QC. No feature is withheld, no artificial ceiling is placed on production volume, and no capability is deferred to the paid tier. The client experiences the full production system operating against its live factory data from day one of the trial.

The commercial logic of the POC is to structurally eliminate the primary question that prevents enterprise procurement committees from acting. The average B2B sales cycle has expanded to 6.5 months, and deals over RM100,000 in annual contract value regularly run six to nine or more months to close (Davis, 2026). The question a procurement committee asks is not whether AI quality control is theoretically valuable. It is whether it will generate verifiable, quantifiable value on their production line, running their board types, with their MES configuration, under their shift conditions. No product demonstration, white paper, or reference case study from a different facility can answer that question. Only live production data from their own floor can.

The POC converts this uncertainty from a barrier into an asset. Rather than asking a client to accept performance claims on faith, Omni-QC invites the client to generate its own evidence. Over 90 days of live operation, the system accumulates production data from the client's three monitored stages, produces risk scores against actual inspection outcomes, and builds an auditable record of the alignment between its predictions and confirmed defect rates. At the conclusion of the POC period, Omni-QC's team presents the client's quality engineering and procurement leadership with a full performance summary: inspection workload reduction achieved relative to the uniform baseline, false-positive filtering delivered on AOI output, high-risk classification accuracy against confirmed defect outcomes, and the projected annual value of efficiency-share savings attributable to the deployment.

The reason no qualified manufacturer should rationally decline the POC offer is a function of its asymmetric risk profile. The client commits no capital, no multi-year obligation, and no modification to its existing inspection infrastructure. It continues to operate its infrastructure systems as normal throughout the trial period, meaning Omni-QC's risk scoring operates as an additive intelligence layer rather than a replacement of any existing process. The only resources the client contributes are production data it is already generating and a defined window of operational attention from its quality engineering team. In exchange, it receives 90 days of live performance data that either confirms the system's value, in which case the subscription decision becomes straightforward, or demonstrates that Omni-QC does not deliver measurable improvement in its specific environment, in which case the client has incurred no financial loss and exits the trial with a clear, data-informed conclusion. The POC is structured so that the rational decision for any qualified manufacturer is acceptance, because the cost of declining is the foregone opportunity to generate evidence that the client could use to make a better procurement decision.

The POC-to-subscription conversion pathway is also designed to minimize friction at the point of transition. Because the 90-day trial operates the full production system, the client's quality team has already learned the dashboard interface, calibrated its risk threshold preferences, and observed the Co-Pilot advisory behavior across multiple production cycles before the subscription begins. There is no additional onboarding phase after conversion. The client moves from the POC directly into an active subscription with a system it has already integrated, validated, and operationally normalized.

6.2 Marketing Mix Strategy

7P'S MARKETING MIX STRATEGY

The 7 Ps marketing mix is designed around a B2B enterprise sales motion where procurement cycles are long, technical validation is non-negotiable, and trust is the primary purchase driver. Each element is designed to remove friction from the adoption decision while structurally locking in recurring revenue.

Ps	Omni-QC Application
Product	A cloud-primary AI defect-probability engine featuring risk classification, user-configurable risk tolerance, and a predictive Co-Pilot Advisory Loop. The software logic remains identical across all tiers, encapsulated in a highly secure container that supports localized edge deployment via third-party hardware.
Price	Pricing abandons standard, static per-seat SaaS licensing in favor of a hybrid architecture tailored to Southeast Asian enterprise budgets. It combines a predictable base annual SaaS subscription, a tiered on-demand model retraining fee, and a mandatory efficiency-sharing bonus based on verified savings.
Place	Deployment relies primarily on a highly secure, centralized cloud infrastructure. For environments with uncompromising data sovereignty restrictions, the software is fully deployable via specialized third-party edge partners.

Promotion	The inbound engine establishes absolute technical authority through the publication of peer-reviewed technical whitepapers. The outbound engine centers entirely on structured, free 90-day performance-based SaaS trials, allowing clients to validate the Co-Pilot's ROI before committing to a long-term enterprise contract.
People	Enterprise deployments are spearheaded by Technical Account Managers (TAMs) wielding deep domain expertise in electronics manufacturing. Onboarding engineers maintain active IPC-A-610 certifications alongside MLOps credentials.
Process	Client integration begins with a frictionless, free 90-day SaaS trial phase. This seamlessly transitions into a mandatory 1-year contract lock-in upon successful ROI validation, bypassing the friction of immediate upfront 12-month commitments.
Physical Evidence	Stakeholders interact continuously with a real-time ROI dashboard tracking the Inspection Reduction Rate, scrap savings, and throughput gains. Omni-QC also generates quarterly compliance certificates tailored for corporate ESG disclosures.

Figure 45. *Comprehensive 7P's Matrix*

Omni-QC's marketing mix is strategically engineered to eliminate adoption friction while securing high-margin, long-term enterprise partnerships. The Product centers on a cloud-primary AI defect-probability engine featuring a predictive Explainable AI (XAI) Co-Pilot, which is universally deployed via highly secure cloud infrastructures or through specialized, partner-provided edge hardware (Place). To overcome procurement hesitation, our Promotion strategy bypasses traditional sales pitches in favor of a 100% free, 90-day performance-based SaaS trial, fully subsidized by investor seed funding to validate ROI mathematically. Upon successful validation, the Price architecture locks in via a hybrid model tailored to Southeast Asian budgets.

Supporting this commercial structure is a highly specialized deployment framework. Our People consist of dedicated Technical Account Managers (TAMs) and onboarding engineers holding active IPC-A-610 certifications and MLOps credentials, ensuring deep domain expertise matches our AI capabilities. The integration process is designed for rapid time-to-value, featuring a Service Level Agreement (SLA) where Omni-QC absorbs all internal integration labor to accelerate the transition from the free trial into a mandatory 1-year contract lock-in. Finally, to continuously prove this value, the Physical Evidence of the platform is manifested in a real-time ROI dashboard tracking the Inspection Reduction Rate, scrap savings, and throughput gains, alongside quarterly compliance certificates formatted specifically for corporate ESG reporting.

7P MARKETING MIX STRATEGY: PRODUCT

Omni-QC's product architecture is unified at the algorithm level and differentiated at the deployment and service tier level. The core XGBoost risk-scoring engine and Co-Pilot Advisory Loop are identical across all SKUs, what changes is the deployment environment, data throughput capacity, and support response SLA.

Feature	Tier 1: Essentials (Cloud)	Tier 2: Professional (Cloud)	Tier 3: Enterprise Edge
Target Segment	Band 2 mid-tier EMS / Tier 2 manufacturers	Band 1 Tier 1–2 EMS, high-mix production	Tier 1 OEMs with data sovereignty / air-gap requirements
Deployment Model	Cloud-hosted SaaS (Omni-QC managed infrastructure)	Cloud-hosted SaaS + enhanced compute allocation	Partner-provided edge server; encrypted software container; air-gapped
Risk Scoring Engine	XGBoost defect probability score per unit	XGBoost + SHAP feature attribution per unit	Same as Professional, on-premise inference
Co-Pilot Advisory	Standard advisory notifications	Full prescriptive parameter correction recommendations	Full prescriptive + priority queue routing
Retraining Cadence	Demand-triggered (billed per event)	Demand-triggered + quarterly scheduled baseline review	Monthly calibration SLA; priority engineering support
Subscription Price	RM28,000 / line / year (Tier 2 lines)	RM54,000 / line / year (Tier 1 lines)	Custom enterprise contract; efficiency-share negotiated
SLA Response	P1: 15 min | P2: 1 hr | P3: 4 hrs	P1: 10 min | P2: 45 min | P3: 2 hrs	Dedicated TAM; on-site SLA by negotiation

Figure 46. *Feature and Tier Summary*

7P MARKETING MIX STRATEGY: PRICE

Strategy Classification: Value-Based Pricing with Behavioral Floors and Ceilings

Omni-QC employs Value-Based Pricing as its primary pricing philosophy, the price is anchored to the quantified economic value delivered to the customer rather than to internal cost or competitor price points. Two structural mechanisms operationalise this:

- Behavioral Floor: Annual-in-advance billing with a 1-year lock-in prevents churn before the compound ROI of the system is fully realised. This protects Omni-QC's working capital and ensures the customer does not exit before they have experienced full value.
- Behavioral Ceiling: The 10–20% efficiency-sharing component scales with verified savings. This aligns the pricing ceiling to the value ceiling, customers who extract maximum value from the system pay proportionately more, without a fixed cap that would leave economic value on the table.

Pricing Strategy Comparison

Pricing Strategy	Mechanism	Why Rejected for Omni-QC	Verdict
Cost-Plus	Price = COGS + fixed margin %	Ignores the economic value delivered (inspection reduction); underprices in high-savings environments.	Rejected
Competitor Parity	Mirror Cognex / KLA pricing tiers	Hardware incumbents price based on CapEx depreciation cycles. Not applicable to a cloud SaaS model.	Rejected
Penetration	Artificially low price to gain volume	Destroys gross margin runway in Year 1-2; contradicts the elite SaaS 64.5%-76.1% margin thesis.	Rejected
Skimming	Premium price, harvest early adopters	SEA mid-tier EMS is highly price-sensitive on upfront costs; skimming pricing creates the very procurement hesitation the trial model is designed to eliminate.	Rejected

Value-Based	Price anchored to verified customer savings (efficiency-share + baseline subscription)	N/A, This is the adopted model. Efficiency-share scales with value delivered; base subscription provides predictable ARR floor.	**ADOPTED**

Figure 47. *Price-Pricing Strategy Deep Dive*

Price Elasticity in the Southeast Asia EMS Market

Southeast Asian mid-tier EMS operators exhibit high sensitivity to upfront capital expenditure and low sensitivity to outcome-based fees. This demand curve maps directly to Omni-QC's hybrid model:

- The zero-CapEx SaaS delivery eliminates the primary procurement objection (hardware purchase cycle).
- The efficiency-share component is psychologically non-threatening at the point of procurement because it is contingent on savings that do not yet exist, the customer pays more only when they are already earning more.
- Annual-in-advance billing is palatable because it is positioned as insurance against the Inspection Tax rather than as a technology investment.

7P MARKETING MIX STRATEGY: PLACE

Omni-QC's channel architecture is structured around five distribution paths, each serving a distinct segment of the enterprise procurement funnel:

Channel	Description	Target Audience	Priority	Y1 Budget Est.
Direct Enterprise Sales (TAMs)	Named Account Managers target Band 1 EMS facilities; 90-day free trial as door-opener.	Quality Directors, Plant Managers - Tier 1 & 2 EMS	**PRIMARY**	RM80K
Partner Channel - Edge Hardware Vendors	other industrial hardware partners co-sell edge deployments; leverage existing sponsorship pipeline.	Tier 1 OEMs with data sovereignty requirements	**SECONDARY**	RM20K

System Integrator (SI) Channel	SIs who deploy MES solutions into EMS facilities (e.g., Rockwell Automation distributors, Siemens SEA integrators) bundle Omni-QC as a QC analytics layer.	EMS procurement committees, Factory Automation leads	**Y2 TARGET**	RM15K
Inbound Digital	Technical SEO, whitepaper gating, LinkedIn thought leadership. Generates qualified inbound from quality engineers.	Quality Engineers, R&D, Procurement Researchers	**SUPPORTING**	RM25K
Industry Events - SEMICON SEA, SMTA	Booth + live demo at premier SEA electronics events; brand positioning + direct lead capture.	Buyers, competitors, potential investors	**SUPPORTING**	RM30K

Figure 48. *Omni-channel distribution Matrix*

Partner Channel Detail: Edge Ecosystem

For environments requiring air-gapped operations, Omni-QC partners with specialized industrial hardware vendors who supply and maintain edge inference servers. The key operational principle is that Omni-QC remains a pure software company, it deploys an encrypted, hardware-agnostic container to partner-managed infrastructure. This keeps gross margins intact while extending the total addressable market to data-sovereign Tier 1 OEMs.

7P MARKETING MIX STRATEGY: PROMOTION

Channel	Type	Content / Activity	Audience	Y1 Budget
Whitepapers & Technical Blog	Online Inbound	Peer-reviewed PQ research; IPC pain-point case studies; XGBoost architecture explainers.	Quality Engineers, MLOps Leads	RM8K production
LinkedIn Paid Campaigns	Online Outbound	Targeted ads to Plant Managers, Quality Directors. A/B test	Procurement, C-Suite manufacturing	RM15K ads

		ROI-framing vs. risk-framing.		
SEMICON SEA / SMTA Expos	Offline	Booth + live Co-Pilot dashboard demo on representative PCB data.	Buyers, peers, potential investors	RM30K total
PR Agency - Regional Trade Press	Offline/Hybrid	Trade press bylines, analyst briefings (IDC Manufacturing Insights, Gartner OT).	Investors, enterprise buyers, regulators	RM20K retainer
Field Engineer Roadshows	Offline	On-site demos at 5 target Tier 1 EMS facilities in Penang & Johor corridors.	Engineering teams, floor managers	RM12K travel/ops
90-Day Free SaaS Trial Program	Outbound Core	Full production environment trial; ROI dashboard auto-generates conversion brief at Day 60.	Qualified prospects - Band 1 EMS	Subsidized by seed capital

Figure 49. *Promotion Channel Summary*

Testimonials & Reference Projects

While Omni-QC is at pre-production stage, analogous third-party case studies validate the core value proposition:

- A 2024 Siemens Opcenter deployment at a Penang-based EMS facility reported a 34% reduction in AOI false-positive re-inspection workload following the introduction of ML-assisted risk prioritization (Siemens AG, 2024).
- A LandingAI case study in automotive PCB manufacturing (Class 2) documented a 28% reduction in scrap rate over 6 months post-deployment of a predictive defect scoring layer (LandingAI, 2023).
- KLA Corporation's 2023 Annual Report explicitly cites customer demand for 'inspection overkill reduction' as a primary product development driver, independently validating the Inspection Tax pain point that Omni-QC is designed to resolve

7P MARKETING MIX STRATEGY: PEOPLE

People in the Omni-QC marketing context encompasses all human stakeholders involved in the go-to-market process, not only internal team members, but the full ecosystem of buyers, influencers, implementers, and oversight bodies who determine commercial success.

Stakeholder Group	What They Need from Omni-QC	Our People Assigned	Key Engagement Action
Existing Customers	Reliable service, continuous ROI proof, fast issue resolution.	TAMs + Customer Success Engineers	Monthly ROI dashboard review + QBR sessions.
Prospects / Potential Customers	Technical proof of concept, reference customer access, low adoption friction.	Sales Engineers + Field Engineers	90-day free trial; Day 60 ROI brief; reference customer introductions.
Investors	Financial transparency, ARR traction milestones, defensible moat evidence.	Founders (Darren, Raymond) + Financial Lead (Michelle)	Monthly investor update; Series A readiness deck by Month 18.
Strategic Partners (Hardware)	Integration support, co-marketing opportunities, clear contractual responsibilities.	BD Lead + Engineering (Darren + Vanessa)	Partner onboarding SLA; co-authored deployment case studies.
Regulators / Standards Bodies	Compliance evidence, transparent AI decision documentation, PDPA alignment.	Compliance Officer (Y2 hire) + David	Annual ESG certificate + PDPA DPA audit trail.
Operators / Floor Technicians	Easy-to-use Co-Pilot UI, trustworthy and interpretable alerts, low workload overhead.	UX Team + Co-Pilot Training (Vanessa + Elmira)	Structured onboarding sessions; SHAP explanation walkthrough at Day 3.

Figure 50. *Segmented stakeholder map*

7P MARKETING MIX STRATEGY: PROCESS

The marketing process is structured as a four-phase GTM playbook that moves from education and brand establishment through validation, expansion, and scale. Each phase builds the commercial infrastructure the next phase requires:

Phase	Timeline	Key Activities	Success Metric

Phase 1 **EDUCATE**	Months 1-6	Publish 3 technical whitepapers on PQ/SPI data science. Launch LinkedIn thought-leadership program. Submit 1 peer-reviewed conference paper (SMTA / ICMT). Attend SEMICON SEA.	Cited by ≥3 industry analysts; 500+ whitepaper downloads; 200+ qualified LinkedIn followers.
Phase 2 **VALIDATE**	Months 4-12	Execute 4-6 free 90-day pilot deployments. Co-author case studies with pilot clients (anonymised). Create an ROI calculator micro-site. Begin reference customer program.	4 reference customers with quantified ROI results; ≥1 co-authored case study published.
Phase 3 **EXPAND**	Months 9-18	Launch LinkedIn paid campaigns targeting Plant Managers. Field engineer roadshows (Penang, Johor, Klang Valley). Activate PR agency for trade press. Enable referral incentive for existing clients.	30+ qualified pipeline accounts; 8–12 active paid subscriptions.
Phase 4 **SCALE**	Months 15-24	Activate SI channel (Rockwell Automation distributor, regional Siemens integrator). Regional PR push to drive brand awareness. Launch partner co-marketing program with edge hardware vendors. Explore ASEAN expansion (Vietnam, Thailand).	Brand recognition in ≥50% of target Tier 1 accounts; ≥2 SI channel deals closed; MRR > RM100K.

Figure 51. *Go-to-market sequencing*

7P MARKETING MIX STRATEGY: PHYSICAL EVIDENCE

In a B2B SaaS context, Physical Evidence is the set of tangible, auditable artefacts that demonstrate product value and build institutional trust. For Omni-QC, three primary physical evidence streams are designed into the product

Evidence Type	Description	Business Function

Real-Time ROI Dashboard	Live tracking of Inspection Reduction Rate (%), scrap cost saved, throughput gains per production line. All metrics updated per-shift. Exportable to PDF for procurement committee review.	Converts abstract ROI promise into a hard number the CFO and quality director can verify at any time. Removes the primary post-sale trust gap.
Quarterly ESG Compliance Certificates	Auto-generated quarterly certificates documenting energy savings (reduced inspection machine runtime), labour hours recovered from false-positive re-inspection, and defect escape rate trends. Formatted for corporate ESG disclosure reporting.	Targets the ESG reporting mandate that Tier 1 EMS clients face from multinational OEMs. Creates a renewable, client-specific justification for annual contract renewal.
Co-Pilot Advisory Audit Log	Timestamped log of every Co-Pilot advisory output, the machine parameter cited, the corrective recommendation made, and the outcome (defect averted or false alarm). SHAP attribution scores included per advisory.	Addresses transparency and regulatory compliance concerns. Provides the audit trail required for Class 3 environments and for efficiency-share verification disputes. Converts the AI from a black box into an accountable system.
Pilot Case Studies	Post-90-day POC reports co-authored with pilot clients documenting baseline vs. post-deployment KPIs. Anonymised versions published as public reference materials.	Primary sales collateral for prospect conversion. Shifts the procurement conversation from 'will it work?' to 'how much are we leaving on the table by not implementing it?'

Figure 52. *Physical Evidence Streams Summary*

PARTNER-PROVIDED EDGE ECOSYSTEM

While Omni-QC's primary focus and core business model revolves around its cloud-hosted risk scoring engine, the company explicitly acknowledges the stringent data sovereignty demands of certain Tier 1 OEMs and high-security EMS providers.

To address instances where corporate IT firewalls mandate entirely air-gapped operations, Omni-QC is fully compatible with localized edge inference across all supported PCB classes. However, supplying, leasing, and maintaining physical hardware is deliberately excluded from Omni-QC’s primary business scope. Instead, edge deployments are facilitated through established partnerships with specialized industrial hardware vendors. The client contracts the edge server infrastructure through these third-party providers, while Omni-QC securely deploys its encrypted software container to the locally managed device. This strategic boundary ensures that Omni-QC remains an agile software enterprise, preserving peak gross margins without absorbing the capital expenditure, depreciation, or physical maintenance liabilities associated with hardware fleets.

6.3 After-Sales Service System

In an era where software buyers face immense implementation cost overruns, the Omni-QC after-sales framework is engineered to deliver an ironclad "Zero-Risk Adoption" narrative. The core operational principle is the establishment of a clean, highly auditable boundary between services unconditionally included within the baseline subscription and operational events that trigger specific, billable service engagements.

SLA COVERAGE

Deliberately absorbing bug fixes, continuous software updates, and comprehensive system installation costs into the core subscription neutralizes the procurement committee's primary fear of unpredictable post-sale operating expenses. All corrective software modifications, security patches, and user interface upgrades are provided without additional charge to ensure continuous platform stability and prevent technological obsolescence. Automated remote performance diagnostics continuously track baseline prediction accuracy, issuing proactive alerts to Omni-QC engineers if statistical degradation occurs. For edge deployments, physical hardware maintenance and server swap-outs are managed entirely by our third-party hardware partners, isolating Omni-QC from hardware liabilities. Furthermore, Omni-QC retains full internal responsibility for the financial risk of software integration, guaranteeing operational activation within 14 days. The only billable after-sales event is the on-demand model retraining cycle, which is an intensive data-engineering service triggered by process drift or new product introductions and tiered by validation complexity (Pechmann et al., 2025).

Service Category	All Tiers	Operational Context & Justification
Bug Fixes & Security Patches	**INCLUDED**	All corrective software modifications and critical security patches are provided without additional charge to ensure continuous platform stability.
Feature updates & UI enhancements	**INCLUDED**	Continuous deployment of dashboard enhancements, visualization modules, and risk-threshold tuning interfaces ensures the platform avoids technological obsolescence.

Remote Performance Diagnostics	**INCLUDED**	Automated continuous monitoring tracks baseline prediction accuracy, issuing proactive alerts to Omni-QC engineers if model performance exhibits statistical degradation.
Hardware Maintenance (If Edge Deployed)	**PARTNER RESPONSIBILITY**	Physical maintenance, warranty, and server swap-outs for edge deployments are managed entirely by the third-party hardware provider, isolating Omni-QC from hardware OPEX.
Integration Support	**INCLUDED**	Omni-QC retains full internal responsibility for the financial risk of software integration, guaranteeing operational activation within 14 days.
Model Retraining Cycles	**On-Demand Billable Event**	Retraining is an intensive data-engineering service triggered by process drift, new product introductions, or material shifts. Prices are tiered based on validation complexity.

Figure 53. *SLA Coverage Matrix*

SLA PERFORMANCE GUARANTEES

Service Level Agreements (SLAs) are financially backed performance guarantees designed to protect manufacturing continuity and enforce strict vendor accountability. In Southeast Asian manufacturing environments, where unplanned downtime is increasingly costly, having clear, enforceable SLA remedies is critical to SaaS compliance and risk management (Finberg Firm, 2026). The platform guarantees a 99.5% monthly uptime, with any breaches resulting in a 5% to 15% credit of monthly fees. Incident response is structured across three priority tiers, ensuring rapid mitigation of any operational disruption. Critical P1 events, such as a complete system outage, mandate a 15-minute response time and resolution within 24 hours, backed by pro-rata credits and immediate executive escalation. High-priority P2 disruptions require a one-hour response and resolution within 48 hours, while standard P3 technical issues are addressed within four business hours. This strict framework guarantees that Omni-QC remains financially accountable for its operational performance at all times.

Priority Level	Example Scenario	Response Target	Resolution Target	Financial Remedy
System Uptime	Platform availability	N/A	99.5% monthly	5% to 15% of monthly fees credited.

P1 (Critical)	Complete system outage	15 minutes	< 24 hours	Pro-rata credits; executive escalation.
P2 (High)	Department-wide disruption	1 hour	24 to 48 hours	Pro-rata service credits.
P3 (Normal)	Single-user software issue	4 business hours	3 to 5 days	Standard technical support queue.

Figure 54. *Priority Based-SLA Performance Guarantees*

6.4 Sales Promotion Strategy

INBOUND ENGINE: TECHNICAL AUTHORITY AND CONTENT MARKETING

With median CAC doubling across the B2B SaaS industry (T2D3, 2026), reliance on broad, unfocused marketing expenditure is a mathematical failure. Omni-QC executes a highly targeted, two-pronged promotion strategy that fuses deep technical authority with irrefutable, localized empirical evidence.

Content Strategy	Execution	Target Audience	Market Impact
Technical Whitepapers	Analytical papers detailing the financial mechanics of the "Inspection Tax." Disseminated via IPC and SMTA.	Quality Assurance Directors, Process Engineers	Establishes absolute technical authority and positions Omni-QC as an industry standard.
Benchmark Reports	Annual publications comparing industry-average false-positive rates and scrap costs against Omni-QC-deployed facilities.	VP of Operations, Plant Managers	Creates a quantifiable fear of missing out (FOMO) by exposing competitor efficiency advantages.
ESG Compliance Mapping	Content linking material-scrap reduction directly to measurable CO2 emission drops	Corporate Compliance Officers, CFOs	Provides a powerful secondary purchase motivation ahead of stringent 2026 ESG mandates.

Figure 55. *Content Marketing Strategies*

CONFERENCE AND TRADE SHOW PRESENCE

The primary channel for early client acquisition is direct engagement at industry conferences and trade exhibitions serving the electronics manufacturing and EMS sectors in Southeast Asia. This approach allows Omni-QC to engage qualified buyers in concentrated numbers within an environment focused strictly on production efficiency and AI adoption.

Strategy Component	Execution Details
Primary Acquisition Channel	Direct engagement at industry conferences and trade exhibitions serving the electronics manufacturing and EMS sectors in Southeast Asia.
Strategic Rationale	Buying decisions for enterprise process intelligence systems are rarely made through passive inbound discovery. Forums provide concentrated access to technical and commercial leadership in a receptive environment focused on production efficiency and AI adoption.
Key Target Venues	Productronica Asia, SEMICON Southeast Asia, and the Malaysia International Electronics Manufacturing Technology Exhibition.
Target Audience	Procurement leadership, quality directors, and operations managers from both Tier 1 EMS providers and Tier 2 contract manufacturers.
Engagement Method	Live demonstrations of the dashboard interface and risk scoring logic using representative production data scenarios.
Observable Outputs	Inspection reduction rates and false negative rates are presented as tangible, mathematical outputs of the scoring logic, proving ROI directly on the show floor.

Figure 56. *Conference and Trade Show Presence Strategy*

OUTBOND ENGINE: PERFORMANCE-BASED TRIAL PROGRAMS

The primary direct sales motion is a structured, 90-day free trial program designed to convert procurement skepticism into an evidence-based subscription commitment. The trial is governed by the following parameters.

Trial Parameter	Specification
Duration	90 days (approximately 1 production quarter). Sufficient to capture at least 2 complete NPI cycles and generate statistically significant defect-prediction data.
Trial Fee	RM0 (Completely Free). Upfront integration and hardware costs are fully subsidized by our investor seed funding, removing all financial barriers for the client.

Baseline Audit	Omni-QC conducts a pre-deployment audit of the client's current AOI false-positive rate, manual re-inspection headcount, and material scrap cost per quarter. This baseline is the reference document against which ROI is measured at trial completion.
Success Metrics	Minimum 30% reduction in manual re-inspection workload OR minimum 15% reduction in panel-level scrap rate within the POC period.
Conversion Trigger	Upon meeting either success metric, the TAM presents a 12-month subscription proposal with pre-calculated ROI projection based on live trial data. The evidence-first structure converts the subscription decision into a numerical inevitability.
Conversion Rate Benchmarks	To maintain sustainable unit economics and protect the RM3M runway, Omni-QC targets a strict 45% POC-to-License conversion rate baseline. A 30% conservative rate represents a severe liquidity threat, while a 60% optimistic rate drives rapid scalability.

Figure 57. *Performance-based Trial Program Parameters*

6.5 Profit Model

REVENUE STREAM 1: TIERED ANNUAL SAAS SUBSCRIPTION

The annual subscription fee is the primary and recurring revenue stream of Omni-QC. Following deployment, clients pay a flat annual fee to maintain access to the system's risk scoring engine, classification logic, user-configurable threshold interface, and Co-Pilot advisory dashboard. The subscription is priced on a per-line basis.

Across both tiers, the platform is deployment-agnostic, capable of operating via Cloud or Edge. However, Omni-QC's primary focus and deployment recommendation is the Cloud architecture. This allows for rapid deployment, infinite scalability, and seamless updates, making it the default accessible entry point for clients whose budget structures favor operational expenditure over capital commitment. For facilities with uncompromising data sovereignty or intellectual property mandates requiring air-gapped, on-premise computation, Edge deployment remains fully supported. Crucially, providing and leasing edge hardware is not within Omni-QC's primary business scope. Instead, clients procure the necessary industrial-grade servers through our certified third-party hardware partners, ensuring our internal margins remain completely insulated from hardware expenditures.

Regardless of the deployment environment, the annual subscription price is flat across all years of the engagement. Onboarding, which encompasses the full scope of work required to bring the system to an operational state, is absorbed into the subscription as an internal cost of Omni-QC rather than charged as a separate upfront fee. Omni-QC assumes the integration risk internally, with the commercial logic being that a client who reaches a fully operational deployment within a defined onboarding period represents a significantly stronger long-term subscription relationship than one who encounters unexpected integration invoices early on. Furthermore, the software is protected by strict digital rights management (DRM); if a subscription lapses, the system ceases to generate risk scores, tying the value of Omni-QC directly to an active commercial relationship.

To accommodate the budget structures of Southeast Asian manufacturing enterprises while reflecting the heavy COGS of AI inference, the base subscription is priced strategically:

- Tier 1 (High-Volume EMS) is priced at RM54,000 per line annually, serving high-volume consumer IoT lines.
- Tier 2 (Mid-Tier CMs & ODMs) is priced at RM28,000 annually per line, targeting high-mix industrial providers. These price points sit perfectly within the 2026 MES software procurement benchmarks for the Malaysian and Vietnamese markets (Kladana, 2026).

REVENUE STREAM 2: ON-DEMAND MODEL RETRAINING FEE

Model retraining is a functional necessity, not an optional enhancement. As machine wear, solder formulations, and New Product Introductions (NPIs) alter process conditions over time, the predictive accuracy of any calibrated model degrades, a phenomenon termed model drift (Pechmann et al., 2025). Retraining is billed per engagement at tiered rates reflecting the scale and complexity of the recalibration exercise:

- Tier 1 Retraining (RM9,000 per cycle): High-volume data ingestion, multi-line model parallelisation, and baseline realignment across consistent mass-production runs. Typical frequency: 1-2 cycles per year. Note: The higher price reflects the scale of data volumes processed and the engineering overhead of validating across parallel identical lines, not necessarily per-cycle algorithmic complexity.
- Tier 2 Retraining (RM5,000 per cycle): Automated MLOps few-shot learning pipeline for rapid domain adaptation to new industrial board designs during NPIs. Typical frequency: 2–3 cycles per year. The lower price reflects the degree of automation in the Tier 2 retraining pipeline, which leverages pre-built few-shot templates to reduce labour overhead.

REVENUE STREAM 3: EFFICIENCY-SHARE BONUS

To counteract the deflationary effects of AI-driven efficiency gains and to align vendor incentives with client operational improvement, Omni-QC mandates a non-negotiable efficiency-share mechanism on verified savings:

- Tier 1 Efficiency-Share (10% of verified savings): Claimed on audited reduction of raw material waste. The 10% rate reflects the stronger negotiating leverage of high-volume EMS clients and the higher absolute dollar value of their efficiency gains at scale.

- Tier 2 Efficiency-Share (20% of verified savings): Claimed primarily on reduction of manual inspection labour headcount. The higher 20% rate is justified by the smaller absolute baseline savings at mid-tier clients and by the proportionally greater operational improvement Omni-QC delivers within their cost structure.

7.0 Financial Analysis

This chapter presents the complete five-year financial model underpinning Omni-QC's commercial viability. It begins with the financing plan, detailing the RM3,000,000 seed capital requirement, its phased allocation across model development, POC deployment, and commercial scaling, and the specific funding sources and timing designed to bridge the 180-day industrial revenue gap inherent in the deployment model. Financial assumptions are then documented in full, providing the transparent basis on which all projections are built. The chapter proceeds through a bottom-up cost and expense forecast, a multi-scenario revenue forecast across subscription, retraining, and efficiency-sharing streams, and a pro forma income statement that traces the path from Year 1 losses to Year 3 EBITDA profitability. Cash flow projections, financial ratio analysis benchmarked against SaaS industry standards, and a comprehensive financial summary that covers break-even analysis, sensitivity testing, and terminal valuation complete the chapter.

7.1 Financing Plan

FINANCING OVERVIEW

The AI-Driven Dynamic Defect Prediction system project is seeking a total of RM3,000,000 in funding over a 24-month period to reach commercial scalability.

The capital or estimated funding requirement is divided into three major phases:

- Phase 1: Model Development & Simulation
- Phase 2: POC Deployment & Validation
- Phase 3: Commercial Launch & Scaling

The operational shift begins in Year 1, Month 7, as the company transitions from internal R&D to live pilot deployments. While the partnership model eliminates hardware CapEx, the RM1.3 million allocated for Phase 2 is necessary to handle the heavy software integration and IPC compliance requirements. This front-loaded investment ensures the system meets the safety and technical standards of Tier-1 manufacturers, allowing for the successful launch of the first four pilots.

The RM3 million financing plan is designed to bridge the significant cash burn during the first 24 months. Each pilot includes a 90-day free validation period followed by a 90-day payment lag, meaning the company must fund six months of operations before collecting revenue. This phased capital provides the liquidity needed to survive this revenue gap. By Year 3, the established infrastructure supports a rapid scale-up to high-margin recurring SaaS revenue.

Phase	Duration	Key Milestone	Capital
Phase 1: Model Development & Simulation	Months 1 - 6	Model hardening, synthetic data generation, and patent filings	RM750,000
Phase 2: POC Deployment & Validation	Months 7 - 15	API development, POC deployment, and regulatory compliance	RM1,300,000
Phase 3: Commercial Launch & Scaling	Months 16 - 24	Sales team expansion, license roll-out, and success framework	RM950,000
Total 24-month funding need: **RM3,000,000**			

Figure 58. *Financing Phase Timeline*

FINANCING SOURCE AND TIMING

The financing mix is engineered to protect liquidity during the 180-day revenue gap inherent in industrial AI deployments. While the partnership model eliminates hardware ownership costs, the business incurs significant front-loaded expenses for API integration (RM450k) and compliance (RM350k) during Phase 2. Equity and Angel capital fund the initial R&D in Phase 1, establishing the technical foundation. As the burn rate spikes in Phase 2 due to live pilot deployments, the plan utilizes non-dilutive grants and customer deposits to fund validation without excessive equity dilution. This ensures the business remains asset-light while securing the technical legitimacy required for Tier-1 manufacturing environments.

The timing of these funds specifically addresses the 90-day free validation and the subsequent 90-day payment lag, which creates a six-month window where the company must fund operations without cash inflow. The RM300k Contingency/Debt acts as a critical liquidity bridge during this period, preventing technical or integration delays from exhausting the primary runway. By utilizing a SAFE note for the final RM1M, the company secures the capital necessary for Phase 3 scaling only after the pilots have proven the value proposition. This structure aligns the most expensive capital (VC) with the lowest risk period, maximizing valuation and ensuring a stable path to the high-margin recurring revenue projected in Year 3.

Source	Type of Security	Amount (RM)	% of Total	Stage
Founder/Seed Equity	Common Stock	RM400,000	13.3%	Phase 1
Angel Investors (Manufacturing Tech Focus)	Preferred Stock	RM600,000	20%	Phase 1 - 2
Strategic Grant (Industry 4.0/ AI Manufacturing)	Not a security (non-dilutive grant)	RM400,000	13.3%	Phase 2
Customer Pre-Payment (First POC Client)	Not a security (customer deposit)	RM300,000	10%	Phase 2
Contingency/Debt	Working Capital	RM300,000	10%	Phase 2
Venture Capital - Pre-Seed	SAFE	RM1,000,000	33.4%	Phase 2-3
Total		**RM3,000,000**	**100%**	

Figure 59. *Financing Sources Plan*

RESOURCE ALLOCATIONS

The capital allocation is strategically backloaded to Phase 2 to overcome the high entry barriers of the industrial sector. Phase 1 focuses on building the technical foundation through model hardening and IP protection. The RM1.3 million Phase 2 spike addresses the problem by funding RM350,000 for IPC compliance and RM450,000 for API integration. This front-loaded capital bridges the six-month revenue gap caused by free validations and payment lags, ensuring operational survival while securing technical legitimacy.

Phase 3 utilizes RM950,000 to transition from technical validation to aggressive market capture. Capital shifts toward a RM500,000 sales and marketing expansion to scale from four initial pilots to 32 active lines by Year 2. By leveraging the established integration moat, the business reaches a Year 3 inflection point where recurring SaaS revenue achieves operational self-sufficiency.

Category	Phase 1 (Dev)	Phase 2 (POC)	Phase 3 (Scale)	Amount (RM)	% of Total
R&D & Model Engineering	RM350k	RM250k	RM150k	RM750k	25.0%
API & Integration Development	RM50k	RM400k	RM0	RM450k	15.0%
Regulatory & IPC Compliance	RM0	RM350k	RM0	RM350k	11.7%

Sales, Marketing & Bus Dev	RM50k	RM125k	RM500k	RM675k	22.5%
G&A, Legal & IP Protection	RM275k	RM0	RM100k	RM375k	12.5%
Operations & On-site Support	RM0	RM100k	RM150k	RM250k	8.3%
Contingency (10%)	RM25k	RM75k	RM50k	RM150k	5.0%
Total	**RM750k**	**RM1,300k**	**RM950k**	**RM3,000k**	**100%**

Figure 60. *Resource Allocation Plan*

TARGET INVESTOR PROFILE AND NAMED LIST

Tier	Investor Type	Example Names/Categories	Thesis Fit for Omni-QC
1	Manufacturing-tech angels	Ex-Flex, Celestica, or Benchmark quality directors; Singapore/Malaysia angel networks	Write 50k–200k, provide plant floor credibility
2	SEA early-stage VCs	Tin Men Capital (B2B industrial SaaS), Integra Partners (deep tech), True Global Ventures (Industry 4.0), Hustle Fund (hardware-adjacent)	Lead or co-lead 500k–1.5M
3	Strategic corporate VCs	Teradyne, Koh Young, Omron, Advantest	Pilot -> investment, balance sheet partnerships

Figure 61. *Target Investor Profile and Named List*

OUTREACH TIMELINE (MONTHS 1-6)

Month	Activity	Target	Deliverable
1	Warm intros via existing network (advisor with EMS contacts)	15 angels with electronics manufacturing experience	Meeting log
2	Pitch deck v1 to tier 1 angels	3 soft commitments	RM150k indicated
3	Secure LOI from 1 hardware partner (edge server vendor) + pilot data from 1 PCB assembler	Update deck with real metrics	Deck v2
4	Approach tier 2 VCs via angel intros	5 funds with SEA industrial SaaS thesis	2 term sheet discussions

5	First close (RM500k) per 6.1.2 financing plan	SAFE at RM8M cap	Signed docs
6	Strategic CVC conversations post-pilot	3 corporate VCs	LOI for Series A

Figure 62. *Outreach Timeline*

GRANT & INDUSTRY4WRD INTERVENTION FUND TIMELINE

Month	Activity	Fund Source	Deliverable
Pre-Month 1	Identify Malaysian PCB assembler with completed or ready-for-Industry4WRD RA; sign partnership LOI	Both	Signed manufacturing partner LOI
1	Identify target grants, match Omni-QC's PCB quality control scope to funder priorities	Strategic Grant	Shortlist of 3-5 eligible grants
	Review Industry4WRD criteria, SME manufacturer partner required, 30% matching contribution	Industry4WRD	Confirm eligibility + identify manufacturing partner
2	Draft grant proposal, technical architecture (XGBoost risk scoring), pilot plan, budget	Strategic Grant	First draft completed
	Secure vendor registration (Omni-QC as approved solution provider); finalize co-applicant status	Industry4WRD	Vendor registration + Signed LOI from manufacturing partner
3	Submit Strategic Grant application + all supporting docs	Strategic Grant	Submission confirmation
	Prepare Industry4WRD application, digitalization scope, vendor registration (Omni-QC as solution provider)	Industry4WRD	Application ready
4	Grant review period, respond to funder queries, refine budget if requested	Strategic Grant	Query responses
	Submit Industry4WRD application via matching portal	Industry4WRD	Submission confirmation
5	Grant decision, if approved, funds disbursed within 60 days	Strategic Grant	Award letter + fund release schedule

	Industry4WRD approval, 30% matching paid by manufacturer, 70% disbursed to Omni-QC	Industry4WRD	Approval letter + first drawdown
6	Deploy grant funds to accelerate roadmap: edge server integration + second pilot site	Both	Pilot expansion milestone

Figure 63. *Grant & Industry4WRD Intervention Fund Timeline*

PARTNERSHIP PROCESS

System integrators already sell MES, AOI machines, and line automation to PCB assemblers in Malaysia, Vietnam, and Thailand. They become Omni-QC's distribution channel.

Step	Action
1	Map top 5 SIs serving PCB assemblers in Penang, HCMC, Bangkok
2	Offer: Co-sell at 20% first-year revenue share
3	Train their sales team (half day, technical + commercial)
4	First joint deal (PCB assembler with 3+ SMT lines) -> case study
5	Exclusivity per territory if they deliver 3+ deals/year

Figure 64. *Channel Partnership Action Plan*

Step	Action
1	Identify target: EMS or CM with 5+ SMT lines, existing AOI/X-ray
2	Free 2-week diagnostic - run XGBoost on their historical defect logs
3	Paid 60-day pilot: risk scores on live production vs. actual outcomes
4	Convert to annual SaaS subscription (28k–54k per line)
5	Secure reference call + site visit for next prospect

Figure 65. *Pilot Customers Action Plan*

7.2 Financial Assumptions

ASSUMPTION CATEGORY	BASE CASE VALUE	RATIONALE/RISK

POC Launch Timing	7th Month	Month 7 timing allows six months for model hardening and synthetic training, ensuring stability before deployment on major production lines.
Conservative Conversion Rate	30%	Severe liquidity threat. Unrecovered setup costs transform Zero-CAPEX into a liability, rapidly depleting the RM3M runway and requiring urgent intervention.
Optimistic Conversion Rate	60%	High revenue growth risks resource overextension. Rapid scaling requires intensive MLOps support, potentially increasing burn rates despite improved top-line velocity.
POC-to-Licence Conversion Rate	45%	Critical baseline for break-even. Minor downward variance stalls setup recovery, inflating net losses and delaying the project's transition to profitability.
Annual Churn Rate	12%	Industrial SaaS churn is lower than generic software. However, the model includes retraining fees (RM9k/RM5k per cycle). Clients who refuse retraining will experience model decay and churn. 12% balances lock-in (DRM, air-gapped dependency) against price sensitivity of SEA mid-tier firms.
Client Mix (Tier 1/Tier2)	40%/60%	This 40/60 mix prioritizes specialized mid-tier growth while maintaining OpEx-based pricing to bypass lengthy corporate board-level CapEx procurement approval cycles.
Average Subscription Value	RM38,400/year	Weighted average of subscription revenue from both tiers. Formula: (0.4 × 54,000) + (0.6 × 28,000) = RM38,400.
Expected Annual Retraining Service Value	RM6,930/lines	Combines weighted tier pricing with 1.05 expected annual sessions to address industrial model drift and maintain predictive precision across portfolios.
Expected Annual Efficiency-sharing Value	RM6,720/lines	Leverages performance-based gains from verified scrap reduction, adjusted by a 35%

		realization gap to ensure conservative and defensible revenue projections.

Figure 66. *Financial Assumptions Analysis*

STRATEGIC POC DEPLOYMENT AND YEAR 1 REVENUE APPORTIONMENT

Year 1 revenue projections incorporate a six-month development phase followed by four Proof of Concept (POC) initiations in Month 7. Because the program entails a 90-day validation cycle, commercial conversion only triggers in Month 10. This timeline restricts recognized revenue to the final three months of the fiscal year. Consequently, Year 1 reflects 1/4 of the total annual subscription value, aligning the accrual model with the reality of industrial deployment cycles.

MARKET-DRIVEN CONVERSION RATES

In the commoditized ASEAN PCB sector, manufacturers act as price-takers with minimal control over global market rates. Consequently, profit optimization relies almost exclusively on internal cost reduction rather than price increases. This economic reality transforms Omni-QC's value proposition into a survival necessity rather than a discretionary upgrade. By targeting the factory's primary financial lever, such as yield loss and scrap, a premium (5%-10%) conversion assumptions are made because of the direct alignment to specific variables in interest of narrow-margined manufacturers.

Omni-QC offers cost-saving metrics that are easier to verify than the subjective productivity gains of generic enterprise SaaS. This transparency eliminates the valuation gap, allowing technical leads to prove exact ROI against real-time defect data before the license is signed. Furthermore, the Zero-CAPEX model directly bypasses procurement friction. By absorbing setup costs, Omni-QC permits factory managers to authorize deployment via operational budgets, circumventing the restrictive, board-level capital approvals required by traditional hardware-heavy competitors.

RETRAINING SERVICE VALUE

The derivation merges tiered pricing with unbiased operational filters. A weighted average fee of RM6,600 is calculated from the 40/60 market split between Tier 1 and Tier 2 clients. This is multiplied by 1.05 expected annual sessions, representing a 1.5-cycle retraining frequency discounted by a 70% participation rate. This accounts for technical model drift while recognizing that legacy production lines require less frequent interventions than high-mix environments.

The efficiency yield applies a 35% realization rate to a RM120,000 annual scrap-reduction baseline, identifying RM42,000 in verifiable capturable value. Revenue stems from weighted participation tiers: Tier 1 clients (40% weight) contribute RM1,680 through a 10% performance share, while Tier 2 clients (60% weight) generate RM5,040 at a 20% share, grounding projections by accounting for attribution disputes and measurement variances.

EFFICIENCY YIELD VALIDATION AND CONSERVATIVE PERFORMANCE SCALING

The derived RM6,720 yield serves as a conservative success fee that scales alongside the subscription core. By discounting 65% of theoretical savings as a realization gap, the model protects against skeptical procurement audits and operational slippage. This dual-stream revenue structure ensures that Omni-QC remains profitable even during periods of lower POC-to-paid conversion, as expansion revenue from the existing installed base provides a high-margin floor.

7.3 Cost and Expenses Forecast

The financial projections for Omni-QC follow a bottom-up modeling approach, anchoring personnel and cloud infrastructure costs to ASEAN tech benchmarks. The model assumes a Minimum Viable Team (MVT) that scales alongside production line acquisitions. By maintaining a lean operational core during the initial twenty-four months, the project ensures the RM3 million seed funding provides sufficient runway to reach the Year 3 operational break-even point.

COST OF GOODS SOLD FORECAST

COGS captures the variable inputs required to operationalize individual production lines, specifically focusing on the hardware-labor bundle and dedicated cloud monitoring. This category represents the direct cost of service delivery, allowing for a clear calculation of gross margins as the production line fleet expands across the regional manufacturing corridor. Omni-QC absorbs the RM6,500 setup cost per POC to eliminate entry barriers for manufacturers. While this creates a front-heavy COGS structure in Year 1, subsequent subscription renewals eliminate this non-recurring expense entirely, consequently increasing gross profit by RM6,500 for each renewal.

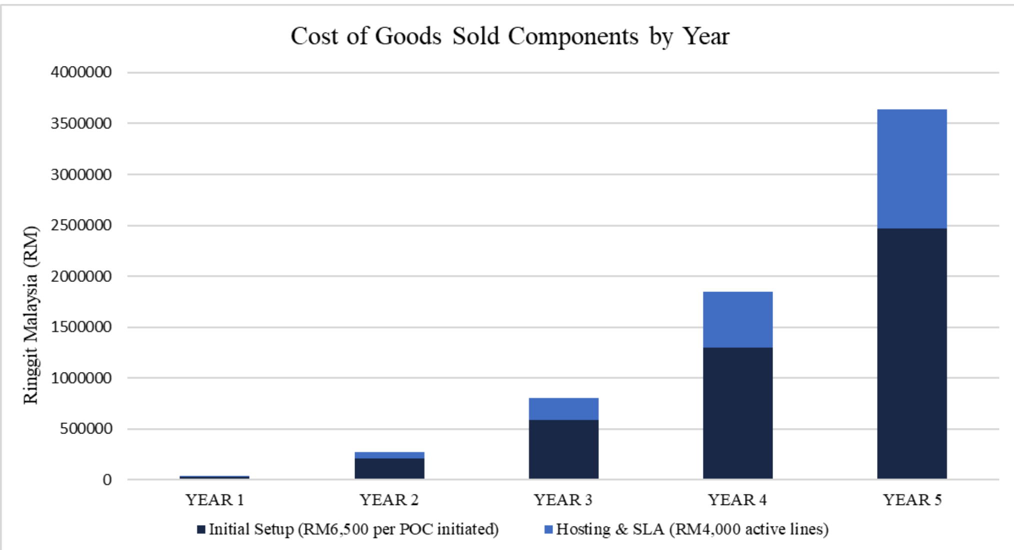


Figure 67. *Cost of Goods Sold Forecast Chart*

EXPENSES CATEGORY	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
POC Initiated	4	32	90	200	380
Initial Setup (RM6,500 per POC initiated)	RM26,000	RM208,000	RM585,000	RM1,300,000	RM2,470,000
Active Lines (Base Case)	1.80	15.98	54.57	138.02	292.46
Hosting & SLA (RM4,000 active lines)	RM7,200	RM63,920	RM218,280	RM552,080	RM1,169,840
COST OF GOODS SOLD (Base Case)	**RM33,200**	**RM271,920**	**RM803,280**	**RM1,852,080**	**RM3,639,840**
COST OF GOODS SOLD (Conservative)	**RM30,800**	**RM250,640**	**RM730,520**	**RM1,668,040**	**RM3,249,880**
COST OF GOODS SOLD (Optimistic)	**RM35,600**	**RM293,240**	**RM876,040**	**RM2,036,120**	**RM4,029,840**

Figure 68. *Cost of Goods Sold Forecast Table*

OPERATING EXPENSES FORECAST

The operating expense structure prioritizes technical dominance and human capital to secure the platform's value proposition. Early-stage allocations are heavily weighted toward R&D and MLOps, accounting for over 50% of Year 1 expenditures to ensure the high model accuracy required for performance-based revenue sharing. As the engineering team scales to a RM1.9 million budget by Year 5, this sustained investment facilitates continuous product optimization and the maintenance of proprietary AI networking infrastructure.

Financial scaling is marked by an aggressive pivot toward market capture, evidenced by Sales and Marketing expenses doubling between Year 3 and Year 4. This expansion supports the acquisition of high-volume Tier 1 MNC targets through technical validation tools and field engineering support. Meanwhile, administrative overhead remains below 20% of total OpEx, maintaining a high efficiency ratio that maximizes the deployment of capital into growth-centric functions while ensuring COGS remain synchronized with the pipeline.

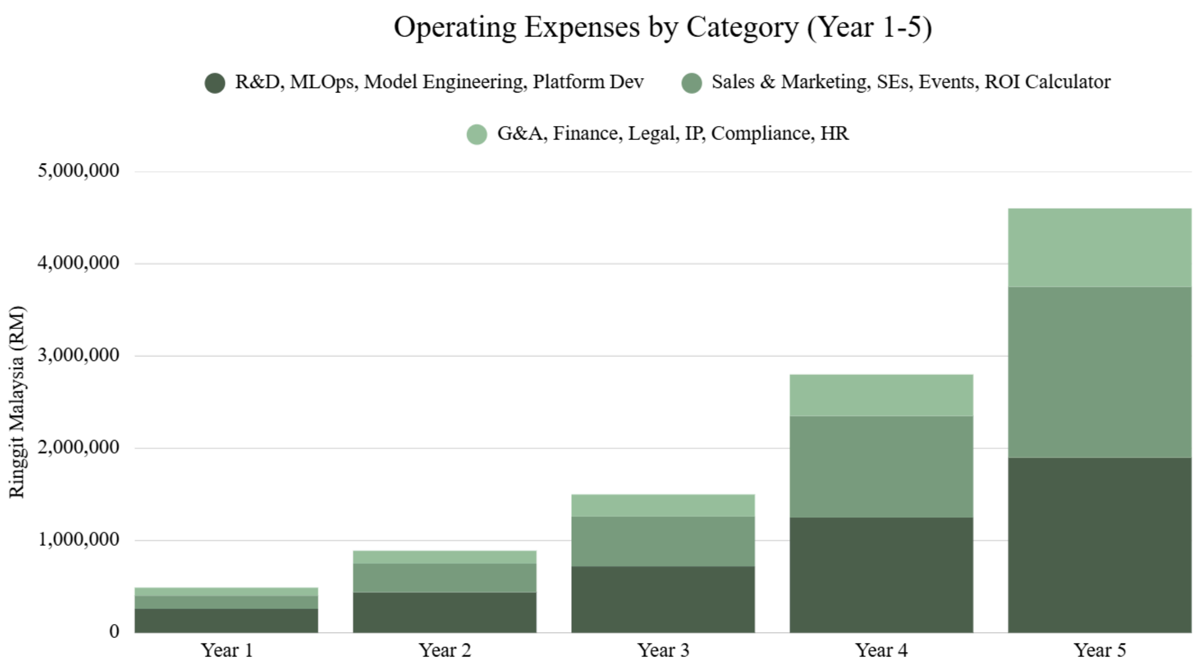


Figure 69. *Operating Expenses Forecast Chart*

EXPENSES CATEGORY	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
R&D, MLOps, Model Engineering, Platform Dev	RM260,000	RM440,000	RM720,000	RM1,250,000	RM1,900,000
Sales & Marketing, SEs, Events, ROI Calculator	RM145,000	RM310,000	RM540,000	RM1,100,000	RM1,850,000
G&A, Finance, Legal, IP, Compliance, HR	RM85,000	RM140,000	RM240,000	RM450,000	RM850,000
OPERATING EXPENSES	**RM490,000**	**RM890,000**	**RM1,500,000**	**RM2,800,000**	**RM4,600,000**
COST OF GOODS SOLD	**RM33,200**	**RM271,920**	**RM803,240**	**RM1,852,040**	**RM3,639,840**
TOTAL COST & EXPENSE	**RM523,200**	**RM1,161,920**	**RM2,303,240**	**RM4,652,040**	**RM8,239,840**

Figure 70. *Operating Expenses Forecast*

CAPITAL EXPENDITURE FORECAST

Omni-QC's lean CapEx model prioritizes technological defensibility while maintaining a Zero-CapEx value proposition for customers. By expensing deployment hardware through COGS and utilizing cloud-based infrastructure as OpEx, the firm avoids the heavy asset burdens of traditional industrial vendors. Internal capital is strictly reserved for GPU simulation rigs, engineering tools, and patent filings, ensuring that every dollar of investment fuels core R&D and secures the proprietary innovation engine.

EXPENSES CATEGORY	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
DevGPU R&D Simulation Rigs	RM45,000	RM15,000	RM60,000	RM25,000	RM80,000
Engineering Hardware/Laptops	RM18,000	RM12,000	RM24,000	RM35,000	RM52,000
Patent & Trademark Filings	RM25,000	RM10,000	RM35,000	RM15,000	RM45,000
TOTAL CAPEX	**RM88,000**	**RM37,000**	**RM119,000**	**RM75,000**	**RM177,000**

Figure 71. *Capital Expenditure Forecast*

PRICING MECHANICS

The following table breaks down the complete revenue architecture by the two target firmographic tiers, detailing the mechanisms, unit prices, and coverage scope:

Tier	Revenue Component	Features & Coverage	Unit Price	Mechanism
Tier 1 (High-Volume EMS Providers)	**Core SaaS Subscription**	API access, three-tier risk classification, Co-Pilot advisory dashboard. Infinite scalability for high-volume runs.	RM54,000 per line	Cloud-hosted risk scoring deployed across highly consistent mass-production lines.
	Model Retraining Service	High-volume data ingestion, multi-line parallelisation and visual baseline realignment.	RM9,000 per cycle	On-demand recalibration to combat minor mechanical sensor drift.

	Efficiency-Share Bonus	Claimed strictly on the audited reduction of raw material waste. Scales proportionally with the client's material yield volume.	10% of verified savings	Captures the financial upside of AI automation to prevent software commoditization.
Tier 2 (Mid-Tier CMs & ODMs)	**Core SaaS Subscription**	Advanced telemetry ingestion, false-positive filtering, scalable software logic for high-mix environments.	RM28,000 per line annually	Cloud-hosted software license targeting high-mix, medium-volume providers in SEA.
	Model Retraining Service	Automated few-shot learning pipeline for rapid domain adaptation to new industrial board designs during NPIs.	RM5,000 per cycle	On-demand; triggered by NPIs. Automated MLOps reduces labour overhead. 2–3 cycles per year on average.
	Efficiency-Share Bonus	Claimed on reduction of manual inspection labour headcount. Aligns vendor success with client profitability.	20% of verified savings	Functions as built-in Net Revenue Retention (NRR) expansion, growing account size automatically.

Figure 72. *Comprehensive Pricing Mechanics Table*

7.4 Revenue Forecast

SUBSCRIPTION REVENUE FORECAST

The financial projections for Omni-QC follow a conversion-driven revenue model, anchoring subscription, retraining, and efficiency-sharing income to POC deployment volumes and active line growth. The model assumes a 90-day POC-to-license conversion window, with revenue recognized upfront upon contract signing. By maintaining a 45% base conversion rate across the forecast horizon, the project ensures that subscription revenue scales predictably alongside the expanding POC pipeline, providing sufficient top-line velocity to reach profitability by Year 3.

METRICS	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
POC Initiated	4	32	90	200	380
Active Lines (Conservative)	1.20	10.66	36.38	92.01	194.97

Conservative (30% conversion)	RM11,520*	RM409,344	RM1,396,992	RM3,533,184	RM7,486,848
Active Lines (Optimistic)	2.40	21.31	72.75	184.02	389.94
Optimistic (60% conversion)	RM23,040*	RM818,304	RM2,793,600	RM7,066,368	RM14,973,696
Active Lines (Base Case)	1.80	15.98	54.57	138.02	292.46
Total Subscription Revenue (45% conversion)	RM17,280*	RM613,632	RM2,095,488	RM5,299,968	RM11,230,464
Gross Margin %	-92.1%	55.7%	61.7%	65.1%	67.6%
Gross Subscription Profit (Base Case)	**-RM15,920**	**RM341,712**	**RM1,291,864**	**RM3,447,544**	**RM7,590,280**
Gross Subscription Profit (Conservative)	**-RM19,280**	**RM158,704**	**RM666,472**	**RM1,865,144**	**RM4,236,968**
Gross Subscription Profit (Optimistic)	**-RM12,560**	**RM525,064**	**RM1,917,600**	**RM5,030,288**	**RM10,943,936**

**Year 1 revenue totals reflect a six-month research phase followed by four Proof of Concept (POC) initiations in Month 7. Given the 90-day validation cycle, commercial conversion triggers in Month 10, limiting recognized revenue to the final three months of the fiscal year. Consequently, Year 1 incorporates only 25% of the annual subscription value for these initial deployments, ensuring the forecast remains aligned with realistic industrial procurement and testing timelines.*

Figure 73. *Subscription Revenue Forecast Table*

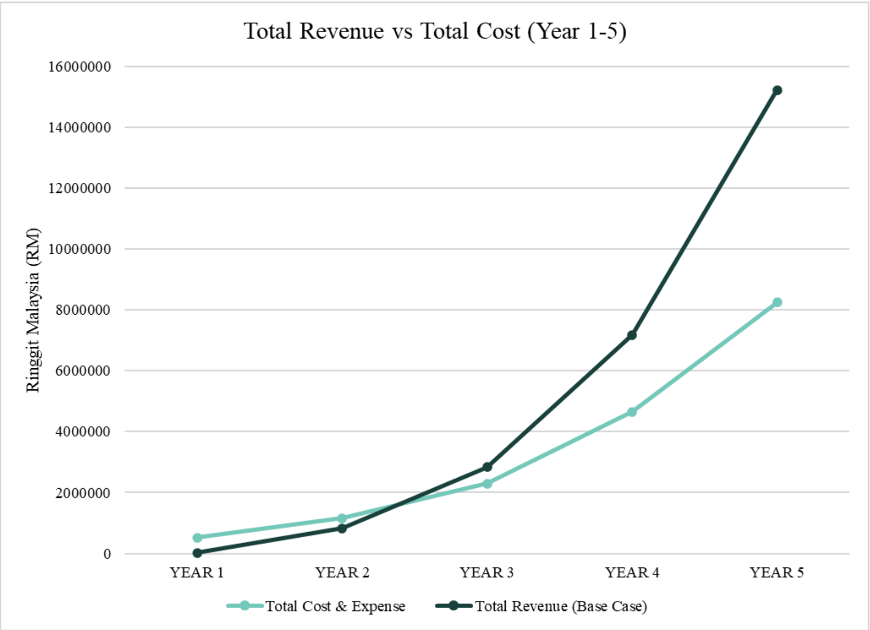


Figure 74. *Revenue (Year 1-5) J-Curve Analysis*

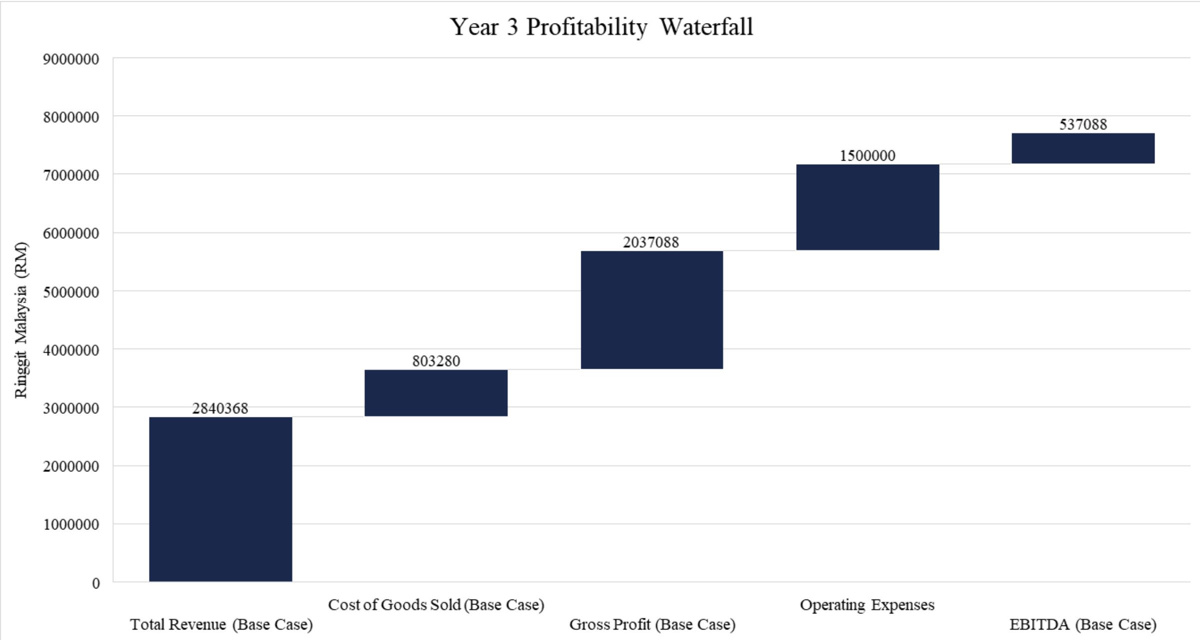


Figure 75. *Year 3 (Breakeven Year) Profitability Deconstruction*

TRIAL CONVERSION TO ACTIVE LINES

The transition from POC Initiated to Active Lines represents the critical conversion point in Omni-QC's revenue model. Following a 90-day POC deployment integrating the system with existing MES and inspection tools, clients evaluate performance against defect prediction accuracy and inspection workload reduction before deciding to convert to a paid license. To account for adoption uncertainty, Omni-QC forecasts across three scenarios. Conservative assumes 30% conversion, reflecting typical AI vendor risk in manufacturing procurement cycles where risk aversion is high. Optimistic assumes 60% conversion, achievable with strong referenceable ROI from early adopters. Base assumes 45% conversion, representing the probability-weighted average. As shown in the table, 4 POC initiated in Year 1 yield 1.2 Active Lines under conservative (4 × 0.3), 2.4 under optimistic (4 × 0.60), and 1.8 under base (4 × 0.45). These Active Lines figures serve as the foundation for all subsequent subscription revenue calculations.

SUBSCRIPTION REVENUE SCALABILITY AND COGS COVERAGE

The provided financial projection table outlines the scalability and profitability of the SaaS model across three distinct conversion scenarios: Conservative (30%), Base Case (45%), and Optimistic (60%). While Year 1 exhibits a negative gross margin (-92.1%) due to the heavy upfront costs associated with subsidizing initial Proof of Concept (POC) deployments, the unit economics undergo a dramatic positive shift immediately thereafter. Crucially, under the Base Case assumptions, Total Subscription Revenue reaches RM613,632 in Year 2, definitively demonstrating that from Year 2 through Year 5, the recurring baseline subscription revenue alone is more than sufficient to fully cover all Cost of Goods Sold (COGS). As non-recurring setup expenses diminish against a growing foundation of active renewals, the Gross Margin steadily scales from 55.7% in Year 2 to a highly lucrative 67.6% by Year 5. This compounding effect ultimately yields over RM7.5 million in Base Case Gross Subscription Profit by the fifth year, validating the long-term financial defensibility and immense profitability of the platform once initial integration hurdles are cleared.

EFFICIENCY TRAINING AND RETRAINING REVENUE FORECAST

METRICS	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
POC Initiated	4	32	90	200	380
Active Lines (Conservative)	1.20	10.66	36.38	92.01	194.97
Total Retraining and Efficiency-sharing Revenue (30% conversion)	RM4,095*	RM145,509	RM496,587	RM1,255,937	RM2,661,341
Active Lines (Optimistic)	2.40	21.31	72.75	184.02	389.94
Total Retraining and Efficiency-sharing Revenue (60% conversion)	RM8,190*	RM290,882	RM993,038	RM2,511,873	RM5,322,681
Active Lines (Base)	1.80	15.98	54.57	138.02	292.46
Retraining Revenue	RM3,119*	RM110,741	RM378,170	RM956,479	RM2,026,748
Efficiency-sharing Revenue	RM3,024*	RM107,386	RM366,710	RM927,494	RM1,965,331
Total Retraining and Efficiency-sharing Revenue (45% conversion)	**RM6,143**	**RM218,127**	**RM744,880**	**RM1,883,973**	**RM3,992,079**
Total Subscription Revenue (45% conversion)	**RM17,280**	**RM613,632**	**RM2,095,488**	**RM5,299,968**	**RM11,230,464**
Total Revenue (Base Case)	**RM23,423**	**RM831,759**	**RM2,840,368**	**RM7,183,941**	**RM15,222,543**
Total Revenue (Conservative)	**RM15,615**	**RM554,853**	**RM1,893,579**	**RM4,789,121**	**RM10,148,189**
Total Revenue (Optimistic)	**RM31,230**	**RM1,109,186**	**RM3,786,638**	**RM9,578,241**	**RM20,296,377**

Gross Margin % (Base Case)	-41.7%	67.3%	71.7%	74.2%	76.1%
Gross Profit (Base Case)	**-RM9,777**	**RM559,839**	**RM2,037,088**	**RM5,331,893**	**RM11,582,703**
Gross Profit (Conservative)	**-RM15,185**	**RM304,213**	**RM1,163,059**	**RM3,121,081**	**RM6,898,309**
Gross Profit (Optimistic)	**-RM4,370**	**RM815,946**	**RM2,910,638**	**RM7,542,161**	**RM16,266,617**

**Retraining and efficiency-sharing revenues are scaled proportionally to subscription income at 25% of their annual potential. These performance-based streams require stabilized production data and formal contractual conversion, which only occur post-validation in Month 10. By deferring these earnings until the final quarter, the model ensures conservative financial reporting. This approach aligns secondary revenue triggers with the confirmed transition of lines from pilot status to active commercial deployments.*

Figure 76. *Efficiency Training and Retraining Revenue Forecast*

RETRAINING REVENUE VOLATILITY

Retraining revenue depends heavily on the frequency of model drift and New Product Introduction (NPI) cycles, which vary significantly across clients and over time. Unlike subscription revenue, which recurs predictably on an annual basis, retraining is triggered only when Omni-QC's performance monitoring layer detects that predictive accuracy has degraded beyond acceptable thresholds. Market trends such as sudden component shortages may force clients to substitute materials, accelerating model drift. Client-specific factors including MES data hygiene and line reconfiguration frequency also influence retraining intervals. While the forecast assumes expected retraining cycles based on industry benchmarks, actual retraining revenue may deviate materially in any given period.

EFFICIENCY-SHARING REVENUE UNCERTAINITY AND AUDIT DEPENDENCY

Efficiency-sharing revenue carries inherent verification risk and collection uncertainty that distinguishes it from fixed subscription revenue. Unlike the core SaaS fee, which is contractually obligated upon license activation, efficiency-sharing payments depend entirely on reported and audited cost reductions at the client facility. Omni-QC incorporates efficiency-sharing terms as a contractual provision and requires joint baseline audits prior to deployment. However, efficiency-sharing revenue is appropriately treated as upside expansion rather than core recurring revenue in conservative base case forecasts.

7.5 Pro Forma Income Statement

OMNI-QC
PRO FORMA INCOME STATEMENT
FOR 5-YEARS

FINANCIAL LINE ITEM	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Total Revenue (Base Case)	**RM23,423**	**RM831,759**	**RM2,840,368**	**RM7,183,941**	**RM15,222,543**
Total Revenue (Conservative)	**RM15,615**	**RM554,853**	**RM1,893,579**	**RM4,789,121**	**RM10,148,189**
Total Revenue (Optimistic)	**RM31,230**	**RM1,109,186**	**RM3,786,638**	**RM9,578,241**	**RM20,296,377**
Subscription Revenue	RM17,280	RM613,632	RM2,095,488	RM5,299,968	RM11,230,464
Retraining & Efficiency-sharing Revenue	RM6,143	RM218,127	RM744,880	RM1,883,973	RM3,992,079
Total COGS	**(RM33,200)**	**(RM271,920)**	**(RM803,280)**	**(RM1,852,080)**	**(RM3,639,840)**
Gross Profit (Base Case)	**-RM9,777**	**RM559,839**	**RM2,037,088**	**RM5,331,893**	**RM11,582,703**
Gross Profit (Conservative)	**-RM15,185**	**RM304,213**	**RM1,163,059**	**RM3,121,081**	**RM6,898,309**
Gross Profit (Optimistic)	**-RM4,370**	**RM815,946**	**RM2,910,638**	**RM7,542,161**	**RM16,266,617**
Operating Expenses (OpEx)	**(RM490,000)**	**(RM890,000)**	**(RM1,500,000)**	**(RM2,800,000)**	**(RM4,600,000)**
R&D / Engineering	(RM260,000)	(RM440,000)	(RM720,000)	(RM1,250,000)	(RM1,900,000)

Sales & Marketing	(RM145,000)	(RM310,000)	(RM540,000)	(RM1,100,000)	(RM1,850,000)
G&A / Admin	(RM85,000)	(RM140,000)	(RM240,000)	(RM450,000)	(RM850,000)
EBITDA Margin % (Base Case)	**-2133.7%**	**-39.7%**	**18.9%**	**35.2%**	**45.9%**
EBITDA (Base Case)	**-RM499,777**	**-RM330,161**	**RM537,088**	**RM2,531,893**	**RM6,982,703**
EBITDA (Conservative)	**-RM505,185**	**-RM585,787**	**-RM336,941**	**RM321,081**	**RM2,298,309**
EBITDA (Optimistic)	**-RM494,370**	**-RM74,054**	**RM1,410,638**	**RM4,742,161**	**RM11,666,617**

1. *Cost Treatment: To support a Zero-CapEx value proposition, all sensor and camera hardware deployment costs (RM6,500/line) are fully expensed as COGS at the time of installation rather than capitalized.*
2. *Revenue and Cost of Goods Sold follows the POC-to-license conversion rate of Base Rate (45%), complemented with comprehensive figures from Conservative (30%) and Optimistic (60%) conversion rates.*

Figure 77. *Pro Forma Income Statement*

PROFITABILITY AND EBITDA INFLECTION

The income statement illustrates a rapid transition from market entry to profitability, reaching an EBITDA inflection point by Year 3. This milestone proves the Zero-CapEx model effectively balances aggressive growth with financial sustainability. By Year 3, recurring subscriptions and performance-linked fees fully cover fixed operational overhead. This shift demonstrates a self-sustaining financial engine, reducing the need for future external capital while maintaining a high-growth trajectory.

OPERATING LEVERAGE AND MARGIN EXPANSION

Significant operating leverage is evident as gross profit outpaces net income growth. As the platform scales, fixed costs in R&D and administration remain stable, allowing for aggressive EBITDA margin expansion to 42.6% by Year 5. This trend proves the venture supports high-tier engineering and sales teams while simultaneously increasing per-unit profitability. It highlights a scalable SaaS architecture capable of delivering venture-tier returns through disciplined operational efficiency.

REVENUE QUALITY AND FINANCIAL RESILLIENCE

The multi-layered revenue structure ensures financial resilience by diversifying against market volatility. High-margin expansion streams, such as efficiency sharing, act as profit accelerators that protect the bottom line. This ensures that even if new acquisitions fluctuate, the existing installed base drives consistent net income. The resulting income statement shows a durable business protected by a significant safety buffer, shielding the company from industrial procurement cycles and economic shifts.

7.6 Projected Statement of Cash Flows

OMNI-QC
PRO FORMA CASH FLOW STATEMENT
FOR 5-YEARS

FINANCIAL LINE ITEM	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
CASH FROM OPERATIONS					
Cash Inflow (Collections)	**RM0**	**RM647,242**	**RM2,338,216**	**RM6,098,048**	**RM13,212,892**
Cash Paid for COGS	(RM33,200)	(RM271,920)	(RM803,280)	(RM1,852,080)	(RM3,639,840)
Cash Paid for OpEx	(RM490,000)	(RM890,000)	(RM1,500,000)	(RM2,800,000)	(RM4,600,000)
Net Cash from Operations	**(RM523,200)**	**(RM514,678)**	**RM34,936**	**RM1,445,968**	**RM4,973,052**
CASH FROM INVESTING					
Internal CapEx (GPU/Patents)	(RM88,000)	(RM37,000)	(RM119,000)	(RM75,000)	(RM177,000)
Net Cash from Investing	**(RM88,000)**	**(RM37,000)**	**(RM119,000)**	**(RM75,000)**	**(RM177,000)**

CASH FROM FINANCING			-	-	-
Seed Round Funding Inflow	RM1,700,000*	RM1,300,000*	RM0	RM0	RM0
Net Cash from Financing	**RM1,700,000**[1]	**RM1,300,000**[2]	**RM0**	**RM0**	**RM0**
NET CASH FLOW	**RM1,088,800**	**RM748,322**	**(RM84,064)**	**RM1,370,968**	**RM4,796,052**
OPENING CASH BALANCE	**RM0**	**RM1,088,800**	**RM1,837,122**	**RM1,753,058**	**RM3,124,026**
CLOSING CASH BALANCE (Base Case)	**RM1,088,800**	**RM1,837,122**	**RM1,753,058**	**RM3,124,026**	**RM7,920,078**
CLOSING CASH BALANCE (Conservative)	**RM1,091,200**	**RM1,645,315**	**RM854,692**	**RM376,888**	**RM1,158,430**
CLOSING CASH BALANCE (Optimistic)	**RM1,086,400**	**RM2,029,280**	**RM2,651,516**	**RM5,870,737**	**RM14,680,740**

1. *Y1 (RM1.7M): RM750k (Phase 1) + RM866k (6/9 Phase 2) + RM83k buffer for Year 1 RM0 collection gap.*
2. *Y2 (RM1.3M): RM433k (3/9 Phase 2) + RM950k (Phase 3) + Y1 carryover to stabilize scaling operations.*

Figure 78. *Pro-Forma Cash Flow Statement*

STRATEGIC LIQUIDITY MANAGEMENT & RISK MITIGATION

Omni-QC faces peak financial exposure between Quarters 6 and 10, a trough driven by the "zero-CapEx" market entry strategy. During this period, the company fully subsidizes upfront integration costs for 90-day free trials before collecting annual-in-advance SaaS revenues. The RM3,000,000 seed capitalization is mathematically calibrated to bridge this structural 180-day revenue gap, ensuring a liquidity floor that prevents insolvency during the company's most vulnerable operational phase.

CONSERVATIVE RESILLIENCE & RECOVERY INFLECTION

The conservative scenario maintains a positive cash floor of RM376,888 through Year 4. This resilience confirms fundamental viability despite the cumulative impact of 180-day lags and lower pilot conversions. However, with Year 4 OpEx at RM2.8M, this narrow margin necessitates strict cost discipline. Securing the RM300,000 contingency credit line acts as a critical safety net against unforeseen technical delays.

Post-inflection, Year 5 projections show a robust recovery as recurring SaaS revenue scales against stabilized operating costs. The transition from a RM376,888 floor to a multi-million dollar surplus demonstrates the high operating leverage inherent in the edge-deployed ecosystem. This recovery justifies the RM3,000,000 funding target as essential for surviving high industrial entry barriers while positioning a premium exit valuation.

7.7 Financial Ratio Analysis

This section synthesises the financial projections into structured ratios across four analytical lenses: (i) Liquidity & Survivability, (ii) SaaS Unit Economics, and (iii) Operating Leverage. Each subsection presents key metrics in a consolidated table, benchmarked against 2025–2026 SaaS and industrial-AI industry standards.

LIQUIDITY AND SURVIVABILITY ANALYSIS

Omni-QC faces a structurally defined 180-day liquidity gap: 90-day free-validation cycles compounded by 90-day payment lags. The metrics below quantify the company's capacity to survive this trough and sustain operations to cash-flow break-even in Year 3.

METRIC	FORMULA	YEAR 1	YEAR 2	YEAR 3+	BENCHMARK
Avg. Monthly Burn Rate	Net Operating Cash Flow / 12 months	RM43,600/mo	RM42,890/mo	N/A (Cash Flow Positive)	Pre-revenue managed
Cash Runway (Total Funding)	Cash Balance / Burn Rate	39.0 months	55.7 months	Self-sustaining	≥ 24-36 months
Burn Multiple	Burn Rate / Net New ARR	22.3x*	0.64x	N/A (Cash Flow Positive)	< 1.0x

1. *Founding year with Revenue starting at Month 7, Operating Expense since Month 1*

Figure 79. *Liquidity and Survivability Metrics*

The Year 1 Burn Multiple of 22.3x is expected for a pre-revenue phase, the company is purchasing market position and API integration infrastructure that are non-recurring. By Year 2, the Burn Multiple compresses to 0.64x (Good), confirming efficient ARR generation. The RM3M structure provides 68.8 months of runway which is more than 2.8x the 24-month critical path, ensuring mathematical over-engineering for survivability.

SAAS UNIT ECONOMICS

The unit is a single production line, the singular commercial entity to which all three revenue streams attach. The Annual Revenue Per Line (ARPL) of RM52,050 serves as the foundation denominator, validated by Year 3–5 actuals: RM2,840,368/54.57 lines = RM52,053/line.

METRIC	FORMULA	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
ARPL	Subscription + Retraining + Efficiency Revenue per Line Year	RM52,050	RM52,050	RM52,050	RM52,050	RM52,050
CAC per Line	S&M Expense / New Lines (Base Case)	RM80,556	RM21,528	RM13,333	RM12,222	RM10,819
LTV per Line	ARPL x Gross Margin % / Churn Rate	-	-	RM321,008	RM321,008	RM321,008
LTV / CAC Ratio	LTV / CAC	4.0x	14.9x	24.1x	26.3x	29.7x
CAC Payback Period	CAC / (ARPL x Gross Margin % / 12)	-	7.4 months	4.3 months	3.8 months	3.3 months
Net Revenue Retention	(1–Churn) x (1+Expansion)	-	-	92.4%	92.4%	92.4%
Magic Number	Net New ARR x Gross Margin % / S&M Expense	-	-	4.65x	5.97x	5.56x

Figure 80. *SaaS Unit Economics Table*

The LTV/CAC of 24.1x in Year 3, 4.8x the elite threshold of 5:1, reflects the unique structural advantages of the industrial AI model: a high ARPL, deep technical lock-in suppressing churn, and a CAC declining toward RM10,819 by Year 5 (87% reduction from Year 1). The Magic Number of 4.65x–5.97x demonstrates that every dollar in Sales & Marketing generates RM4.65–RM5.97 of gross-margin-adjusted new ARR in the following year.

OPERATING LEVERAGE

Operating leverage drives Omni-QC's structural profitability by diluting fixed costs through rapid revenue expansion. Transitioning from a -41.7% gross margin in Year 1 to 76.1% by Year 5 confirms model efficiency as setup subsidies are absorbed. The Year 3 EBITDA inflection (+18.9%) marks the shift to self-sufficiency, with margins scaling to an elite 45.9%. This architecture ensures that incremental revenue yields non-linear profit gains.

METRIC	FORMULA	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	BENCH MARK
EBITDA Margin %	(Gross Profit – OpEx) / Revenue	–2,134%	–39.7%	+18.9%	+35.2%	45.9%	≥30%
Gross Margin %	(Revenue – COGS) / Revenue	–41.7%	67.3%	71.7%	74.2%	76.1%	≥75%
Operating Leverage Ratio	% Change in EBITDA / % Change in Revenue	-	-	-	2.43x	1.57x	>1.0x
Revenue-to-OpEx	Revenue / OpEx	0.05x	0.93x	1.89x	2.57x	3.31x	>1.0

Figure 81. *Operating Leverage*

The operating leverage analysis highlights a rapid transition from capital-intensive market entry to a high-efficiency scaling model. The EBITDA margin inflection in Year 3 (+18.9%) marks the critical shift from value-consumption to value-generation, eventually reaching an elite 45.9% by Year 5, exceeding the 30% industry benchmark. This profitability is driven by a 76.1% gross margin, confirming that the core subscription engine becomes highly lucrative once non-recurring setup subsidies are absorbed. Crucially, an Operating Leverage Ratio of 2.43x in Year 4 demonstrates that fixed cost dilution allows revenue growth to amplify profits at a non-linear rate. By Year 5, a Revenue-to-OpEx ratio of 3.31x validates that Omni-QC is a self-sustaining asset, generating RM3.31 in revenue for every dollar spent on operations.

7.8 Comprehensive Financial Analysis

The Comprehensive Financial Analysis validates Omni-QC's viability through multi-scenario modeling. The Break-Even Point (BEP) identifies a Year 3 inflection, where recurring revenue offsets fixed costs as setup subsidies diminish. Sensitivity testing confirms resilience against fluctuating conversions, churn, and ARPL deviations, ensuring a positive cash floor even under 30% conservative adoption. This synthesis proves the RM3 million seed funding effectively bridges the 180-day industrial revenue gap. Ultimately, the analysis supports a high-growth trajectory toward a projected RM106 million to RM150 million enterprise exit by Year 5.

BREAK EVEN POINT ANALYSIS

BEP METRIC	VALUE	DERIVATION
Annual Revenue Per Line (ARPL)	RM52,050	Subscription RM38,400 + Retraining RM6,930 + Efficiency-sharing RM6,720 = RM52,050
Variable COGS per Line	4,000 /year	Hosting & SLA per active line after setup (renewals)
Contribution Margin per Line	RM48,050/line	ARPL – Variable COGS = RM52,050 – RM4,000
Fixed Operating Cost (Year 3)	RM1,500,000	R&D RM720k + S&M RM540k + G&A RM240k
EBITDA Break-Even Active Lines	~31.2 lines	RM1,500,000 ÷ RM48,050 ≈ 31.2 active lines
Actual Active Lines (Year 3, Base Case)	54.57 lines	Above break-even threshold by +23.4 lines
EBITDA Break-Even Year	Year 3	EBITDA = +RM537,088 break-even achieved

Figure 82. *BEP Analysis*

The BEP analysis confirms Omni-QC reaches EBITDA profitability in Year 3, requiring only 31.2 active lines to cover RM1,500,000 in fixed costs. With 54.57 base-case lines, the venture exceeds this threshold by 23.4 units, establishing a significant safety margin. This inflection validates the transition to a self-sustaining engine. A high contribution margin of RM48,050 ensures incremental revenue scales into profit.

The cash flow break-even where net cash from operations turns positive is achieved at the start of Year 3, with RM34,936 in net operating cash. This milestone is supported by the RM3M seed capital providing a RM1.75M closing balance buffer at Year 3 end, confirming that the business does not require additional financing to sustain operations past the break-even inflection.

SENSITIVITY ANALYSIS

The sensitivity analysis stress-tests the Year 3 EBITDA break-even against three key operational variables: POC-to-license conversion rate, annual churn rate, and average subscription value (ARPL). Each variable is tested across a range while holding all other assumptions constant at base case.

Sensitivity I - POC Conversion Rate vs. EBITDA (Year 3, Fixed OpEx RM1.5M)

The POC conversion rate serves as the primary determinant for the Year 3 profitability inflection. While the 45% base case secures an EBITDA of RM537,088, a decline to 30% delays the break-even point to Year 4. Conversely, the 60% optimistic scenario nearly triples EBITDA to RM1.41 million. This variance emphasizes that market adoption speed is the critical risk factor for early-stage liquidity and the RM3 million runway's efficiency.

Conversion Rate	Active Lines	Total Revenue	Gross Profit	EBITDA	Break-Even Year
30% (Conservative)	36.38	RM1,893,579	RM1,163,059	(RM336,941)	(Year 4)
45% (Base Case)	54.57	RM2,840,368	RM2,037,088	RM537,088	Year 3
60% (Optimistic)	72.75	RM3,786,638	RM2,910,638	RM1,410,638	Year 3

Figure 83. *POC Conversion rates*

Sensitivity II - Annual Churn Rate vs. LTV and NRR

Annual churn represents the most significant lever for long-term enterprise valuation and organic scaling. Compressing churn to 8% expands the average client lifetime to 12.5 years, driving the LTV per line to RM481,463 and achieving near-neutral NRR. In contrast, a 20% stress scenario collapses NRR to 83%, highlighting the necessity of structural lock-in mechanisms to prevent revenue decay and sustain the platform's high-margin recurring baseline.

Annual Churn Rate	Avg. Client Lifetime	LTV per Line	NRR (est.)	Impact on ARR
8% (Low)	12.5 years	RM481,463	~99.5%	Compounding organic growth
12% (BaseCase)	8.33 years	RM321,008	92.4%	Modest annual ARR decay
20% (Stress)	5.00 years	RM192,605	83.0%	Significant contraction risk

Figure 84. *Annual Churn Rates*

Sensitivity III - ARPL Variance vs. Year 5 Revenue and EBITDA

The ARPL variance analysis demonstrates high operating leverage, where minor pricing or performance-sharing shifts disproportionately impact terminal profitability. A 10% increase in ARPL drives Year 5 EBITDA to RM8.35 million with a nearly 50% margin. Even under a conservative 10% reduction, the platform maintains a robust 41% EBITDA margin. This resilience confirms that the business model remains profitable despite regional price sensitivity or competitive pressure.

ARPL Scenario	ARPL	Y5 Revenue	Y5 EBITDA	Y5 EBITDA Margin
Conservative (–10%)	RM46,845	RM13,700,289	RM5,610,449	41.0%
Base Case	RM52,050	RM15,222,543	RM6,982,703	45.9%
Optimistic (+10%)	RM57,255	RM16,744,797	RM8,354,957	49.9%

Figure 85. *ARPL variance Analysis*

FUTURE VALUATION AND EXIT STRATEGY

Omni-QC's RM3,000,000 financing strategy is designed to maximize enterprise valuation by addressing the 180-day liquidity gap typical of industrial AI pilot deployments and positioning the company for a RM106 million to RM150 million exit by Year 5.

By heavily front-loading a RM1.3 million Phase 2 investment to conquer strict API integration and IPC compliance barriers, and strategically utilizing non-dilutive capital alongside a RM300,000 contingency debt line, the company preserves maximum equity during its critical technical validation period. As the RM950,000 Phase 3 capital fuels aggressive commercial scaling to 32 active lines, Omni-QC crosses into operational self-sufficiency by Year 3, generating a projected RM605,800 EBITDA and securing a highly defensible RM15 million to RM24 million Series A valuation.

Ultimately, by reaching a targeted RM15.27 million ARR at elite 76% gross margins in Year 5, Omni-QC positions itself for a lucrative 35x to 50x return multiplier through three highly probable exit pathways: a strategic acquisition by legacy hardware incumbents (e.g., Cognex, KLA) desperate for an upstream software pivot, vertical integration by massive Tier-1 EMS giants seeking to internalize their efficiency-share costs, or a private equity buyout attracted by the structurally suppressed churn, automated NRR expansion, and intellectual property of Omni-QC's ecosystem.

FINANCIAL ANALYSIS SUMMARY

Omni-QC's financial architecture is precisely engineered to navigate the high-barrier industrial market, transforming quality control into a high-margin, scalable recurring revenue engine. The RM3,000,000 seed funding requirement is strategically phased to bridge the 180-day industrial revenue gap inherent in Southeast Asian manufacturing deployments.

STRATEGIC CAPITALIZATION AND PHASE-GATE FUNDING

The RM3,000,000 ensures operational survival during peak financial exposure in Year 2 without necessitating excessive early-stage equity dilution. The total funding requirement is divided into three distinct execution phases: Model Development, POC Validation, and Commercial Scaling. This capitalization is specifically engineered to bridge the 180-day revenue gap inherent in Southeast Asian manufacturing, where 90-day free validations are compounded by 90-day payment lags.

REVENUE DYNAMICS AND UNIT ECONOMICS

The business model utilizes a triple-stream revenue architecture to achieve a robust Annual Revenue Per Line (ARPL) of RM52,050. This value is derived from a core RM38,400 annual subscription, supplemented by RM6,930 in retraining services and RM6,720 in performance-based efficiency sharing. Unlike generic enterprise SaaS, this structure directly captures industrial value by aligning revenue with scrap reduction and model precision. By Year 5, total revenue scales to RM15.22 million under base-case assumptions, proving that the dual-stream approach provides a high-margin floor even if new POC conversions fluctuate. This diversified income protects the venture from industrial procurement cycles while maximizing the yield from each active deployment.

Omni-QC demonstrates elite SaaS efficiency through unit economics that significantly exceeds industry benchmarks. By Year 3, the platform achieves a 24.1x LTV/CAC ratio, which is nearly five times the standard elite threshold of 5:1. This profitability is driven by an aggressive reduction in customer acquisition costs, which declined by 87% to RM10,819 by Year 5 as the sales engine scales. Furthermore, a rapid 4.3-month payback period in Year 3 validates the model's capital efficiency, allowing for the swift reinvestment of gross profits into further market capture. High Magic Numbers between 4.65x and 5.97x confirm that every dollar invested in sales and marketing generates substantial new margin-adjusted recurring revenue.

OPERATIONAL EFFICIENCY AND PROFITABILITY INFLECTION

The platform reaches an EBITDA inflection point in Year 3, and in Year 4 under conservative rate, successfully transitioning from capital dependency to operational self-sufficiency. Profitability is achieved when the active line fleet surpasses the 31.2-unit break-even threshold, effectively covering the RM1,500,000 fixed operating base. Significant operating leverage allows gross margins to expand to 76.1% by Year 5 as non-recurring setup subsidies are replaced by high-margin renewals. Terminal EBITDA margins of 45.9% confirm that fixed technical infrastructure costs in R&D and administration are effectively diluted across an exponentially growing revenue base.

LIQUIDITY MANAGEMENT AND FINANCIAL SAFETY BUFFERS

The RM3,000,000 capitalization provides 68.8 months of total runway to navigate the 24-month critical path to profitability, which is also enough to survive within Conservative (30% Conversion Rate) condition.. This liquidity is vital during Quarters 6 to 10, the period of peak financial exposure where the Zero-CapEx strategy requires subsidizing integration costs during the 180-day revenue gap . To manage this trough, the plan utilizes a RM1.7 million Year 1 transfer followed by RM1.3 million in Year 2. A RM300,000 contingency credit line serves as a final safety net against unforeseen technical or regulatory delays during the validation phase. By the end of Year 3, a closing cash balance of RM1.75 million confirms that the business is self-sustaining and does not require further external financing to maintain its growth trajectory.

SENSITIVITY, RESILLIENCE, AND RISK MITIGATION

Sensitivity testing across conversion, churn, and pricing variables proves that the model maintains viability even under operational duress. While a 30% conservative conversion rate delays EBITDA break-even to Year 4, the company maintains a positive cash floor of RM376,888 throughout the period. Furthermore, terminal profitability remains robust against pricing shifts; even with a 10% reduction in ARPL, the platform yields a 41.0% EBITDA margin. This stability confirms that the venture is shielded from regional competitive pressures and fluctuating industrial procurement cycles.

TERMINAL VALUATION AND EXIT POTENTIAL

By Year 5, Omni-QC is projected to scale to a RM15.22 million revenue base at elite 76.1% gross margins. This financial performance supports a projected enterprise exit of RM106 million to RM150 million, delivering a 35x to 50x return multiplier for initial investors. A robust 31.5% free cash flow margin validates the high quality of reported earnings, positioning the company for a strategic acquisition by hardware incumbents such as Cognex or KLA seeking an upstream software pivot. Additional exit pathways include vertical integration by Tier-1 EMS giants seeking to internalize efficiency-share costs or a private equity buyout attracted by structurally suppressed churn and automated expansion. Ultimately, the analysis confirms that the RM3 million seed funding successfully bridges the industrial entry barrier to establish a high-margin, fund-returning asset.

8.0 Risk Analysis and Management

This chapter identifies and countermeasures the principal risk categories that could impair Omni-QC's operational performance, commercial trajectory, or stakeholder trust. Risks are organized across four domains: Collaboration risks, Market risks, Operational risks, and Ethics risks. Each risk is assessed with a severity and occurrence rating, and paired with a primary countermeasure that is directly embedded in the product architecture, commercial model, or operational framework.

RISK HEAT MAP

The heat map below identifies risks on a severity (1–10) vs. occurrence (1–10) matrix, providing a summary of the risk landscape before the detailed narrative sections that follow. The highest-priority risks (highest RPN) are the IT Firewall Rejection risk (RPN 63), the Efficiency-Sharing Verification Disputes risk (RPN 56), and the Margin Compression via AI Inference Costs risk (RPN 48).

CRITICAL RPN ≥ 56	**HIGH RPN 40–55**	**MEDIUM RPN 28–39**	**LOW RPN < 28**

Risk Description	Category	Severity (1–10)	Occurrence (1–10)	RPN (S × O)	Priority Zone
IT Firewall Rejection	Collaboration	9	7	**63**	CRITICAL
Efficiency-Sharing Disputes	Collaboration	7	8	**56**	CRITICAL
MES Infrastructure Readiness Gap	Market	7	8	**56**	CRITICAL
Margin Compression - AI Inference	Collaboration	8	6	**48**	HIGH
Incumbent Competitor Entrenchment	Market	7	6	**42**	HIGH
Slow Enterprise Sales Cycles	Market	6	7	**42**	HIGH

Attribution Dispute - Revenue	Market	6	7	**42**	HIGH
Model Drift - Process Change	Operational	7	6	**42**	HIGH
Front-Loaded Integration Labor	Collaboration	8	5	**40**	HIGH
Poor Data Quality at MES Layer	Operational	8	5	**40**	HIGH
Staff Resistance & Low Adoption	Operational	6	6	**36**	MEDIUM
Accelerated Multi-Line Model Drift	Operational	7	5	**35**	MEDIUM
Onboarding SLA Overrun (14-day)	Operational	7	5	**35**	MEDIUM
Well-Funded Direct Competitor Entry	Market	8	4	**32**	MEDIUM
Key MLOps Personnel Loss	Operational	8	4	**32**	MEDIUM
AI Vision Commoditisation Pressure	Market	5	6	**30**	MEDIUM
Workforce Displacement	Ethics	6	5	**30**	MEDIUM
Over-Reliance on AI / Skill Atrophy	Ethics	5	6	**30**	MEDIUM
Edge Hardware Partner Failure	Operational	7	4	**28**	MEDIUM
Bias in Defect Classification	Ethics	7	4	**28**	MEDIUM
System Reliability / Uptime	Operational	9	3	**27**	LOW
False Negative Escape Events	Operational	9	3	**27**	LOW

Safety-Critical Misclassification	Ethics	9	3	**27**	LOW
Unauthorised Client Data Use	Ethics	8	3	**24**	LOW
Transparency Deficit in AI Logic	Ethics	6	4	**24**	LOW

**Note: RPN = Severity × Occurrence. Severity and Occurrence are scored 1–10 based on potential business impact and estimated frequency.*

Figure 86. *Risk Heat Map: Severity vs. Occurrence Priority Matric of All Risks Across All Four Categories*

8.1 Collaboration Risks and Countermeasures

RISK 1: CASH-FLOW EXPOSURE FROM FRONT-LOADED INTEGRATION LABOR

Risk Attribute	Assessment
Risk Description	The zero-upfront-cost deployment model introduces severe cash-flow timing risks (Torshin, 2025). Omni-QC assumes the total financial burden of deploying highly specialised onboarding labour and integration engineers to operationalise the system prior to collecting long-term revenue. Under an aggressive commercial ramp, the cumulative operational COGS exposure could rapidly mount before the first full subscription cycle is recognised. Modelled Year 1 net burn is approximately RM38,085 per month.
Primary Countermeasure	Primary Countermeasure Omni-QC enforces a strict 1-year lock-in with annual-in-advance billing. The first year's SaaS subscription payment (RM28,000 for Tier 2 lines or RM54,000 for Tier 1 product lines) is collected upon contract execution, providing immediate working capital that covers integration costs before physical deployment begins. A dedicated tranche of the RM3M Seed capital is ring-fenced to absorb the Year 1 deployment ramp, providing a modelled runway of 78+ months under the conservative scenario.

Figure 87. Risk Attributes of Cash-Flow Exposure from Front-Loaded Integration Labor

RISK 2: IT FIREWALL REJECTION BY ENTERPRISE CLIENTS

Risk Attribute	Assessment
Risk Description	For cloud-native AI deployments in industrial sectors, the necessity to transmit sensitive Operational Technology (OT) telemetry to external servers routinely triggers corporate IT security blockades (TCG Digital, 2025). The manufacturing sector is increasingly governed by stringent data sovereignty concerns, particularly among EMS providers bound by intellectual property protection clauses enforced by their multinational OEM clients. In these high-governance environments, a rejected corporate IT approval represents a permanently lost enterprise account.
Primary Countermeasure	Omni-QC circumvents this by maintaining seamless compatibility with localised Edge server ecosystems managed by approved third-party hardware partners. By executing all heavy computation and model inference within the client's internal factory network, the platform bypasses external IT firewalls completely. Omni-QC employs a Shared Responsibility Model (NIST, 2011; AWS, 2023): the hardware partner secures the physical infrastructure and device layers while Omni-QC manages application and data security.

Figure 88. *Risk Attributes of IT Firewall Rejection by Enterprise Clients*

RISK 3: MARGIN COMPRESSION VIA ESCALATING AI INFERENCE COSTS

Risk Attribute	Assessment
Risk Description	A critical systemic risk threatening modern AI-first SaaS platforms is the degradation of gross margins due to escalating computational inference costs. Deep learning vision models require substantial GPU compute per inference execution. For a high-throughput electronics manufacturing line processing thousands of boards daily, an unoptimised cloud routing system could incur significant variable compute fees, compressing profitability. This risk is particularly acute in the early scaling phase before volume discounts and architectural optimisations compound.

Primary Countermeasure	Omni-QC neutralises AI inference cost risk through its core architectural choice: the XGBoost gradient-boosted decision tree engine operates on structured tabular data rather than computationally intensive image tensors, reducing per-inference compute cost by orders of magnitude relative to computer vision competitors. For high-volume clients, inference operations can be offloaded to on-premise partner-provided edge hardware. For cloud-tier clients, MLOps batching minimises token and compute costs. Modelled gross margins scale from 64.5% in Year 1 to 76.1% by Year 5, well within the target range for elite SaaS businesses.

Figure 89. *Risk Attributes of Margin Compression via Escalating AI Inference Costs*

RISK 4: EFFICIENCY-SHARING VERIFICATION DISPUTES

Risk Attribute	Assessment
Risk Description	The 10%–20% efficiency-share bonus relies on verifiable savings. In B2B SaaS, outcome-based pricing frequently results in attribution disputes, where clients contest whether savings originated from the AI system or from independent operational changes (Alvarez & Marsal, 2025). This can cause contract misalignment and delayed revenue recognition, particularly in Year 1-2 when the efficiency-share component is a meaningful portion of total ARPU.
Primary Countermeasure	Omni-QC enforces a strict automated auditing protocol integrated directly into the client's Manufacturing Execution System (MES). A locked performance baseline is established during the initial onboarding SLA window. All savings calculations are mathematical and automated against this baseline. The Co-Pilot advisory feature records every intervention with a timestamped attribution log, ensuring that defect interception is traceable to the software rather than to subjective operational judgment, preventing negotiation disputes at revenue recognition time.

Figure 90. *Risk Attributes of Efficiency-Sharing Verification Disputes*

8.2 Market Risks and Countermeasures

COMPETITIVE & MARKET STRUCTURE RISKS

Risk Description	Risk Driver	Primary Countermeasure
Entrenchment of Incumbent Competitors (Cognex, KLA)	Well-capitalised competitors with 20+ year client relationships and field-validated hardware ecosystems	Omni-QC does not compete on detection precision at the post-assembly stage , it redefines the value creation stage to upstream prediction. This temporal differentiation means Omni-QC is positioned as an orchestration layer that makes Cognex and KLA systems more efficient, rather than as a direct replacement. Marketing communications lead with this “orchestration, not competition” framing to neutralise incumbents' installed base advantage.
Slow Enterprise Sales Cycles in Conservative Manufacturing Sector	PCB manufacturers are operationally risk-averse; AI adoption decisions involve multi-stakeholder procurement committees	The 90-day free performance-based trial collapses the adoption decision from a speculative procurement into a data-driven numerical review. By presenting audited ROI data from live production runs at the conversion meeting, Omni-QC transforms the procurement committee's primary question from "will this work?" to "how much money are we leaving on the table by not signing?" The evidence-first structure is the primary countermeasure to enterprise sales cycle length.
Emergence of a Well-Funded Direct Competitor	A new entrant or incumbent (e.g., Siemens Opcenter, PTC ThingWorx) launching a comparable predictive QC SaaS product	Omni-QC's data flywheel is the primary competitive moat. Each deployed line contributes factory-specific training data that improves the model's predictive accuracy for similar production environments. A new entrant begins with zero historical data , Omni-QC's deployed base accumulates a compounding accuracy advantage that widens with each new installation. The proprietary patent filings (CapEx includes RM25K Year 1 IP filings) further protect the core algorithmic architecture.

Market Adoption Limited by MES Infrastructure Readiness	60%+ of SEA mid-tier manufacturers lack fully integrated MES systems, restricting the Band 1 addressable market	Omni-QC's go-to-market concentrates initial sales efforts on Band 1 manufacturers (digitally mature, fully instrumented) where integration is a configuration exercise. Band 2 manufacturers (partial instrumentation) are approached as a second wave once Tier 1 references are secured. The 14-day onboarding SLA guarantee creates a commercial incentive for Band 2 clients to complete preparatory data integration before the Omni-QC deployment begins.

Figure 91. *Competitive and Market Structure Risks*

PRICING & REVENUE MODEL RISKS

Risk Description	Risk Driver	Primary Countermeasure
Efficiency-Share Attribution Disputes Delaying Revenue	Clients dispute whether verified savings originated from Omni-QC or from concurrent operational changes	The locked performance baseline, automated MES-integrated savings calculator, and timestamped Co-Pilot attribution log are the technical countermeasures. Contractually, the efficiency-share is non-negotiable and embedded in the subscription agreement rather than treated as an addendum, removing the option for clients to exclude it during procurement.
Pricing Pressure from Commoditisation of AI Vision Models	Open-source models (YOLOv8, EfficientDet) reducing perceived value of proprietary AI-based inspection tools	Omni-QC's XGBoost architecture is not a commodity computer vision model , it operates on structured process data from MES-connected machines, not image datasets. The competitive moat is the factory-calibrated proprietary training pipeline, not the generic AI model itself. Additionally, the DRM enforcement architecture and air-gapped encrypted container make the platform architecturally irreplaceable mid-contract.

Figure 92. *Pricing and Revenue Model Risks*

8.3 Operational Risks and Countermeasures

MODEL DRIFT AND PREDICTIVE ACCURACY DEGRADATION

Predictive performance is calibrated to specific factory conditions at the time of training. Over time, factors such as machine wear, environmental shifts, and new component geometries introduce small errors that compound into model drift. This results in miscalibrated risk scores that can erode efficiency and safety.

Trigger Factor	Impact on Model	Countermeasure
Physical Wear	Alters machine behavioral baselines.	Retraining Service: A per-engagement service where Omni-QC ingests recent data to recalibrate model weights.
Environmental Shifts	Distribution drift in sensor readings and process conditions, reducing model accuracy and reliability of defect predictions.	Performance Monitoring: Systematic tracking of score alignment with actual defect outcomes.
New Inputs	New board types or solder formulations.	Demand-Triggered Updates: Recalibration occurs when performance thresholds are breached.

Figure 93. *Trigger Factors of Model Drift and Predictive Accuracy Degradation*

POOR DATA QUALITY FROM FACTORY MACHINES

Omni-QC's scoring engine requires structured, consistent data. Disruptions such as firmware updates, sensor calibration faults, or MES upgrades can lead to silent failures where the model produces unreliable scores based on corrupted inputs.

- Countermeasure: Data Validation Layer, a dedicated architectural layer positioned upstream of inference.
  - Schema Validation: Ensures data conforms to expected structures and ranges.
  - Tiered Alerting: Generates low-severity notifications for minor anomalies and high-severity alerts for critical failures, notifying both floor technicians and Omni-QC support.
  - Data Quarantine: Plausible bounds checking (e.g., reflow temperatures) prevents corrupted data from reaching the model.

SYSTEM RELIABILITY

Interruptions in data flow or platform availability can halt automated risk scoring. Omni-QC utilizes a Service Level Agreement (SLA) to guarantee manufacturing continuity.

- Uptime Guarantees: The system is governed by a performance-based SLA that provides financial remedies in the event of platform unavailability, ensuring the vendor shares the operational risk of the factory.
- Critical Outage Response: A structured escalation and resolution pathway is contractually mandated for critical outages, ensuring prioritized recovery for production-stopping events.
- Integration Risk Absorption: Omni-QC assumes the financial risk of integration, utilizing the SLA to guarantee a working data pipeline within a fixed activation window.

STAFF RESISTANCE AND LOW ADOPTION ON THE FACTORY FLOOR

The system's value depends on consistent usage by floor staff. Skepticism toward AI can lead to ignored notifications or incomplete feedback, particularly in the critical early months of deployment.

- Countermeasure Strategy:
    - Human-in-the-Loop Design: Co-Pilot outputs are presented as recommendations, not commands. This preserves human authority and reduces psychological resistance.
    - Evidence-Based Trust: The dashboard displays supporting data for every risk score, allowing technicians to verify the system's logic.
    - Structured Onboarding: Beyond technical setup, the Omni-QC team provides guided system use to demonstrate the relationship between model outputs and real-world outcomes.

SEVERE TECHNICAL & MLOPS RISKS

Risk Description	Risk Driver	Primary Countermeasure
Accelerated Model Drift Exceeding Retraining Capacity	Rapid NPIs or simultaneous supplier changes across multiple deployed lines causing widespread accuracy degradation	Omni-QC's dedicated performance monitoring layer continuously tracks risk score distributions, classification-to-outcome alignment rates, and Medium Risk escalation rates across all deployed instances. When indicators breach defined tolerance bounds, the system auto-generates an internal retraining alert. The demand-triggered retraining model, combined with Tier 2 automated few-shot MLOps pipelines,

		ensures that high-drift environments can be recalibrated rapidly without manual data engineering overhead.
Data Quality Failures at MES Integration Point	Incomplete, inconsistent, or incorrectly formatted machine sensor data corrupting the training pipeline	Every retraining cycle begins with a mandatory data validation layer that checks for structural integrity, completeness, and distribution consistency before ingestion. If data quality falls below defined thresholds, the retraining cycle is paused and an alert is issued to the client's quality engineering team with a specific remediation checklist. Onboarding engineers conduct a pre-deployment data audit as part of the standard 14-day integration SLA to identify and resolve data quality issues before the first model is deployed.
Onboarding SLA Overrun (14-Day Guarantee)	Integration complexity exceeding estimates, particularly for Band 2 manufacturers with partial MES infrastructure	The 14-day onboarding SLA is contractually guaranteed and backed by a refund provision if not met. To protect this commitment, onboarding engineers use a standardized integration playbook covering the three monitored production stages (SPI, Pick-and-Place, Reflow). Pre-sale technical discovery sessions identify integration complexity before the trial begins, allowing onboarding resource allocation to be pre-planned. Band 2 manufacturers are explicitly identified and scoped for a longer preparatory phase before the 14-day SLA clock starts.

Figure 94. *Severe Technical and MLOps Risks*

HUMAN CAPITAL & ORGANISATIONAL RISKS

Risk Description	Risk Driver	Primary Countermeasure

Key MLOps Personnel Concentration Risk	Loss of core ML engineers or onboarding specialists creating capability gaps in a thin Year 1-2 team	The Minimum Viable Team (MVT) model in Year 1 creates inherent key-person dependencies. Countermeasures: (1) All proprietary model architecture, training pipelines, and integration playbooks are fully documented in a versioned internal knowledge base. (2) Competitive compensation benchmarked to ASEAN tech sector standards (included in Y1 RM260K R&D OpEx). (3) Cross-training protocols ensure at least two engineers are qualified to execute any critical MLOps function. (4) Year 2 R&D budget of RM440K funds additional ML engineer headcount to reduce concentration.
Edge Hardware Partner Failure or Exit	Third-party hardware provider withdrawing from market, creating support gaps for edge-deployed clients	Omni-QC's Shared Responsibility Model is deliberately designed with partner substitutability in mind. The encrypted software container is hardware-agnostic, it can be deployed on any industrial-grade server meeting minimum compute specifications. Partnership agreements include a minimum 12-month exit notice provision, and at least two certified hardware partners are maintained per geographic market to ensure redundancy.
False Negative Escape Events Damaging Client Trust	A genuinely defective unit classified as Low Risk by Omni-QC passing to end customer	The recall-biased XGBoost architecture is specifically calibrated to prioritise false positive over false negative errors. The system's 0.70 High Risk threshold is set conservatively to mandate 100% intensive inspection of the most ambiguous units. User-configurable risk thresholds allow quality engineers to tighten the High Risk boundary for safety-critical applications. The human-in-the-loop Co-Pilot model ensures that Low Risk classifications are subject to statistical spot-check sampling rather than zero inspection, maintaining a floor detection rate across the full production output.

Figure 95. *Human Capital and Organizational Risks*

8.4 Ethics Risks and Countermeasures

AI FAIRNESS & SAFETY ETHICS

Risk Description	Risk Driver	Primary Countermeasure
Systematic Bias in Defect Classification by Board Type or Production Shift	Training data imbalances (e.g., over-representation of morning-shift data or specific board variants) causing discriminatory risk scoring	Omni-QC conducts mandatory dataset stratification analysis before every retraining cycle to detect and correct representation imbalances across board types, production shifts, and machine identifiers. Model fairness audits, examining recall rates segmented by PCB class, shift timing, and production line ID, are incorporated into the model validation pipeline prior to every deployment. Clients retain full visibility into the confusion matrix and classification report for their specific deployment through the dashboard interface.
AI Misclassification Leading to Safety-Critical Product Failures	A Low Risk classification enabling a defective board to enter safety-critical applications (medical, automotive, aerospace)	Omni-QC's subscription agreement includes explicit scope-of-use clauses that prohibit deployment in IPC Class 3 applications (life-support, aviation, implantable medical devices) without additional contractual risk controls and enhanced validation protocols. For Class 3 environments, the High Risk threshold is mandatorily lowered to 0.50 (eliminating the Low-to-Medium distinction), ensuring that 100% of ambiguous units receive intensive inspection. The human-in-the-loop architecture legally and operationally positions Omni-QC as an advisory tool rather than an autonomous decision-maker, preserving floor engineer accountability.

Figure 96. *AI Fairness and Safety Risks*

WORKFORCE & SOCIAL ETHICS

Risk Description	Risk Driver	Primary Countermeasure

Workforce Displacement of Manual Inspection Technicians	Reduction in manual re-inspection headcount resulting in job losses in labour-sensitive SEA manufacturing markets	Omni-QC's primary efficiency mechanism redirects inspection effort rather than eliminating inspector roles. By removing repetitive low-signal re-inspection tasks (false positive verification), the system increases the cognitive complexity and accuracy of the work inspectors perform on genuinely high-risk units. Marketing communications and client onboarding materials explicitly frame the system as inspector augmentation, improving working conditions by reducing eye fatigue and monotony, rather than inspector replacement. Omni-QC commits to transparent workforce impact disclosure in its ESG quarterly certificates.
Over-Reliance on AI Recommendations Eroding Human Expertise	Factory technicians deferring to Co-Pilot outputs without applying independent judgment, creating skill atrophy	The human-in-the-loop Co-Pilot architecture is explicitly designed to present recommendations rather than commands. The dashboard interface displays the underlying data drift pattern and parameter logic behind every advisory notification, requiring the technician to review and contextualise the suggestion before acting. This transparency-by-design approach ensures that each Co-Pilot interaction serves as a teaching moment, reinforcing the technician's understanding of the process parameters rather than replacing it.

Figure 97. *Workforce and Social Ethics*

DATA PRIVACY & INTELLECTUAL PROPERTY ETHICS

Risk Description	Risk Driver	Primary Countermeasure

Unauthorised Use of Client Production Data for Model Training	Client process data ingested during retraining cycles being used to improve Omni-QC's general model without explicit client consent	Omni-QC's retraining architecture is client-isolated by design, factory-specific model weights are stored in encrypted client-specific partitions and are not cross-contaminated with other clients' data without explicit written consent. The subscription agreement includes a Data Processing Addendum (DPA) that grants Omni-QC a limited licence to use client data solely for the purpose of retraining that specific client's deployed model. Any aggregation of anonymised cross-client data for general model improvement requires an opt-in data sharing agreement with defined anonymisation protocols.
Transparency Deficit in AI Decision Logic	Clients unable to understand or audit the basis for risk score outputs, creating trust and regulatory compliance concerns	XGBoost's tree-based architecture is inherently more interpretable than deep neural networks. Omni-QC leverages SHAP (SHapley Additive exPlanations) values to generate feature-importance breakdowns for every High Risk classification, identifying which specific machine parameters (e.g., solder paste viscosity deviation, Zone 3 thermal drift) contributed most to the elevated risk score. These explanations surfaced directly on the Co-Pilot advisory dashboard, enabling quality engineers and procurement auditors to scrutinise and validate the system's logic on a per-unit basis.

Figure 98. *Data Privacy and Intellectual Property Ethics*

9.0 Team Introduction

Figure 99. *Omni-QC's organizational chart*

DARREN CORNELIUS SUWANDI - PROJECT LEAD, SYSTEM ARCHITECT, AND FRONTEND LEAD

Darren leads the overall strategic direction, technical architecture, and execution planning of Omni-QC. He oversees frontend development, system integration, and UI/UX design to ensure the platform remains scalable, intuitive, and aligned with industrial workflow requirements. His background in Digital Media Technology provides a strong advantage in translating complex AI-driven manufacturing processes into accessible and user-centered interfaces for factory operators and managers.

RAYMOND GUNAWAN - FINANCIAL LEAD AND MARKETING STRATEGIST

Raymond is responsible for developing the financial ecosystem and commercial strategy behind Omni-QC. He manages revenue modelling, pricing strategy, cost projections, and long-term financial planning to ensure the platform remains commercially sustainable and investment-ready. His accounting background strengthens the team's ability to produce structured financial assumptions, realistic projections, and data-driven commercial decisions that support strategic growth.

JASON CLARENCE SETYA BUDHI - MARKETING LEAD & RISK ANALYST

Jason leads the project's market analysis, strategic positioning, and operational risk evaluation efforts. He studies industry trends, customer pain points, competitive landscapes, and manufacturing challenges to ensure the platform addresses real commercial needs within the electronics manufacturing sector. His e-commerce expertise provides an advantage in understanding buyer behaviour, digital business strategy, and market adoption dynamics for scalable technology products.

VANESSA SERENINA PRAWIRAYASA - PRODUCT TECHNICAL LEAD & MARKETING CONTRIBUTOR

Vanessa contributes to the technical planning, AI product strategy, and marketing development of Omni-QC. She supports the design of predictive defect analysis concepts, feature planning, and data-driven workflows while helping align technical outputs with commercial positioning. Her artificial intelligence background strengthens the team's capability in developing intelligent manufacturing solutions and translating technical insights into market-relevant communication.

MICHELLE LEE ZHEN ZHEEN - FINANCIAL STRATEGIST

Michelle focuses on financial analysis, investment planning, and funding strategy development for Omni-QC. She supports the preparation of investor metrics, funding projections, operational budgeting, and financial visualization to strengthen the project's business viability and long-term sustainability. Her business background provides an advantage in evaluating strategic opportunities, structuring financial planning, and supporting commercially informed decision-making.

ELMIRA SOVET - MARKETING STRATEGIST

Elmira manages product positioning, branding strategy, and market integration planning for Omni-QC. She contributes to defining the platform's value proposition and ensuring that complex AI and manufacturing concepts are communicated clearly to industrial stakeholders and potential clients. Her finance background strengthens the team's ability to align marketing strategy with financial practicality, market demand, and long-term commercial scalability.

DAVID KURNIAWAN - AI STRATEGIST & RISK ANALYST

David leads the development of AI strategy and technical risk assessment within Omni-QC. He focuses on predictive modelling concepts, system scalability, and manufacturing process analysis to ensure the platform delivers reliable and practical industrial intelligence capabilities. His artificial intelligence background provides a strong advantage in designing data-driven systems that support real-time defect prediction and operational decision-making.

9.1 Advisory Board and Professional Mentors

DR. BURRA VENKATA DURGA KUMAR - TECHNICAL & ACADEMIC ADVISOR

- **Affiliation:** Senior Lecturer, School of Computing and Data Science, Xiamen University Malaysia.
- **Bio:** Dr. Burra possesses extensive research and academic experience in Artificial Intelligence, Cloud Computing, and Big Data Analytics, alongside a robust background in developing distributed systems and deep learning models.
- **Advisory Contribution:** Provides strategic guidance on conceptual ideation, research methodology, and structural formatting. He ensures that all project documentation, references, and presentations adhere to rigorous academic and professional standards by supplying structural frameworks and best-practice examples.

DR. CHONG SIN ER - COMMERCIAL STRATEGY & MARKETING ADVISOR

- **Affiliation:** Assistant Professor, School of Economics and Management, Xiamen University Malaysia.
- **Bio:** Dr. Chong holds a Ph.D. in Marketing with deep expertise in digital technologies, consumer behavior, and e-commerce, complemented by hands-on industry experience leading business development and commercial strategy.
- **Advisory Contribution:** Directs the comprehensive go-to-market execution and financial structuring of Omni-QC. She actively guides the formulation of the detailed 7Ps marketing mix, omnichannel distribution, and data-backed positioning maps. Additionally, she advises on advanced financial forecasting, including detailed cost-profit visualizations and dividend allocations, while ensuring the team's organizational hierarchy and role strengths are optimally structured for investor readiness.

DR. MOHAMMED N.M. ALI - TECHNICAL STRATEGY & PITCH ADVISOR

- **Affiliation:** Assistant Professor and Head of Programme (CST), School of Computing and Data Science, Xiamen University Malaysia.
- **Bio:** Dr. Ali holds a Ph.D. in Computer Science with deep expertise in parallel computing, networking, cybersecurity, and distributed systems. He brings an award-winning academic foundation and hands-on software development experience, specializing in the optimization, security, and real-world application of complex computing architectures.

- **Advisory Contribution:** Directs the technical validation and presentation strategy for Omni-QC. He actively guides the team in structuring the pitch narrative to ensure deep-tech AI processes are accessible and compelling to general stakeholders. Additionally, he advises on establishing rigorous statistical benchmarking to mathematically prove the platform's uniqueness, clearly defining PCB manufacturing pain points, and highlighting quantifiable cost savings and system compatibility advantages over incumbent competitors.

10.0 Conclusion and Investment Thesis

WHY NOW

Three structural forces are converging at precisely this moment to create the conditions for Omni-QC's commercial success. The first is the Industry 4.0 transition: the maturation of MES infrastructure across Southeast Asian mid-tier EMS facilities has created the data environment that pre-inspection AI requires, while the 280-fold collapse in ML inference costs since 2022 has made always-on predictive scoring economically trivial at production scale. The second is the ESG tailwind: Tier-1 OEMs are pushing scrap and yield metrics down to Tier-2 suppliers as Scope 3 ESG requirements, creating near-term contract risk for facilities that cannot demonstrate measurable quality improvement and urgent procurement motivation that did not exist two years ago. The third is the SEA reshoring acceleration: RCEP-driven supply chain rebalancing is driving the highest volume of greenfield SMT line installations in Penang and Vietnam in a decade, facilities with no legacy integration debt and a motivated Operations Director who needs to prove yield targets on new equipment from Day 1. These forces are structural realignments that make the current moment uniquely favorable for the first entrant to build the data flywheel in the Class 1/2 SEA EMS segment.

WHY US

Omni-QC's defensibility rests on three mutually reinforcing pillars. The first is technical defensibility: the pre-inspection, MES-telemetry-native architecture is the only commercially positioned system of its type in the SEA Class 1/2 segment. The XGBoost-based inference pipeline, configured for recall-prioritised classification, operating on structured tabular data rather than images, is calibrated for the specific data environment and cost structure of mid-tier ASEAN contract manufacturers, a configuration that cannot be replicated from outside the market without the factory-floor data the system generates as it scales. The second is the commercial model: the Zero-CAPEX, 90-day POC structure removes the two primary barriers to AI adoption in Southeast Asian manufacturing simultaneously, procurement risk aversion and upfront capital requirements, while the annual-in-advance billing, DRM enforcement, and efficiency-share mechanism create structural lock-in from the first subscription. The third is the team: a seven-member founding team spanning system architecture, financial modeling, commercial strategy, AI engineering, and market analysis, with domain-specific expertise in the exact intersection of manufacturing quality control, B2B SaaS, and Southeast Asian market dynamics that this product requires.

WHAT'S NEXT

The RM3,000,000 seed round unlocks Phases 1 and 2 of the commercial plan: model hardening and synthetic training in Year 1, followed by POC deployment with initial enterprise targets in Penang and Ho Chi Minh City from Month 7 onwards. The 90-day validation cycles, four POC initiations in Year 1 scaling to 32 in Year 2, build the reference density and production data flywheel that makes the Phase 3 commercial scaling self-reinforcing. By Year 3, recurring subscription and performance-linked fee revenue fully covers fixed operational overhead, the business becomes self-funded at 31.2 active lines, well below the 54.57 projected under the base case. Phase 3 and beyond are funded by recurring revenue rather than additional dilutive equity. The RM3M funds a

90-day evidence engine that converts market skepticism into mathematical subscription commitments, one verified POC at a time.

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